RECERT OBJECTIVES

Note: Some objectives can be found in *The Chart of the Nuclides* book (given during testing) and will be noted as such.

1.01.01 – 1.01.12 Basic Math and Algebra: Add, Subtract, Multiply, Divide, and Convert fractions, decimals, percents, and scientific notation. Solve square roots, variables, and logarithms.

1.02.01 Identify the commonly used unit systems of measurement and the base units for mass, length, and time in each system.

English system (English Engineering System) base units: EES: foot (ft.), pound (lb.), and second (sec.).

International System SI: Base Units: MKS (meter, kilogram, second). CGS (centimeter-gram-second)

1.02.02 Identify the values and abbreviations for SI prefixes.

Table 4. International System (SI) Units

Physical Quantity	Unit	Symbol	Dimensions
Base Units			
Length	meter	m	m
Mass	kilogram	kg	kg
Time	second	s	s
Temperature	kelvin	K	K
Electric current	ampere	A	A or (C/s)
Luminous intensity	candela	cd	cd
Molecular quantity	mole	mol	mol
Selected Derived Units			
Volume	cubic meter	m^3	m^3
Force	newton	N	$kg \cdot m/s^2$
Work/energy	joule	J	$N \cdot m$
Power	watt	W	J/s
Pressure	pascal	Pa	N/m^2
Electric charge	coulomb	C	$A \cdot s$
Electric potential	volt	V	J/C
Electric resistance	ohm	Ω	v/A
Frequency	hertz	Hz	s^{-1}
Activity	becquerel	Bq	dps
Absorbed dose	gray	Gy	J/kg
Dose equivalence	sievert	Sv	$Gy \cdot wR$

Table 2. SI Prefixes

Prefix	Factor	Symbol
yotta	10^{24}	Y
zetta	10^{21}	Z
exa	10^{18}	E
peta	10^{15}	P
tera	10^{12}	T
giga	10^{9}	G
mega	10^{6}	M
kilo	10^{3}	k
hecto	10^{2}	h
deka	10^{1}	da
deci	10^{-1}	d
centi	10^{-2}	c
milli	10^{-3}	m
micro	10^{-6}	μ
nano	10^{-9}	η
pico	10^{-12}	p
femto	10^{-15}	f
atto	10^{-18}	a
zepto	10^{-21}	z
yocto	10^{-24}	y

Found on page 82 in *Chart of the Nuclides* book.

1.02.03 Perform conversion of units.

Examples: Convert nano to kilo, Convert miles to centimeters, Convert years to seconds, etc.

1.02.04 Temperature Conversions

K = °C + 273.15

°F = 1.8(°C) + 32

°C = (°F – 32) / 1.8

Found on page 88 in *Chart of the Nuclides* books.

1.03.01 Define the following terms as they relate to physics:

***Work*:** A force acting through a distance. Units in joules. W=Fd.

***Force*:** A force is a push or pull. Any action on an object to change speed or direction. Units in Newtons.

***Energy*:** The ability to do work. It takes energy to do work and work can generate energy.

1.03.02 Identify and describe four forms of energy:

1. *Kinetic energy*: Energy of motion an object possesses.

   For example, a rolling ball has kinetic energy (energy of motion) and it uses that energy to perform work (force the ball to keep rolling across a distance).

$$Kinetic\ Energy\ = \left(\frac{1}{2}\right) mass * velocity^2$$

$$E_k = \left(\frac{1}{2}\right) m * v^2$$

2. *Potential energy*: Indicates how much energy is stored because of the position or the configuration of an object.

   For example, a ball held in the air has higher potential energy than a ball resting on the ground due to its position/configuration. If the ball in the air is released, all the potential energy will perform work and force the ball to travel across a distance (fall to the ground). As the ball gains motion, potential energy is converted to kinetic energy.

$$Potential\ Energy\ =\ mass\ *gravity\ freefall\ acceleration\ *height\ vertical\ distance$$

$$E_p = m * g * h$$

3. *Thermal energy*: The energy that results from the random motion of molecules.

   When a substance is heated, its temperature rises because its molecules move faster, gaining thermal energy. For example, placing water over a heat source causes the water molecules to begin rapidly and randomly moving/vibrating which increases the temperature of the water.

   Units in calories (CGS) or BTUs (English).

   a. The CGS uses calories where:
      Calorie= amount of heat needed to raise the temperature of 1 gram of water by 1 °C

   b. The English system uses BTU where:
      BTU= amount of heat needed to raise the temperature of 1 pound of water by 1 °F

4. *Chemical energy*: The energy that is derived from atomic and molecular interactions in which new substances are produced.

   For example, dry wood contains chemical energy and when the wood is burned in a fireplace, the chemical energy is released (generating light and heat) and the wood is converted into a new substance – ash. This is different than heating water because a new substance is produced that cannot be reversed.

Even if water is boiled completely into steam/vapor it is reversible since the steam/vapor can be cooled back to the liquid state of water; therefore, boiling water does not produce a new substance. Ash cannot be returned to the state of dry wood so burning wood uses chemical energy for a chemical reaction.

- *Other Forms of Energy:* Other forms of energy, electrical and nuclear. Energy may also appear as acoustical (sound) or radiant (light) energy.

1.03.03 State the law of Conservation of energy:

The Law of Conservation of Energy states "The total amount of energy in a closed system remains unchanged."

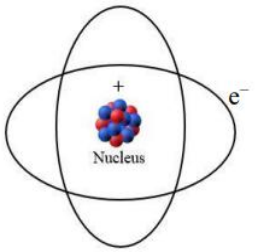
Stated in other terms: If no energy enters or leaves the system, the amount of energy in the system will be the same, although it can be converted from one form to another.

1.03.04 Distinguish between a solid, liquid and a gas:

1. Gas: Indefinite shape and indefinite volume.
2. *Liquid:* Definite volume and indefinite shape.
3. *Solid:* Definite shape and definite volume.

1.03.05 Identify the basic structure of the atom, including the characteristics of the particles:

The atom consists of protons and neutrons in the nucleus and electrons in structured orbits. Typically shown with Bohrs Model.



Proton: Charge +1
Mass 1.007276470 amu.
Elements are determined by the number of protons in the nucleus.
All atoms of the same element have the same number of protons.

Neutron: Charge 0 (Neutral)
Mass 1.008665012 amu.
Number of neutrons determine the Isotope of an element.

Electron: Charge -1
Mass .00054858026 amu.
The number of electrons in the outer shell determine the chemical properties of the atom.
Atoms with the same number of protons as electrons cancel charges and are neutral atoms.

1.03.06 Define the following terms:

- *Atomic number:* The number of protons in the nucleus of an element. Represented by the symbol Z.

- *Mass number:* The total number of protons and neutrons in the nucleus of a particular isotope.

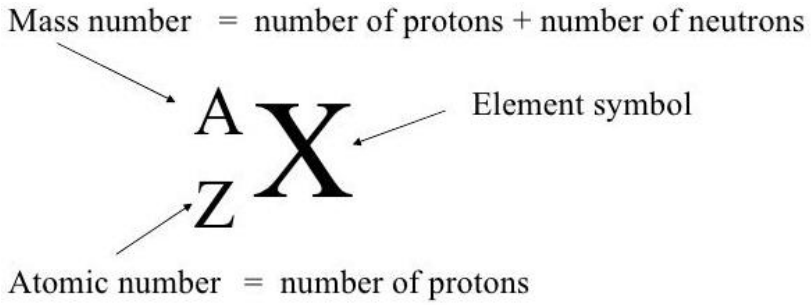
- *Atomic mass:* The actual mass of an atom of a particular isotope.

  Expressed in amu (atomic mass units).

- *Atomic weight:* The weighted average of the isotopic masses of an element based on the percent abundance of its naturally occurring isotopes, expressed in amu units**.**

  Stated in other terms: Atomic Weight is the overall average weight of an element based on all natural isotopes of that element, whereas Atomic Mass is the specific mass of just one **isotope.**

1.03.07 Identify what each symbol represents in the X^A_Z notation:



A = number of protons + number of neutrons

Z = number of protons

A – Z = number of neutrons

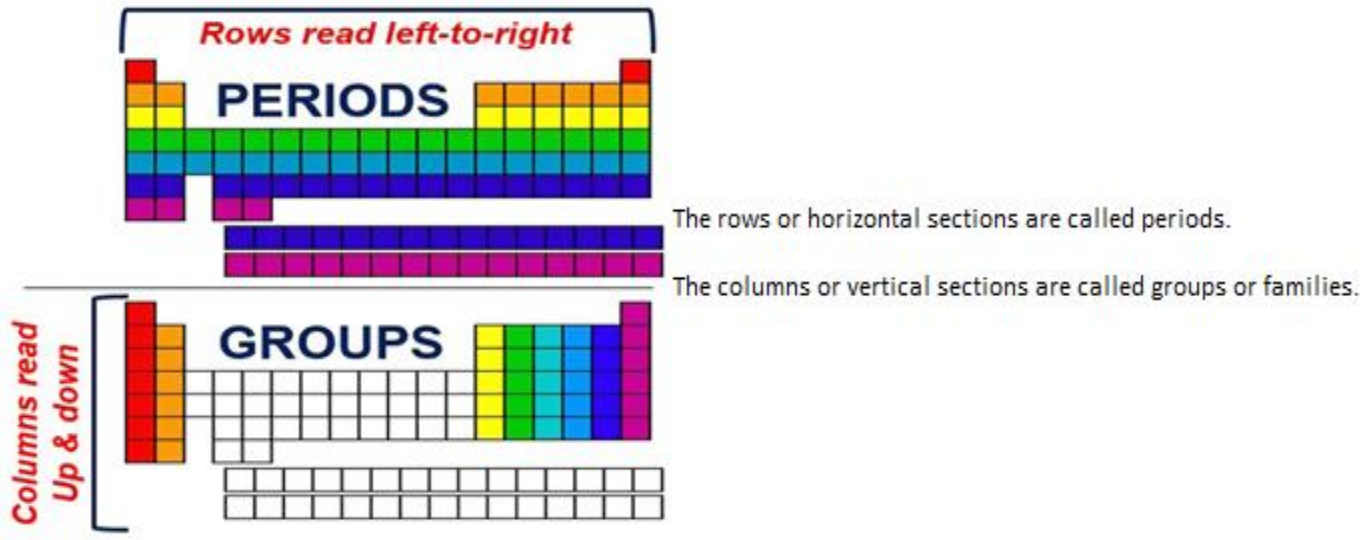
Number of neutrons = Mass Number – Atomic Number*

Found on page 10 in *Chart of the Nuclides* book (described in a text paragraph).

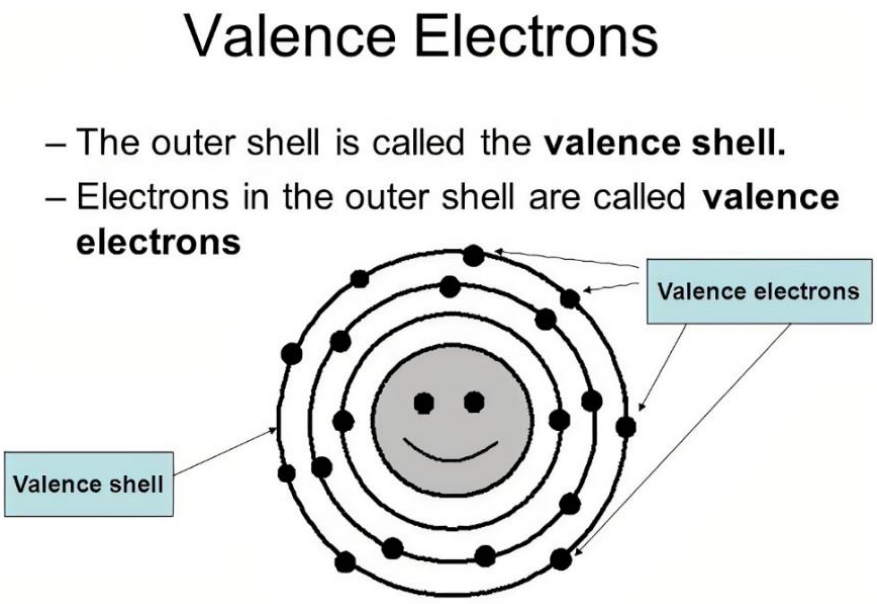
1.03.08 State the mode of arrangement of the elements in the periodic table:

The periodic table is an arrangement of the elements in order of increasing atomic number.

1.03.09 Identify periods and groups in the periodic table in terms of their layout:



1.03.10 Define the terms as they relate to atomic structure:



Valence shell: The highest occupied energy level in a ground state atom.

Valence electron: an electron in the valence shell.

1.04.01 Define the following terms:

Nucleon: A particle within the nucleus of an atom (a proton or a neutron).

Nuclide: Species of atom characterized by the specific constituents (particles) of its nucleus, which is specified by its number of protons, number of neutrons, and energy content.

Descriptions of each nuclide and the differences between them can be seen in *The Chart of the Nuclides.*

Isotope: Nuclides with the same number of protons but differing numbers of neutrons.

1.04.02 Identify the basic principles of the mass-energy equivalence concept:

$E = mc^2$

E = Energy
m = mass
c = speed of light

Mass and energy are interchangeable.
All Matter contains energy by virture of its mass.
Pair annihilation is an example of Mass to Energy conversion.

1.04.03 Identify the definitions of the following:

Mass defect: The total mass of an atom is less than the sum of its individual protons, neutrons, and electrons.

Binding energy: The energy equivalent of the mass defect.

Binding energy is the energy used by the individual pieces of an atom to hold it together. The nucleons (protons and neutrons inside the nucleus) convert a portion of their mass into energy that binds them together. For this reason, the Atomic Mass listed for an isotope is often less than the mass calculated by adding together the parts of the atom (protons + neutrons + electrons). This difference in mass is known as the Mass Defect and is equivalent to binding energy.

Binding energy per nucleon: The binding energy of a nucleus divided by the total number of nucleons. This is the average energy that must be supplied to remove a nucleon from the nucleus.

1.04.04 Identify the definitions of the following:

Fission: The splitting of a nucleus into two or more smaller nuclei accompanied by a release of energy. Usually this also releases 2 to 3 neutrons.

Criticality: The condition in which the neutrons produced by fission are equal to the number in the previous generation.

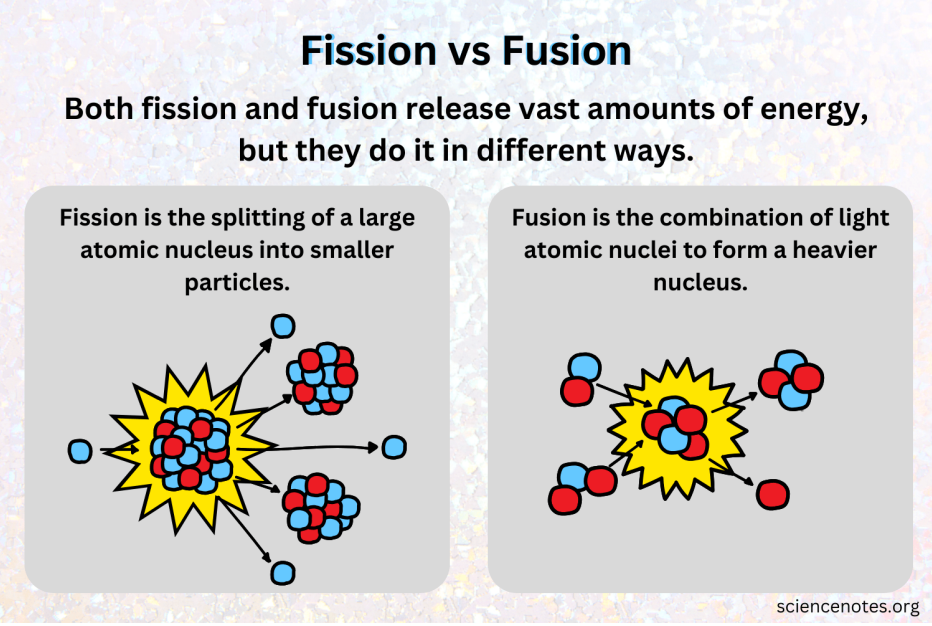
Fission occurs when a neutron slams into a larger atom, forcing it to excite and split into two smaller atoms. When the atom splits, it releases energy and neutrons. The released neutrons are now free to slam into other atoms of the same type, repeating the fission process. This creates a chain reaction of fission events.

The number of neutrons that are released (and therefore the expected number of chain reactions) is expressed using the *effective multiplication constant,* K_{eff}.

If released neutrons escape the system or do not result in fission, the system is *subcritical,* and the chain reaction will eventually stop. If the population of neutrons released remains constant and continues to produce the same number of fission events per neutron released, then the system is *critical,* and the chain reaction will be sustained. If each fission produces multiple neutrons which go on to each fission and create multiple neutrons the number of neutrons increases exponentially, and the system is *supercritical.*

K_{eff}	Effective multiplication constant
$K_{eff} < 1$	subcritical
$K_{eff} = 1$	critical
$K_{eff} > 1$	supercritical

Fusion: The act of combining or “fusing” two or more atomic nuclei, which builds atoms by combining nuclei into a larger nucleus with an accompanying release of energy.



1.05.01 Identify the following sources of natural background radiation including origin, radio nuclides, and contribution to exposure:

1. *Terrestrial:* 20 mrem / yr.
   K40, Rb87, UR, Th from soil and rocks.

2. *Cosmic:* 30 mrem / yr.
   Elevation to space
   Galactic and solar rays from the atmosphere.

3. *Internal Emitters*: 30 mrem / yr.
   K40, Ra87, from food chain.

4. *Radon:* 230mrem / yr.
   Ur238, Th from the Earth's crust.

1.05.02 Identify the following sources of synthetic radiation and the dose received from each:

1. *Nuclear Fallout*: Term applied to debris that settles to the earth as the result of a nuclear blast.

2. *Medical Exposure*: 300 mrem / yr
   Computed Tomography ~150 mr/yr
   Nuclear Medicine ~ 75 mr/yr
   Radiography/fluoroscopy ~ 75 mr/yr

3. *Consumer Products:* 10 mr / yr
   Ra226, H3, C14 from TVs, Smoke detectors, tobacco, 70% of which is from building materials.

4. *Nuclear Facilities:* Current estimates of the yearly average dose equivalent in the U.S. from environmental release is <1 mrem.

The average annual total background radiation is 620mr/yr.
310mr/yr (half) from manmade sources and 310mr/yr (half) from natural sources.

Sources of Radiation

Terrestrial	20 mR/yr
mainly uranium and thorium with the Earth (soil and rocks)	
Atlantic Ocean and Gulf of Mexico: 15-35mrem	
the Colorado Plateau region: 75-140 mrem	
remainder of the country: 35-75 mrem	

Cosmic	30 mR/yr
Primary: galactic cosmic rays (87% proton, 11% alpha & 1% heavier nuclei)	
Geomagnetically trapped & Solar cosmic rays (Charged Particles ≥ 1GeV)	
Secondary: interactions that occur when the primary rays reach the earth's atmosphere; pions, muons, electrons, photons, protons, & neutrons are produced as well as up to 108 secondary radiations from each primary ray (75% muons)	

Internal Emitters	30 mR/yr
some activity is transferred to humans through the food chain	
Human Body: ^{40}K, ^{87}Rb, ^{226}Ra, ^{238}U, ^{210}Po, and ^{14}C	

Radon	230 mR/yr
Radon (^{222}Rn) & Thoron (^{220}Rn) from ^{238}U & Thorium decay chain	

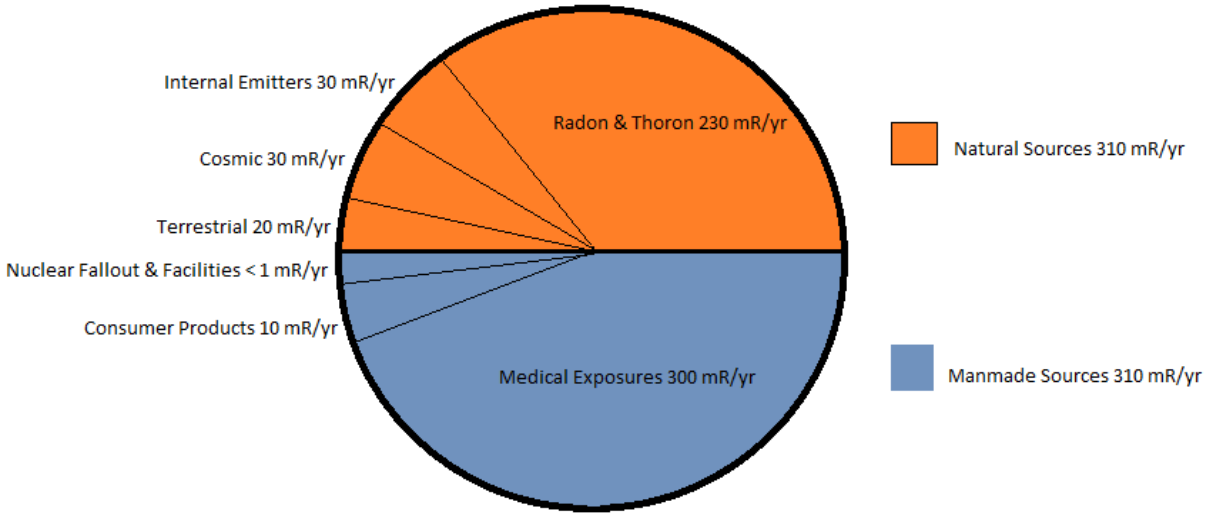
Medical Exposures	300 mR/yr

Computed tomography	150 mR/yr
Nuclear medicine	75 mR/yr
Radiography/fluoroscopy	75 mR/yr

Consumer Products	10 mR/yr
Building materials - 70%, Tobacco Products, & Airplane travel	

TOTAL **620 mR/yr**

Nuclear Fallout < 1 mR/yr & Nuclear Facilites < 1 mR/yr



1.06.01 Identify how the neutron to proton ratio is related to nuclear stability:

The higher the neutron to proton ratio (n:p ratio) the more unstable atoms become.
If n:p ratio is higher than 1.5:1 the nuclei will not be stable.

Low atomic numbers have the greatest stability. Above Z = 20, the n:p ratio increases until Z = 83 (bismuth) where n:p exceeds 1.5:1. At Z > 83 there are no completely stable nuclei.

There are three main forces acting in the nucleus:

Force	Interaction	Range
Gravitational	**Weak force among all nucleons**	**Relatively long**
Electrostatic	**Strong repulsive force among like-charged particles (protons)**	**Relatively long**
Nuclear force	**Strong attractive force among all nucleons**	**Extremely short**

1.06.02 Identify the definition of the following terms:

Radioactivity: The property of certain nuclides to spontaneously emit radiation.

Radioactive decay: The process by which a nucleus spontaneously disintegrates (or transforms) by one or more discrete energy steps until a stable state is reached.

1.06.03 and 1.06.04: Identify the characteristics of α, β, γ, radiations:

Alpha decay: +2 charge (an alpha particle is two protons)

Stopped by sheet of paper or few cm's of air.

Mass of 4 AMU

$\alpha^4{}_2\ Ra^{226}{}_{88} \rightarrow Rn^{222}{}_{86} + \alpha^4{}_2$ $\qquad {}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}X + {}^{4}_{2}\alpha$

Beta decay: –1 charge (same as an electron)

0.000549 amu (same mass as an electron)

β^- or β symbol is used to designate a beta particles.

Expected when n:p ratio is high. Travels 50 ft. in air.

$Pb^{210}{}_{82} \rightarrow Bi^{210}{}_{83} + \beta^- + v^-$ $\qquad {}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + \beta^- + \bar{\nu}$

v^- = antineutrino

Positron decay: Low n:p ratio (excess protons)

β^+ symbol used to designate a positron.

Positron emission results in the changing of a proton into a neutron.

$Ni^{57}{}_{28} \rightarrow Co^{57}{}_{27} + \beta^+ v^-$ $\qquad {}^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + \beta^+ + \nu$

Electron capture: Low n:p ratio.

This occurs when a nucleus captures an electron from the orbital shell of the same atom (usually from the K-shell because it is closest to the nucleus). Also referred to as K-capture. The captures electron combines with a proton to form a neutron, followed by the emission of a neutrino.

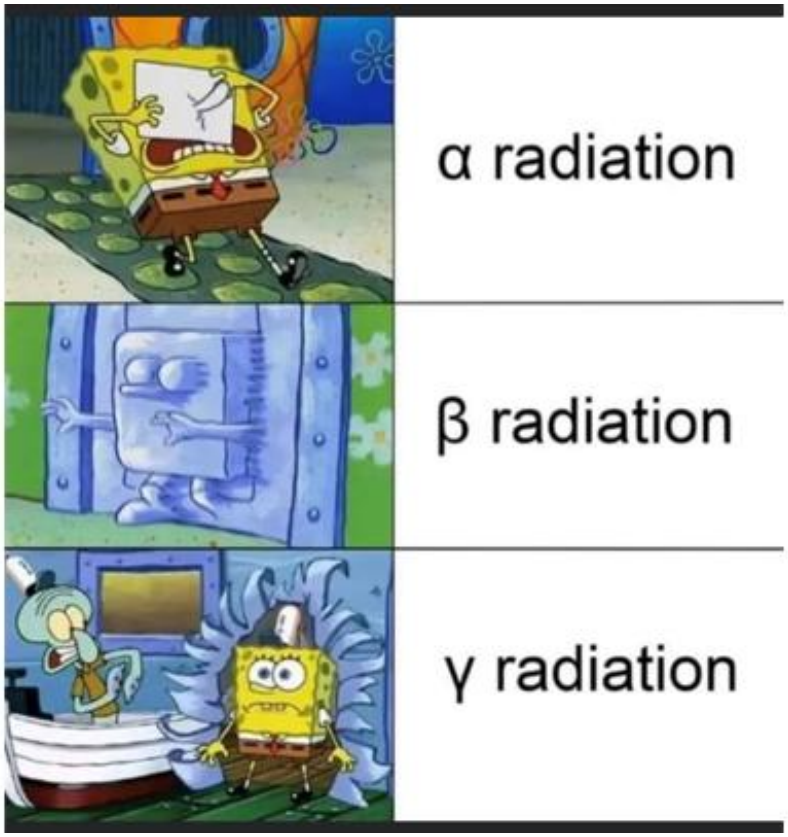
$Ni^{57}{}_{28} + e^- \rightarrow Co^{57}{}_{27} + v$ $\qquad {}^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + \nu$

Gamma emission: Nuclei in an excited state reach a ground state by gamma emission.

γ's are usually higher energy than x-rays.

X-rays originate in the electron shells while gamma rays originate in the nucleus.

${}^{198}_{79}Au^* \rightarrow {}^{198}_{79}Au + \gamma$ $\qquad {}^{A}_{Z}X^* \rightarrow {}^{A}_{Z}X + \gamma$



1.06.05 Identify two aspects associated with the decay of a radioactive nuclide:

Each radionuclide, artificial and natural, has its own characteristic of decay**.** There are four main aspects associated with this pattern:

1. Decay Modes.
2. Emission Types.
3. Energies of the emissions involved.
4. Decay rate.

1.06.06 Identify the differences between natural and artificial radioactivity:

Natural is naturally occurring in nature.
Artificial is occurring from manmade sources.

1.06.07 Identify why fission products are unstable:

The products resulting from fission tend to have a high n:p ratio making them unstable.

1.06.08 Identify the three naturally occurring radioactive families and the end product of each.

1. *Uranium* series: $U^{238}_{92} \rightarrow Pb^{206}_{82}$

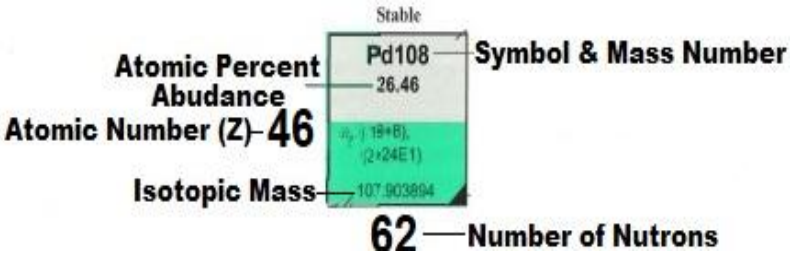
2. *Actinium* series: $U^{235}_{92} \rightarrow Pb^{207}_{82}$

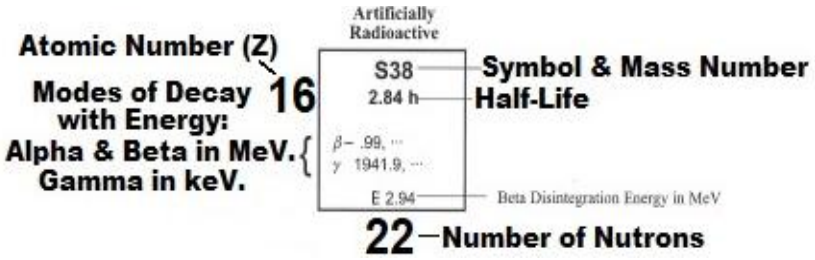
3. *Thorium* series: $Th^{232} \rightarrow Pb^{208}_{82}$

Note that each series ends in a different stable isotope of Lead (Pb).
Found on page 27 in *Chart of the Nuclides* book.

1.06.09 Given a nuclide, locate its block on the chart of the nuclides and identify the following:

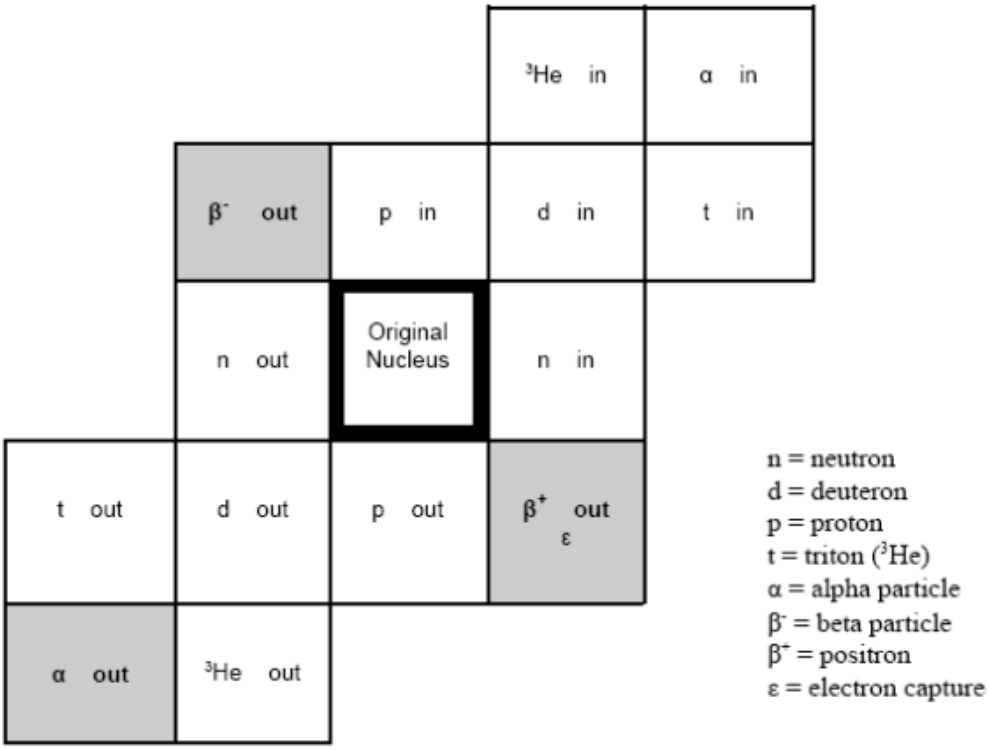
a. Atomic number.
b. Atomic mass.
c. Natural percent abundance.
d. Stability.
e. Half life.
f. Types and energies of radioactive emissions.





Found on page 44 in *Chart of the Nuclides* book.

1.06.10 Given the chart of the nuclide, trace the decay of a radioactive nuclide & identify the stable end product:



Found on page 23 and page 44 of the *Chart of the Nuclides* book.

1.06.11 Identify the definition of the following units:

Curie: 3.7 x 10^{10} disintegrations per second or 2.22 x 10^{12} dpm.
The curie is the historical unit for activity.

Becquerel: One disintegration per second (1dps).
The becquerel is the SI (International System of Units) unit for activity.

1.06.12 Identify the definition of specific activity:

Specific activity: Activity per unit mass (activity/mass). Reported in curies per gram (Ci/g) or becquerels per kilogram (Bq/kg).

1.06.13 Identify the definition of half-life:

Half-Life = $T^{1/2}$ = the time required for the activity present to be reduced to one-half.

1.06.14 Calculate Activity, time of decay, and half-life using the formula for rad decay:

$A_t = A_o e^{-\lambda t}$

A_t = activity after time

A_o = original activity

$\lambda = \frac{.693}{T^{1/2}}$

t = time (same units as $T^{1/2}$ above)

$A_t = A_o (.5)^n$

$n = \frac{t}{T^{1/2}}$

n = number of half lives

t = time elapsed

$T^{1/2}$ = half life

1.06.15 and 1.06.16 Identify the definition of the following terms and units:

Exposure: A measure of the ability of a photon to produce ionization in air, the traditional unit is roentgen (R),
1R = 2.58 x 10^{-4} coulombs/kg of air
(coulomb is a unit of electrical charge)

Absorbed dose: Dose = measure of the amount of radiation energy deposited per unit of mass.
Old unit: Rad = 100 ergs deposited in one gram in any material.
New unit: Gray = 1 joule deposited per kilogram (1J/kg).
1 Gray = 100 Rad

Equivalent Dose (H): The absorbed dose multiplied by the quality factor.
Old unit: Rem = Rad x quality factor
New unit: Sievert = 100Rem

Radiation Weighting Factor: Used to calculate the equivalent dose from the average tissue absorbed dose.
The absorbed dose is multiplied by the appropriate radiation weighting factor.
This is done to relate the absorbed dose to the biological damage caused to the tissue.

Roentgen: Unit of exposure defined as the change per unit of mass of air.
1R = 2.58 x 10^{-4} coulombs/kg of air

Rad/gray: Rad is the traditional unit of absorbed dose. It is an acronym for “radiation absorbed dose.”
Gray (Gy) is the SI derived unit of absorbed dose.

Rem/sievert: Rem is the unit of equivalent dose. It is an acronym for “roentgen equivalent man.”
Sievert (Sv) is the SI derived unit of equivalent dose.

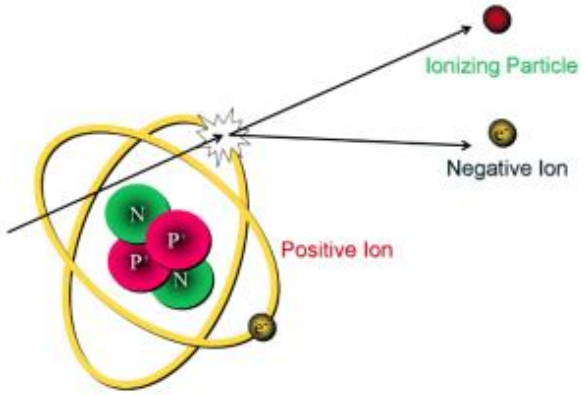
$H_T = D_{T,R}\ w_R$

H_T = equivalent dose
$D_{T,R}$ = average absorbed dose
W_R = radiation weighting factor

Type and Energy Range	Radiation Weighting Factor
Photons, electrons, & muons, all energies	1
Neutrons <10 keV (thermal)	5
Neutrons 10 to 100 keV	10
Neutrons >100 keV to 2 MeV	20
Neutrons >2 to 20 MeV	10
Neutrons >20 MeV	5
Protons other than recoil protons >2 MeV	5
Alpha particles, fission fragments, heavy nuclei	20

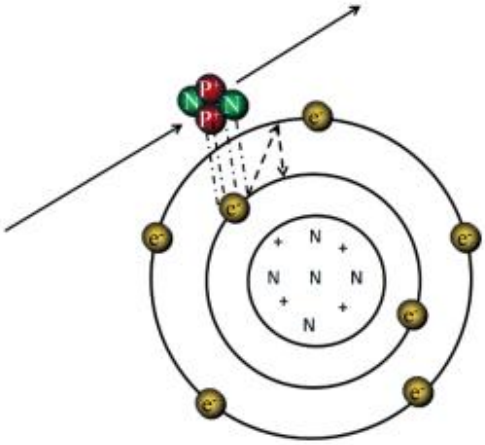
1.07.01 Identify the definitions of the following terms:

Ionization: Any process that results in the removal of a bound electron (negative charge) from an electrically neutral atom or molecule by adding enough energy to overcome its binding energy.



Excitation: Any process that adds enough energy to an electron of an atom or molecule so that it occupies a higher energy state than its ground state.

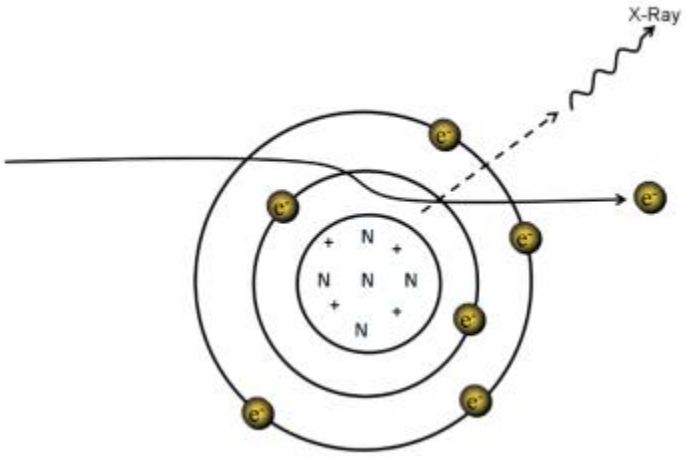
Stated another way: An electron is attracted to a passing positive particle (such as an alpha particle) and "jumps" to a higher shell in the electron cloud (higher energy state).



Bremsstrahlung: The radioactive energy loss of moving charged particles as they interact with the matter through which they are moving.

Also known as “braking radiation.”

Stated another way: As an electron passes through an atom it is attracted to the protons in the nucleus which causes it to slow down (“brake”) when they are nearby. Since the electron becomes slower it needs less energy so the excess energy is released as an X-ray (gamma rays and x-rays are the same type of energy but since it was created outside the nucleus it is called an x-ray).



1.07.02 Identify the definition of the following terms:

Table 1 - Summary of Energy Loss Terms and Units

Term	Abbr.	Definition	Units
W-Value	W	the average amount of energy needed to produce an ion pair in a given medium	eV/ion pair
Specific Ionization	S.I.	(average number of) ion pairs produced (by a charged particle) per unit distance traveled in an absorbing medium	Ion pairs/cm
Linear Energy Transfer	LET	(average value of) energy locally deposited (by a charged particle) in an absorbing medium per unit distance	keV/cm
Range	R	average distance traveled by a radiation in an absorbing medium	cm
Stopping Power	S	For a given absorber, the average energy lost by a charged particle per unit distance traveled	keV/cm

1.07.03 Identify the two major energy transfer mechanisms for alpha radiation:

1. *Ionization*
2. *Excitation*

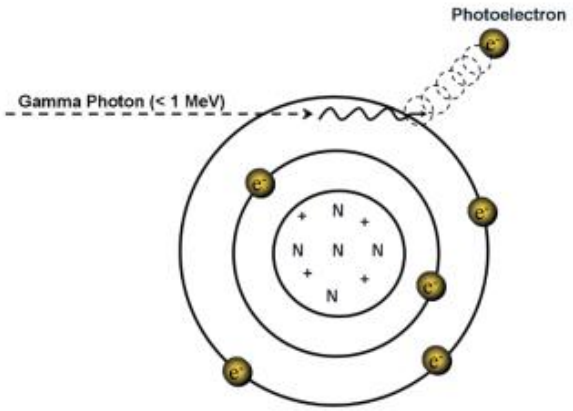
1.07.04 Identify the three major energy transfer mechanisms for beta radiation:

1. *Ionization*
2. *Excitation*
3. *Bremsstrahlung*

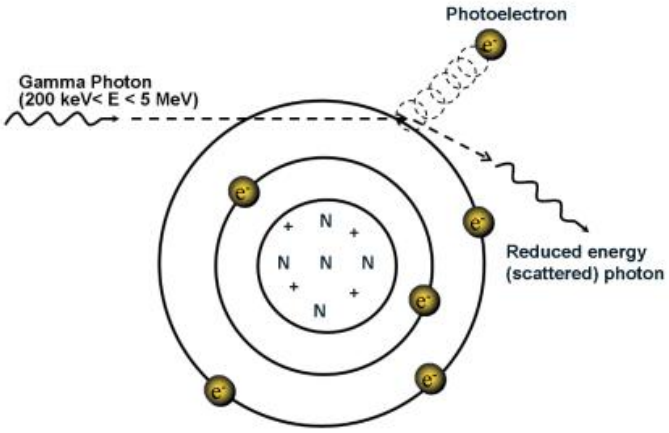
1.07.05 Identify the three major mechanisms by which gamma photon radiation interacts with matter:

1. *Photoelectric effect:* An *all-or-none* energy loss. The mechanism in which a photon interacts with matter and imparts all its energy to an orbital electron which is then ejected at a velocity that matches the energy.

Generally only photons less than 1 MeV.

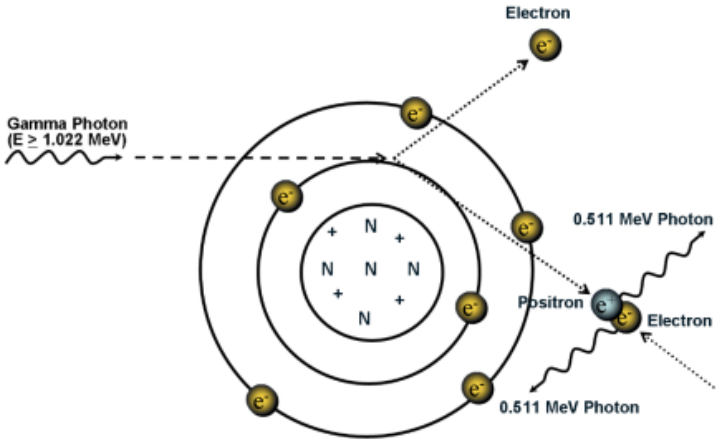


2. *Compton scattering:* A *partial* energy loss. A photon interacts with an atom's orbital electron and only part of its energy is transferred to the electron resulting in the electron ejecting at a lower velocity and the remaining unabsorbed energy bouncing off and continuing in a different direction as a lower-energy photon.



3. *Pair production:* Occurs when all a photon's energy is converted to mass. In pair production, a gamma photon comes near the vicinity of a nucleus and immediately converts to a positron-electron pair. That is, the gamma energy simply disappears and in its place a positron-electron pair appear as the mass replacement.

Pair production is only possible if the gamma possesses energy >1.022MeV (the minimum needed to create the mass of the positron-electron pair)



1.07.06 Identify the 4 main categories of neutrons as they are classified by kinetic energy for interaction in tissue:

Table 2. Neutron Energy Categories

Category	Energy Range
Thermal	≈ 0.025 eV (< 0.5 eV)
Intermediate	0.5 eV - 10 keV
Fast	10 keV - 20 MeV
Relativistic	> 20 MeV

1.07.07 Identify three possible results of neutron capture for slow neutrons: (n,γ)

Radiative capture with gamma emission is the most common reaction type for slow neutrons.

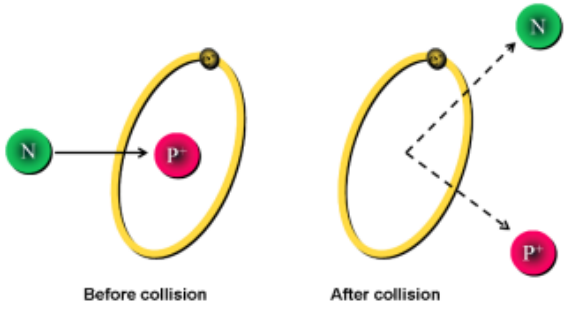
1. *Neutron activation:* The process of converting a stable nucleus to its radioactive counterpart by neutron bombardment. Introducing (bombarding an atom with) neutrons causes it to become unstable which "activates" the atomic reaction.

2. *Charged particle emission:* A reaction in which a proton, a deuteron, or an alpha is ejected from the target nucleus.

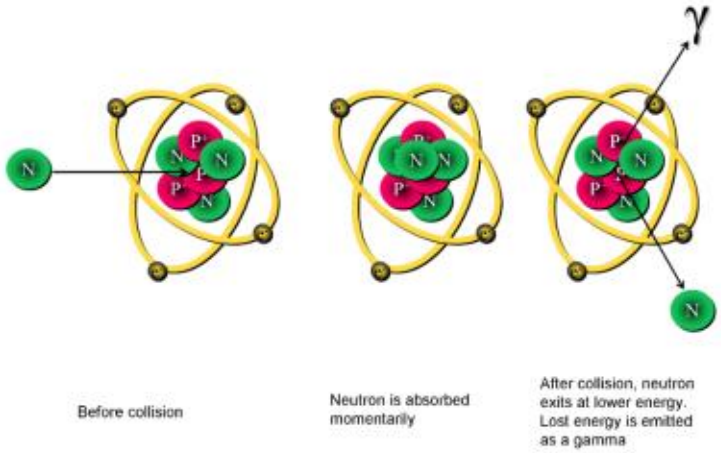
3. *Fission:* The nucleus splits into two smaller nuclei and two to three neutrons are expelled with an accompanying release of energy. Occurs when one of several very heavy elements absorbs a slow neutron.

1.07.08 Identify elastic and inelastic scattering interactions for neutrons:

Elastic: A neutron strikes a nucleus of approximately the same size, transfers most of its energy, no γ is emitted.

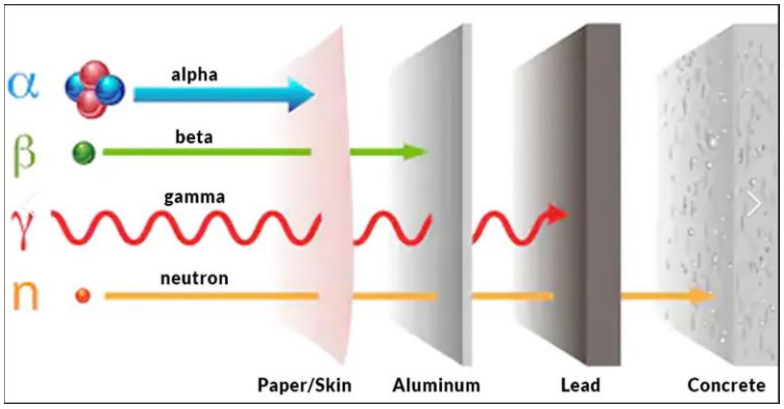


Inelastic: Occurs when a neutron strikes a large nucleus. The neutron penetrates the much larger nucleus for a short period of time, transfers energy to a nucleon inside, and then exits with a small decrease in energy. The nucleus is left in an excited state and emits gamma radiation.



1.07.09 Identify the characteristics of materials best suited to shield:

Radiation	Typical Shielding Characteristics
Alpha	Thin amounts of most materials (paper, unbroken layer dead skin cells, few cm of air)
Beta	Low-Z and low-density materials (rubber, plastic, aluminum, glass, wood)
Gamma	High-Z and high-density materials (lead, steel, depleted uranium, tungsten)
Neutron	Hydrogenous material for moderation (oil, polyethylene plastic, water) and capture material for absorption (boron, cadmium)



1.08.01 and 1.08.02 Identify the function of the following cell structures:

Cell membrane: Helps regulate the concentration of water, salts, and organic matter that form the interior of the cell. It is capable of "active transport" and regulates all food entering the cell and waste products exiting the cell.

The cell membrane is 1 millionth of cm thick and is a living, functional part of the cell.
3000 to 5000 Rads will rupture the cell membrane causing it to leak.
Rupturing the membrane will cause cell death compared to drowning.

Cytoplasm: The jelly-like substance encased within the cell membrane in which the nucleus is suspended.

An aqueous solution of soluble proteins and salts.
Radiation has negligible effect on cytoplasm compared to rest of cell.

Mitochondria: The "power plant" of the cell which uses oxygen to produce ATP (the energy used for all cell activities, including reproduction and repair).

"A few thousand rad" (>2000 rad) will disrupt the mitochondria function and cut off food to the cell.
Destroying the mitochondria will cause cell death compared to starving.

Lysosomes: Contain digestive enzymes that break down large molecules, such as fats and proteins, into smaller pieces that can be oxidized by the mitochondria.

500 to 1000 Rad will rupture the lysosomes and release enzymes that will begin to dissolve the cell.

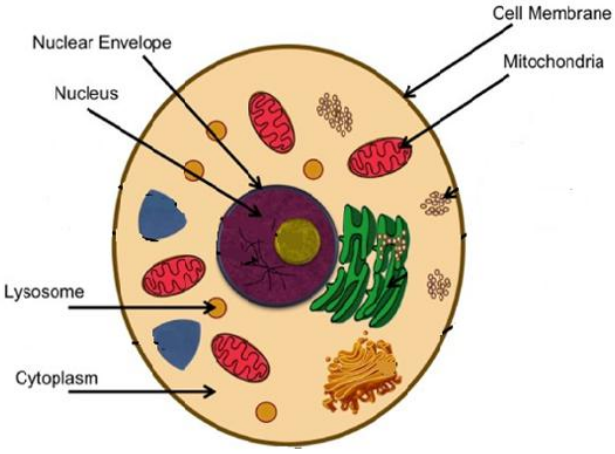
Nucleus: Control center of the cell that directs cellular activity and contains hereditary factors (DNA/genes) responsible for traits and necessary for reproduction.

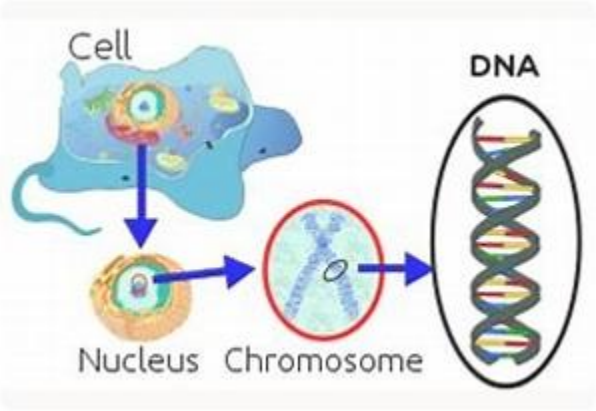
Most radiologically sensitive part of the cell. Due to varying sensitivity quantifying dose ranges is difficult.
If the nucleus is damaged duplicate chromosomes cannot be manufactured and the cell eventually dies.

DNA (Deoxyribonucleic acid): Most important part of the chromosomes and serves as the blueprint for cell.

Most damaging effect from rad in the nucleus is on DNA.

Chromosomes: Highly convoluted super coils of DNA and associated protein.





1.08.03 Define the law of Bergonie and Tribondeau:

"The radiosensitivity of a tissue is directly proportional to its reproductive capacity and inversely proportional to its degree of differentiation"

Cells most active in reproducing themselves and cells not fully mature will be most harmed.

1.08.04 Identify factors which affect the radiosensitivity of cells:

Cells are *radiosensitive* if they:
1. Have a high division rate.
2. Have a high metabolic rate.
3. Are of a non-specialized type.
4. Are well nourished.

1.08.05 Given a list of types of cells, identify which are most or least radiosensitive:

- *Radiosensitive:*

1. Germinal (Reproductive) cells of the ovaries and testis
2. Hematopoietic (Blood forming) tissues: red bone marrow, spleen, lymph nodes, thymus
3. Skin basal cells
4. Gastrointestinal (GI) tract epithelium. (Interstitial crypt cells)

- *Radio resistant:* (BLaCK MaN are resistant to Radio)

1. Bone
2. Liver
3. Cartilage
4. Kidney
5. Muscle
6. Nervous tissue

1.08.06 Identify primary and secondary reactions on cells produced by ionizing radiation:

Primary: Damage caused to the tissue from the incident radiations interaction with the atoms in the cells.

Secondary: Damage caused to the tissue from the free radicals and ions from the initial incident radiation.

- Free radicals are atoms or molecules with an unpaired electron in its outer shell. When radiation ionizes an atom and causes an electron to leave the atom, that atom becomes a free radical.
- Free radicals are very reactive chemically, and when combing can produce hydrogen peroxide (H_2O_2), which is a chemical poison and is the most harmful free radical product.
- Radiation directly damages tissue by damaging the atoms in the cells (primary) leaving behind damaged atoms (free radicals) that can go on to create more interactions and cause more damage (secondary).

1.08.07 Identify the following definitions and give examples of each:

Stochastic: Stochastic effects are those in which the probability within a population of the effect occurring increases with dose, without threshold (cancer and genetic effects).

Deterministic: **Deterministic** (non-stochastic) effects are those in which the severity of the effect varies with the dose. A threshold dose exists. That is, if dose remains below the threshold, these effects are unlikely to occur. Effects are cataracts, skin ulcerations/burns, blood-forming cell depletion in bone marrow, and fertility impairment.

1.08.08 Identify the LD 50/60 value for humans:

LD (Lethal Dose) 50/60:
300 to 500 Rad will kill 50% of the humans exposed within 60 days without any medical treatment.
Usually stated as 450 RAD or 4.5 Gray

1.08.09 Identify the possible somatic effects of chronic exposure to radiation:

1. Cancer (latent periods of 10 to >30 years)
2. Cataracts (600 – 900 rad may produce cataracts, symptoms can take years to show)
3. Shortness of lifespan

1.08.10 Distinguish between the three types of the acute radiation syndrome, and identify the exposure levels and the symptoms associated with each:

Hematopoietic syndrome: 200 to 1000 Rads, death results from damage to hematopoietic (blood forming) organs, such as red bone marrow, lymph nodes, spleen, and thymus.

Prodromal Phase: NVD (nausea, vomiting, diarrhea) occurs within 1 to 5 days
Latent Phase: Asymptomatic period with no or minimal symptoms lasts one to three weeks.
Illness: Severe fever, NVD, fatigue, anemia, loss of hair, anorexia, hemorrhaging.
Death: If it occurs, normally two to six weeks after exposure.

Gastrointestinal tract syndrome: 1000 to 5000 Rads, survival is impossible.
Death occurs from GI tract lining damage.

Prodromal Phase: NVD, loss of appetite, upset stomach, apathy. Occurs within a few hours.
Latent Phase: Asymptomatic period with no or minimal symptoms lasts one to seven days.
Illness: Severe fever, NVD, fatigue, apathy, anorexia, loss of appetite.
Death: Occurs within three to twelve days after exposure.

Central nervous system syndrome: >5000 Rads, survival impossible
Death occurs from respiratory failure and brain edema within thirty hours.

1.08.11 Identify risks of radiation exposure to the developing embryo and fetus:

- With respect to various stages of human development, the most sensitive to least sensitive in order:
  Embryo > Fetus > Child > Adult
  The embryo is the most sensitive, especially during development. Adults (fully developed) are least sentitive.
- During pregnancy, most major organs are developed between the second and sixth week of pregnancy. Doses as low as 25 Rad have been shown to produce developmental changes during this time.
- 400 to 600 Rads during the first trimester will cause fetal death and abortion.

1.08.12 Distinguish between the terms Somatic and Heritable as they apply to biological effects:

Body cells are called somatic cells and while they contain all genes most are not used for production (i.e., damaging the eye color gene in a skin cell has no impact as skin cells contain but do not use the eye color gene).

Germ cells are cells that reside in the testes or ovaries and are used for reproduction.

Somatic: Somatic effects come from damaged somatic cells (body cells). Somatic effects occur in the exposed individual but are not passed on to future generations.

Heritable: Heritable effects occur when germ cells (reproductive cells) are damaged. If a change occurs in a germ cell, no visible injury will be sustained by the individual, but the effect may appear (be passed on to) future generations.

Somatic effects affect the exposed individual while *heritable* affect the offspring.

1.09.01 Identify the role of advisory agencies in the development of recommendations for radiological control:

In 1925, at the First International Congress of Radiology, the International Commission in Radiological Units and Measurements *(ICRU)* was formed. In 1928, this group adopted the definition of an international unit, the roentgen.

The 3 organizations, ICRU, ICRP, & NCRP, have figured prominently in the development of current radiation protection practices.

The FRC was abolished in 1970, & its responsibilities were transferred to the newly formed U.S. Environmental Protection Agency (EPA)

1.09.02 Identify the role of regulatory agencies in the development of standards and regulations for radiological control:

The AEC, established in 1954, provided regulatory roles until 1974, when it was replaced by the ERDA and NRC

1974 NRC (license and regulatory functions) & ERDA (R & D) replaced AEC

1977 DOE replaced ERDA

1996 DOE is in full compliance with 10 CFR 835

The role of regulatory agencies with respect to standards and regulations for radiological control is to enforce regulations and requirements

1.09.03 Identify the purpose and scope of 10 CFR part 835:

Purpose: To move from a policy-based management approach to a rule based management approach, DOE orders were transitioned to codified by the 1993 approval of 10CFR 835.

Scope: The scope of 10CFR 835 establishes radiation protection standards, limits and program requirements for protecting individuals from ionizing radiation resulting from the conduct of DOE operations.

1.09.04 Identify the definition of the terms "shall" and "should" used in DOE documents:

Shall: Requirements that DOE considers mandatory unless prior approval of an alternative approach is obtained from DOE headquarters. Must have approval of the lead program secretarial official.

Should: Contractor has the responsibility to either follow the provision or demonstrate technical equivalency by an alternate solution.

1.10.01 Describe the assumptions on which the current ALARA philosophy is based:

ALARA: As Low As Reasonably Achievable.

Based on the theory that exposure to certain agents could cause undesirable effects. The concept also implies that there is a relationship between the amount of exposure and the possibility of an effect - that there is a risk involved in receiving exposure.

The basis for the ALARA philosophy is: reduce your exposure to certain agents to reduce the potential risk of an unwanted effect. DOE, through the CFR, mandated that there should not be any occupational exposure of workers to ionizing radiation without the expectation of an overall benefit from the activity causing exposure.

1.10.02 Identify the ALARA philosophy for collective personnel exposure and individual:

Collective as well as Individual exposure must be minimized.

Individual dose: The total dose received by an individual radiological worker due to occupational exposures.

Collective dose: The total of all individual doses in a group or population.

1.10.03 Identify the scope of an effective radiological ALARA program:

1. The scope of ALARA applies equally to internal and external exposure.
2. ALARA must be used in all function's routine or not.
3. Must have the commitment of all facility personnel.

Everyone involved in radiological work must demonstrate responsibility and accountability through an informed, disciplined, and cautious attitude toward radiation and radioactivity.

1.10.04 Identify the purpose of conducting pre and post job ALARA reviews:

PRE-JOB ALARA review

1. Proposed protective measures are evaluated, and costs are justified.
2. Applicable ALARA practices are factored into the plans and procedures.

Post job ALARA review

1. Ensure effectiveness of job planning and implementation.
2. Investigate unusual exposure events.
3. Recommendations made; corrective actions considered.

1.10.05 Identify RCT responsibilities for ALARA implementation:

1. Should participate in the ALARA review process.
2. Observe workers and ensure that Rad control requirements are followed.
3. Correct bad radiological practices on the spot.
4. Assist workers in maintaining their exposures ALARA.

1.11.01 Identify the basic methods for minimizing personnel external exposure:

1. Time
2. Distance
3. Shielding
4. Source reduction

1.11.02 Calculate γ exposure using 6CEN:

I = 6CEN or I =(0.5)CEN (Use 6 for answer in feet or use .5 for answer in meters)

I = Exposure rate in R/hr @ 1ft
C = Activity of the source in curies
E = γ energy in MeV
N = Number of γ's per disintegration (decimal form percentage)

1.11.03 Identify source reduction techniques for minimizing personnel external exposure:

1. Allow source to naturally decay.
2. Move source to another location (decontaminate it, flush it away, physically remove it, etc.)

1.11.04 Identify time saving techniques for minimizing personnel external exposure:

1. Mock up's
2. Pre-job briefings / review job history data
3. Pre stage tools and equipment / preassemble equipment outside area
4. Time limiting devices (stopwatch, alarming dosimeters)
5. Experienced personnel / communication devices
6. Rotate a team of workers instead of one worker taking all the exposure.

1.11.05 Using the stay time equation, calculate allowable stay time or dose equivalent:

$$\text{Stay time} = \frac{H_{\text{allowable}} - H_{\text{received}}}{\text{(Equivalent Dose rate)}}$$

1.11.06 Identify "distance to radiation sources" techniques for minimizing personal External exposure:

1. Remote handling tools
2. Remote observation
3. Remove work to new location
4. Maximize distance between work and source
5. Use of extendable instruments
6. Posting of areas.

1.11.07 Using the point source equation, calculate the exposure rate or distance from a point source:

$I_1(d_1)^2 = I_2(d_2)^2$

I_1 = Exposure rate at first distance
I_2 = Exposure rate at second distance
d_1 = first (known) distance
d_2 = second (unknown) distance

Note: This point source equation is also referred to as the "Inverse Square Law" equation.

1.11.08 Using the line source equation, calculate the exposure rate or distance for a line source:

Line source equation is used up to L/2, then point source is used for distances greater than L/2.

$I_1(d_1) = I_2(d_2)$

I_1 = Exposure rate at first distance
I_2 = Exposure rate at second distance
d_1 = first (known) distance
d_2 = second (unknown) distance

1.11.09 Identify how exposure rate varies depending on the distance from a plane source and identify examples of plane sources:

As the distance from a plane source increases, the exposure rate drops off at a Rate approaching $1/d^2$

If the source is very close, then the rate falls off at 1/d.

1.11.10 Identify the definition and units of "mass attenuation coefficient" and "linear attenuation coefficient":

Linear attenuation coefficient: The probability of a photon interaction per path length. Units are cm^{-1}

Dividing the linear attenuation coefficient by the density of the material it is attenuating (penetrating/interacting with) gives the mass attenuation coefficient.

Mass attenuation coefficient: The quantity obtained by dividing the linear attenuation coefficient by the physical density of the material. Units are cm^2/mg

1.11.11 Identify the definition and units of "density thickness":

Density thickness: A value equal to the product of the density of the absorbing material and its thickness. Units are mg/cm^2.

Density-thickness is how dense (mass per volume) a material is compared to how thick (distance from end to end) the material is. Materials with similar density-thickness will attenuate radiation in a similar matter. For example, a piece of Mylar used in an alpha detector window with a density-thickness of 7 mg/cm^2 will attenuate radiation similarly to the skin of the human body.

1.11.12 Identify the density thickness values, in mg/cm^2, for the skin, lens of the eye, and the whole body:

Tissue	Depth (cm)	Density-Thickness (mg/cm^2)
Skin	0.007	7
Lens of the eye	0.3	300
Whole body	1.0	1,000

1.11.13 Calculate shielding thickness or exposure rates for gamma/x-ray radiation:

$I = I_0(1/2)^n$
I = shielded exposure rate
I_0 **=** unshielded exposure rate
n = number of HVL's = $\frac{\text{shielded thickness(cm)}}{\text{HVL thickness(cm)}}$

$I = I_0(1/10)^n$
I = shielded exposure rate
I_0 **=** unshielded exposure rate
n = number of TVL's = $\frac{\text{shield thickness(cm)}}{\text{TVL thickness(cm)}}$

1.11.14 Identify the Inverse Square Law as it relates to a point source of radiation

Radiation from a source is emitted equally in all directions. Thus, the photons spread out to cover a greater area as the distance from the point source increases.

The Inverse Square Law states that as the distance from a point source changes (further or closer) the dose rate decreases or increases by the by the square of the ratio of distances from the source.

Note: The Inverse Square Law becomes inaccurate when too close to the source.
The equation is: $I_1(d_1)^2 = I_2(d_2)^2$ (covered in 1.11.07 "Point Source Equation")

1.12.01 Identify 4 ways in which radioactive materials can enter the body: and
1.12.02 Given a pathway for radioactive materials into the body, identify one method to prevent or minimize entry by that pathway:

1. *Ingestion* (through the mouth, swallowing)

   Method to prevent: Proper radiological controls and work practices.

2. *Inhalation* (through air that is breathed)

   Method to prevent: Engineering controls, respiratory protection.

3. *Absorption* (through skin contact)

   Method to prevent: Assessment of conditions and protective clothing.

4. *Entry through wounds* (penetration or injection)

   Penetration: passively through existing wounds not covered.

   Injection: forcefully through wounds incurred on the job

   Method to prevent: Not allowing contamination near wounds, proper radiological controls if an injury occurs in a contaminated area.

1.12.03 Identify the definition and distinguish between the terms "ALI" and "DAC":

ALI (Annual limit on intake): The quantity of a single radionuclide, which if inhaled in a year would irradiate a person to the limiting value for control of the workplace (5 Rem).

DAC (derived air concentration): The quantity obtained by dividing the ALI for any given radionuclide by the volume of air breathed by the average worker during a working year. ($2.4 \times 10^3\ m^3$)

1.12.04 Identify the basis for determining the ALI:

ALI is based on known metabolic processes for isotopes involved and the reference man.

1.12.05 Identify the definition of "reference man":

"Reference man" defines the physiological makeup of an average man in terms of factors required for dose calculations, and includes such items as height, weight, size, and mass of organs.

1.12.06 Identify a method of using DAC's to minimize internal exposure potential:

By minimizing concentrations to below DAC values workers cannot exceed the ALI.
By posting areas at 10% of a DAC the potential for internal is reduced.

1.12.07 Identify three factors that govern the behavior of radioactive materials in the body:

- Chemical form – Solubility.
- Location – Pathways.
- Body's need – Intake and incorporation vs. elimination.
- *Intake*: The amount of radionuclide taken into the body
- *Uptake*: The amount of radionuclide deposited in the body

1.12.08 Identify the two natural mechanisms which reduce the quantity of a radionuclide in the body:

1. Biological Elimination – Exhalation, perspiration, urination, and defecation.

   Each element has a measurable *biological half-life* which is the time required to reduce the amount of material in the body to one-half its value via biological elimination).

2. Radioactive decay – Each elements specific decay rate and *physical half-life*.

1.12.09 Identify the relationship between the physical, biological, and effective half-lives:

Biological half-life: The time required to reduce the amount of material in the body to one half its original amount via biological elimination (normal biological functions and excretion).

Physical half-life: The amount of time required for one half the material in the body to decay naturally.

Effective half-life: The combined process of biological elimination and radioactive decay.

The *Effective* $T_{1/2}$ will always be shorter than the Physical $T_{1/2}$ or the Biological $T_{1/2}$ as it is a combination of both.

1.12.10 Given the physical and biological half-lives, calculate the effective half-life:

$$T_E = \frac{T_R \times T_B}{T_R + T_B}$$

T_E = effective half-life
T_R = radiological half-life
T_B = biological half-life.

1.12.11 Given a method used by medical personnel to increase the elimination rate of radioactive materials from the body, identify how and why that method works:

Blocking agents: Saturates the metabolic process in a specific tissue with the stable element and reduces uptake of the radioactive forms. Must be administered almost immediately after uptake to be effective.

Diluting agents: Compound which includes a stable form of the nuclide of concern. By introducing many stable atoms, the statistical probability of the body incorporating the radioactive atoms is reduced. "Displacement therapy" example: administer calcium to reduce strontium deposits.

Mobilizing agents: A compound that increases the natural turnover process, releasing some forms of radioisotopes from body tissues. Most effective within two weeks of exposure.

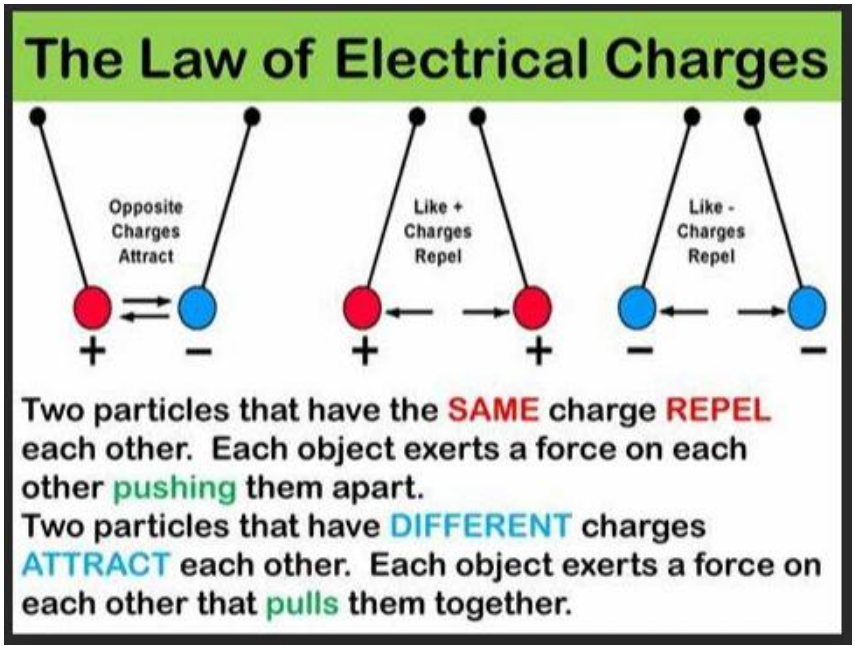
Chelating agents: A compound which acts on insoluble compounds to form a soluble complex ion which can be removed through the kidneys. By making the ion soluble it can be processed and excreted through the kidneys. Most effective immediately after exposure.

Diuretics: Increase urinary excretion of sodium and water. Limited use.

Expectorants and Inhalants: Used to increase the flow of respiratory tract excretions. Not yet proven successful.

Lung Lavage: This method involves multiple flushing of lungs with appropriate fluid to remove radioactive materials in the lungs. Limited applications.

1.13.01 Identify the three fundamental laws associated with electrical charges:



1- Opposite electric charges of equal value cancel.
2- Opposite electric charges attract.
3- Like electric charges repel.

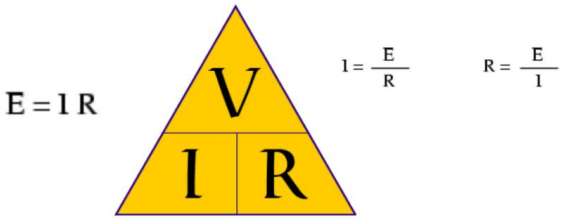
1.13.02 Identify the definition of current, voltage, and resistance with units:

Ohm's Law: $I = \frac{E}{R}$

E = *Voltage*: The electrical potential difference that causes electrons to flow in a circuit. Measured in volts (V).

I = *Current:* The movement or flow of electrons past a given point in a circuit. measured in amperes (A).

R = *Resistance:* The electrical quantity that opposes electron flow in a circuit measured in ohms (Ω).



1.13.03 Select the function of the detector & readout circuitry components in a radiation measurement system:

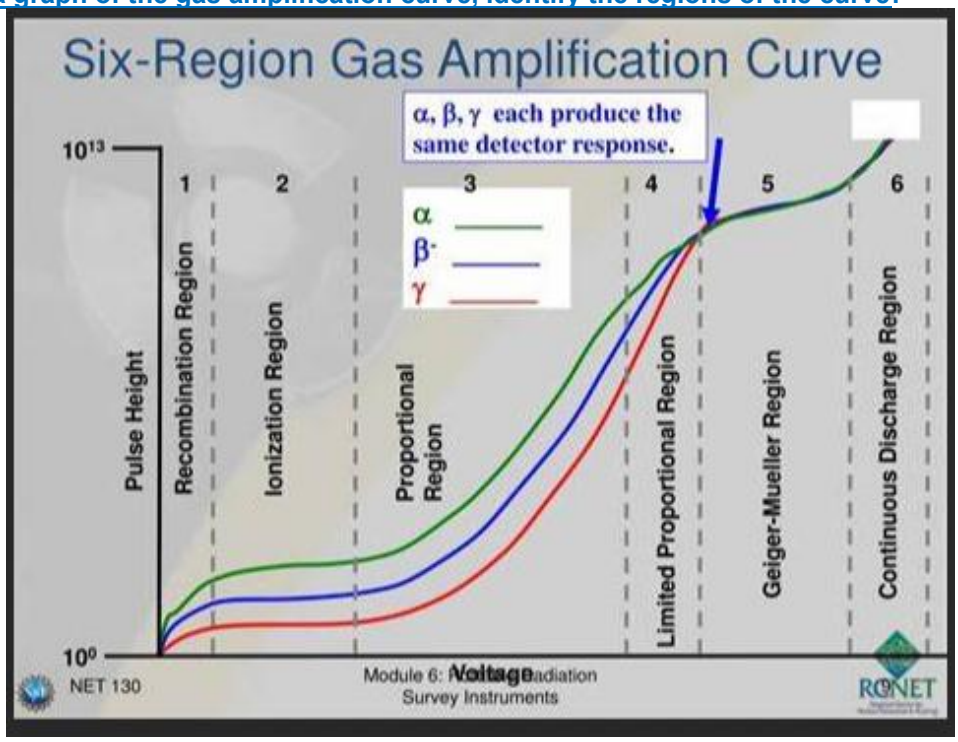
Detector function: The incident radiation interacts with the detector material to produce an observable effect.

Readout circuitry: Measures and analyzes the produced effect and provides a usable output indication.

1.13.04 Identify the parameters that affect the number of ion pairs collected in a gas filled detector:

- *Type of radiation*: Differing specific ionization (e.g., alpha, beta, gamma).
- *Energy of radiation*: The more energy an incident radiation possesses, the higher potential for interactions.
- *Quantity of radiation*: As the number of radiation events striking the detector increases, the overall probability of interaction increases.
- *Detector size*: A physical quality offering more "targets" to the incident radiation.
- *Type of gas*: Characteristics of the absorber (in this case, the gas) effects the amount of energy expended to create ion pairs.
- *Detector gas pressure*: There is a practical limit to detector size. Instead of increasing detector size, increasing the gas pressure (more gas housed within the same volume) increases the number of target atoms for interaction.
- *Voltage potential across electrodes:* When voltage potential is applied across the electrodes, a field is created in the detectors, and the ion pairs will be accelerated toward the electrodes where they can be counted and displayed to the user. The stronger the field, the stronger the acceleration. As electron velocity increases, the electron may cause one or more ionizations on its own which is known as *secondary ionization.*

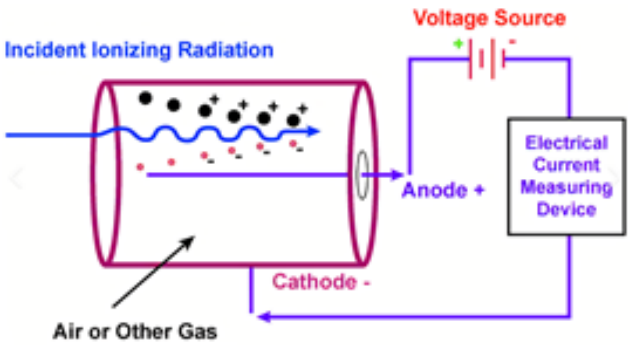
1.13.05 Given a graph of the gas amplification curve, identify the regions of the curve:



"Rest In Peace Little Gray Cat"

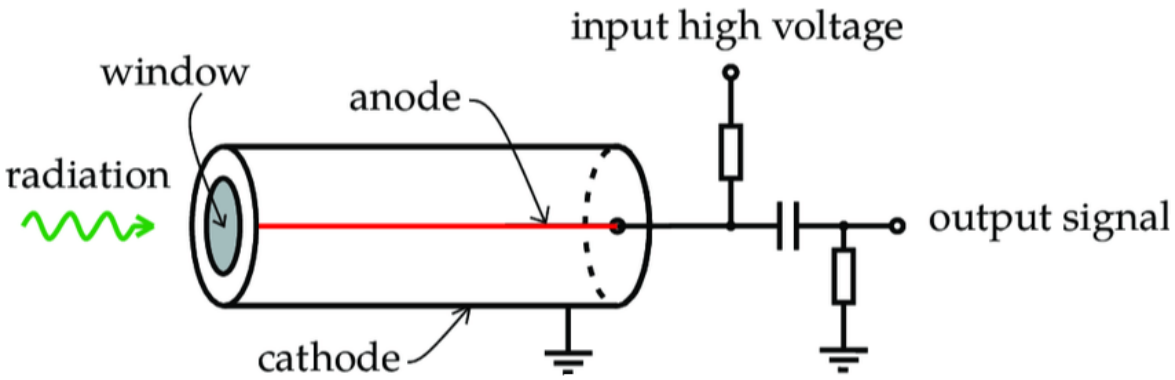
1.13.06 Identify the characteristics of a detector operated in each of the useful regions of the curve:

- ***Ion chamber:*** All ions are collected. No secondary ionization or gas amplification occurs.



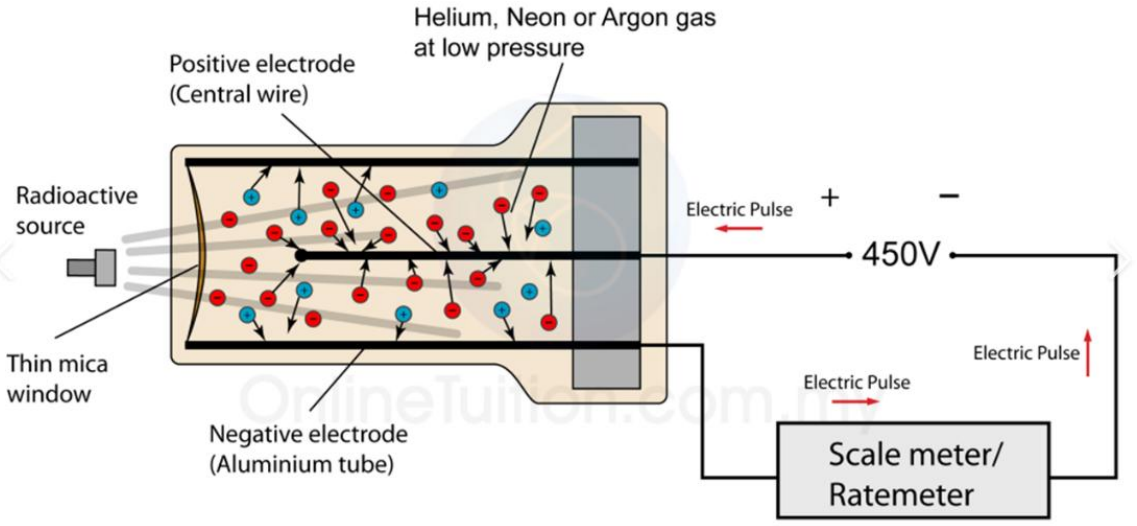
Advantages	Disadvantages
Output current is independent of detector voltage	Only primary ion pairs created by each radiation event are collected, the output currents are small.
Since the number of primary ion pairs is a function of the energy deposited in the detector by the incident radiation, its response is directly proportional to the exposure rate.	The effect that humidity can have on the instrument response.
Since exposure (X) is defined in terms of ionization of air by photons, an air-filled ion chamber, when used for photon radiation, yields true exposure rate.	Anything that changes the density of the gas affects the response, changes in barometric pressure and/or ambient temp can affect instrument response.

- ***Proportional:*** Voltage increase gives primary ions enough energy to cause secondary ionization.



Advantages	Disadvantages
Discriminate between types of radiation	The need for a highly regulated power supply
A larger output signal is created & therefore a single ionizing event being recorded improving sensitivity	
When measuring current output, the output signal is proportional to the energy deposited by ionization & therefore proportional to the dose rate	

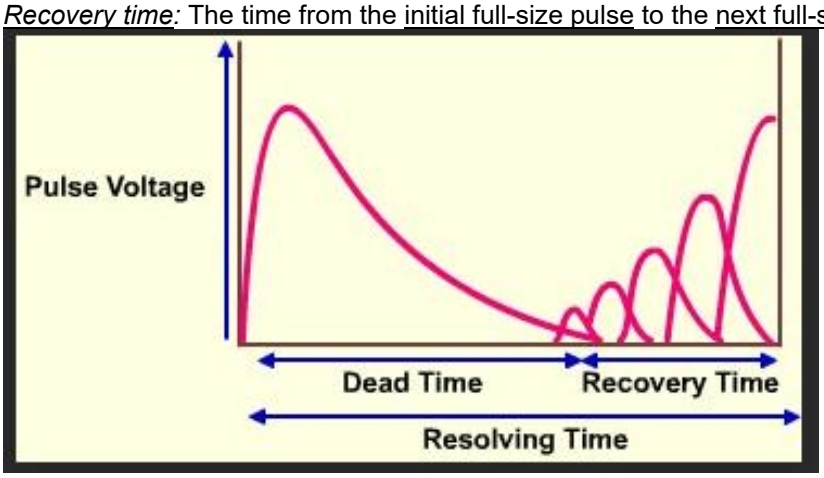
- ***GM:*** Secondary and tertiary ionization occurs causing avalanche effect.



Advantages	Disadvantages
Relative independence of pressure & temp	Large recovery times limits the use in extremely high radiation fields
Pulse repetition being measure instead of pulse height allowing for a less regulated power supply	Inability to discriminate against different types of radiation or energies
	detector response is not related to energy deposited

1.13.07 Identify the definition of the following terms:

- *Resolving time*: The time from the initial measured pulse until another pulse can be measured by the electronics.
- *Dead time:* The time from the initial pulse until another pulse can be produced by the detector.
- *Recovery time:* The time from the initial full-size pulse to the next full-size pulse produced by the detector.



1.13.08 Identify the methods employed with gas-filled detectors to discriminate between various types of radiation and radiation energies:

- Physical discrimination – Shielding (most common method).
- Detector gas fill – Appropriate gas type and pressure for the specific ionization of each radiation.
- Electronic discrimination – Analyzing pulse heights is the primary form of electronic discrimination.

1.13.09 Identify how a scintillation detector and associated components operate to detect and measure radiation:

Scintillation is the process by which a material emits light when excited.
Each scintillation detector compromises two major components: the phosphor or fluor, and photomultiplier tube. Radiation interacts with the phosphor, which emits light. The photomultiplier tube detects the scintillations and provides an output signal proportional to the number of scintillations which is converted to a usable number (and reported to the user) by the detector circuitry.

1.13.10 Identify how neutron detectors detect neutrons and provide an electrical signal:

Because neutrons do not interact with material to form ions, neutrons must be detected indirectly.

Slow Neutron Detection:

- Boron activation: $n^1_0 + B^{10}_5 \rightarrow Li^7_3 + He^4_2$ the alpha produces ionization which can be measured.
- Fission chamber – A slow neutron will cause an atom of ^{235}U to fission.
- Scintillation – Interaction with boron or lithium produce alpha particles which produce ionization/scintillations.
- Slow neutron thermo luminescence – Incorporates lithium-6 in the crystal.
- Activation foils – By measuring the radioactivity of thin foils such as gold, silver, or indium, it is possible to determine the number of neutrons to which the foils were exposed.

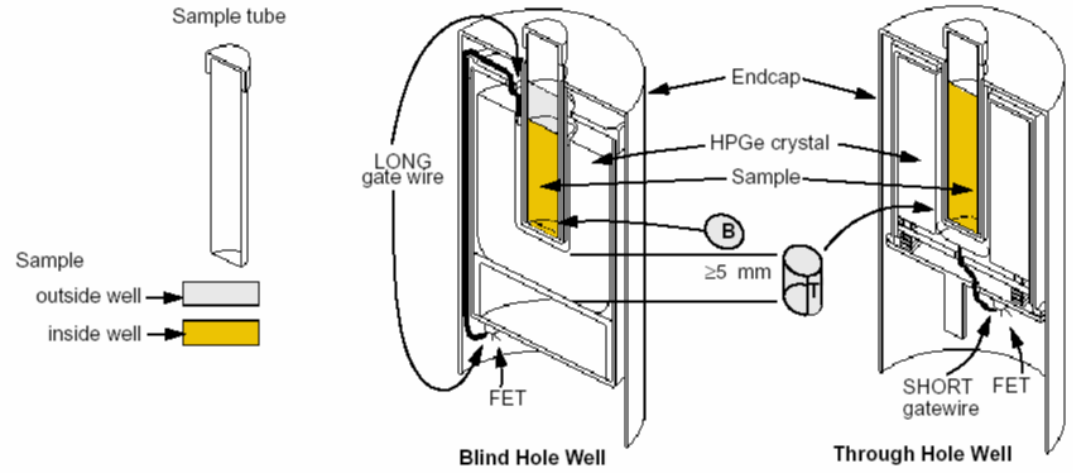
Fast Neutron Detection:

- Proton Recoil (Ion Chamber/Proportional) – Fast neutrons undergo elastic scattering, frequently striking hydrogen atoms with enough force to knock away the proton nucleus which goes on to produce ionization.
- Thermalization (Slowing Down Fast Neutrons) – A thickness of paraffin or other good moderator will reduce the speed of fast neutrons converting them to slow neutrons which can then be measured using slow neutron detection.

1.13.11 Identify the principals of detection, advantages, & disadvantages of a Ge(Li) detector & an HPGe detector:

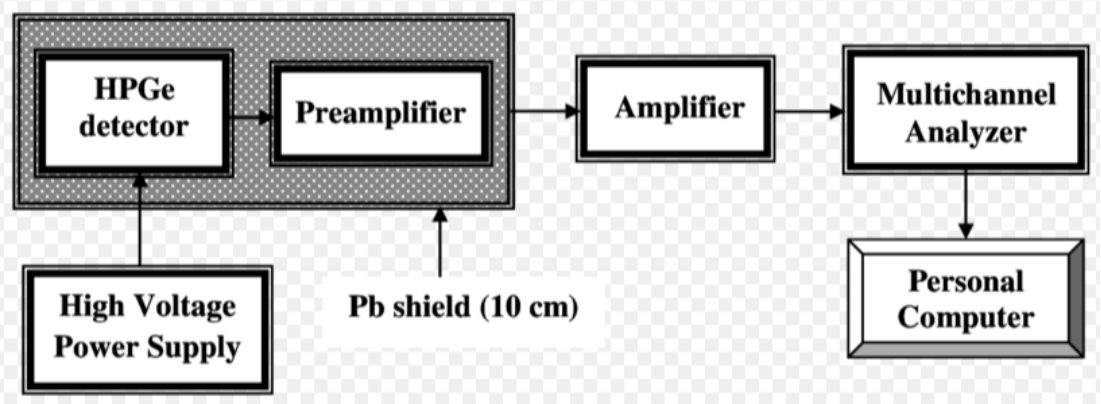
- *Radiation interacts with atoms in the intrinsic region to produce electron hole pairs. The presence of ion pairs in the depletion region causes current flow.*

- **Ge(Li):** detector system consists of a vacuum-enclosed Ge(Li) crystal that is coaxial in shape and attached to a copper cold finger through an agate insulator.



Advantages	Disadvantages
High resolution	Only used for Gamma detection
Small size crystals offer the best resolution, but the efficiency of detection is lower	Liquid nitrogen is a initial and continuing operational cost
	Manufactures no longer produce or service the detectors if the crystals lose their lithium drift.
	Short response time and a more linear response than NaI(T1)

- **HPGe:**



Advantages	Disadvantages
High resolution	Even though the dewar is smaller, they still require liquid nitrogen cooling and tend to be expensive
The dewar used for liquid nitrogen can be smaller	
Allowed to rise to room temp, the detector need only be returned to and stabilized at liquid nitrogen temp to be used	

2.01.01 List the standards that must be met by the RCT when completing record documentation in accordance with the RadCon Manual.

1. Identification of the facility, purpose, specific location, function and process, as appropriate.
2. Signature or other identifying code of the preparer and date.
3. Legible entries that meet company recordkeeping standards (e.g., using blue or black ink only, no shorthand).
4. Corrections identified by a single line out, initialed and dated.
5. Supervisory signature to ensure review and proper completion of forms.

2.01.02 State the purpose of the health physics field logbook:

Logbooks provides a quick and accurate account of the activities occurring during a work shift.
Logbooks are permanent legal records that must be controlled and disposed of according to procedure.
Logbooks provide a method of communication where RadCon employees and supervisors can find the necessary information to perform assignments or initiate corrective actions.

2.01.03 Describe the RCT's responsibilities for maintaining the RadCon field logbook:

- Immediately after shift change, review the log for entries made in the Previous 25-hour period.
- Back-shift personnel should initial in the margin upon completion of their review.
- All RCTs are required to review their own entries for completeness, accuracy, and legibility then sign & date.
- Sequentially numbered pages, facility location, beginning and ending period.
- Only make entries for the position the worker is qualified/authorized for (i.e., RCT, RFLM, etc.)
- Make all entries in chronological order with no blank lines.
- Maintain and physically control logbooks, storing them in an approved location.

2.01.04 List information that should be included as a logbook entry:

- Abnormal events that occurred during the shift
- Corrective actions written
- Delinquent routine surveillances
- Follow-up action on previous entries
- Instructions for follow-up actions
- Out-of-service instrumentation
- Radiological posting changes
- Respiratory protection requirements at specified locations
- RWP changes issues
- Safety issues
- Status of jobs in progress that impact oncoming shifts (where shifts are used)

2.01.05 List examples of items requiring action or special attention in the RadCon field logbook:

- Emergency response actions taken
- Safety or compliance issues
- Radiological posting changes
- Status of jobs that may impact oncoming shifts
- Delinquent radiological routine surveillances
- Routine surveillance action levels exceeded
- Survey reports unable to be completed by the end of shift

2.01.06 Describe the method of denoting items requiring action or special attention in the RadCon field logbook.

Emphasize important entries and entries requiring action with red arrows or asterisks in the left margin.

Entries requiring action should include at least the following information:

- Date and shift covered by the entry
- Survey number
- Actions taken
- RCFLM that was notified
- Any pertinent factual comments about the issue that may provide a more complete picture

2.01.07 Identify the purpose of and requirements for documenting radiological survey reports

- To provide a method for tracking and trending changes in radiological conditions.
- Aid in maintaining radiation, contamination, and airborne levels ALARA.
- To provide a legal and quality assurance record.
- Records must contain sufficient detail to be meaningful after the originator is no longer available.
- RSRs should include date, time, general location, specific locations, purpose, facility code, applicable RWP, instrument types/correction factors used, any applicable maps, and name/ID of the surveyor RCT.

2.01.08 Objective REMOVED and number marked as "RESERVED" (for future use):

N/A

2.01.09 Describe the process for initiating and submitting an RSR:

- Log out the RSR number on the Report Log.
- The appropriate RSR from Site Forms may be used if Survey Simple is unavailable, but the data must be uploaded when the program becomes available.
- Completed form is signed by the technician and submitted for review and approval.

2.01.10 Describe the requirement for an RSR that must be rewritten or is lost:

All missing report forms must have a Record of Missing/Void/Incomplete Radiation Reports/Documents completed as follows:

1. Record RSR/document number.
2. Record facility code.
3. Indicate type of record that is missing/void/incomplete.
4. Circle the reason (missing/void/incomplete).
5. Give explanation.
6. Sign and date form.
7. The RCFLM must review and sign the form.

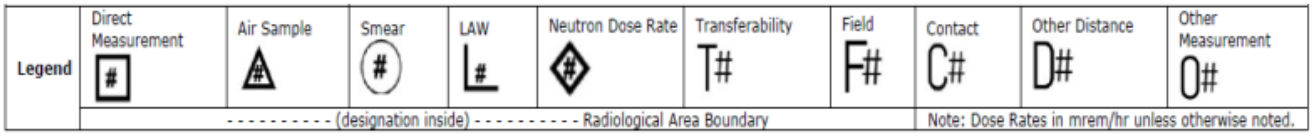
2.01.11 Describe requirement for using the symbol <D on the RSR:

- <D is only used as a verification (qualitative) survey.
- Verification surveys are qualitative surveys typically performed in areas not expected to be contaminated.
- They are not quantitative measurements and are recorded for information only.
- "<D" signifies that there was no observable meter deflection or audible response above background.
- If a response above background is observed, more rigorous contamination surveys are performed to quantify the contamination levels.
- Verification surveys cannot be used to determine posting status of an area or for clearance of M&E.

2.01.12 Describe the use of legends on the RSR:

The standard legend symbols at the bottom of the report should be used in conjunction with the maps.

The symbols are used to indicate where 100cm^2 swipes, large area wipes, air samples, and dose rate readings were taken. Other symbols used should be defined in the map or comments.



2.01.13 Discuss why and how required radiological surveillances are performed

A required radiological surveillance is performed to check the radiological conditions in each work area on a routine basis.

The required radiological survey frequencies are normally one of the following: each shift, daily, weekly, monthly, quarterly, semi-quarterly, or annually.

Survey frequency depends on the type of work, frequency of work, and potential for radiological hazards.

2.01.14 Describe the purpose of the Integrated Contractor Assurance system (iCAS):

Provides a means to document various issues that either violate a requirement or offer an opportunity for improvement, including radiological issues. Establishes a way to report problems to management for action.

2.01.15 List the guidelines that are used to identify a radiological problem:

- Violations of requirements contained in manuals, procedures, etc.
- Inadequate implementation of radiological controls
- Poor housekeeping in rad areas
- Access control violations
- Containment or other engineered control failure
- Contamination spread outside posted contamination areas
- Dose rates exceeding an area radiological posting
- Disregarding RadCon instructions
- Improper dress / undress
- Improper rad postings
- Management direction
- Out-of-spec routine and surveillance items, when required
- Poor rad work practices
- RSR not completed to standards
- RWP violations
- Use of stop work authority
- Work package violations – radiological criteria

2.01.16 State who may initiate a Corrective Action Report:

Any worker can initiate a corrective action report upon identification of a concern or deficiency. Corrective action reports are records and must meet the records standards for quality and accuracy.

2.02.01 Explain the importance of good communication:

Accurate communications are essential for the safe and efficient operation of facilities. Clear, concise communication eliminates confusion and prevents misunderstanding.

2.02.02 Identify two methods of communication and determine different types of each:

1. *Verbal* methods of communication include talking directly, telephone, voice mail, and video call.
2. *Nonverbal* methods of communication include signs, letters, signals, documents, and email

2.02.03 Identify characteristics of the following types of communication equipment:

- *Public address systems (PAX):* loudspeakers, public address system notifications
- *Telephones:* means for point-to-point communication, semi-private, ability to leave voice messages
- *Cellular Telephones:* same as land line, ability to leave text messages
- *Radios:* provide mobility and communication access while at remote locations
- *E-mail:* written and used without any expectation of privacy, designed as an informal delivery system and not a permanent file system or repository, however, signed documents that are emails are as valid as originals

2.02.04 Describe the Federal Communication Commission and Department of Energy restrictions regarding proper use of communication systems:

- Use profane, indecent, or obscene language.
- Willfully damage or permit radio equipment to be damaged.
- Cause malicious interference with any radio communication.
- Intercept and use or publish the contents of any radio message without the permission of proper authorities.
- Make unnecessary or unidentified transmissions.
- Transmit without first making sure that the transmission will not cause harmful interference.
- Make any adjustments, repairs, or alterations to a radio transmitter without being licensed by the FCC or accepted equivalent.
- Transmit a call sign, letter, or numeral that has not been assigned to you station.
- Rebroadcast another transmission (e.g., radio station music).

2.02.05 Describe general attributes of good communication with any form of communication system:

- Minimize abbreviations and acronyms; only use approved abbreviations and acronyms.
- Make all oral instructions clear and concise; do not include multiple actions that can be confused.
- Identify yourself and your position.
- Use clear and precise terminology, no slang, avoid words that sound alike (e.g., “increase” and “decrease”).
- Utilize the phonetic alphabet.
- Repeat back messages either paraphrased or verbatim; orders must be repeated verbatim.
- Speak distinctly and deliberately.
- Acknowledge all communications and use three-way communication.

Phonetic Alphabet and Numbers

A - Alpha	N - November	1 - One
B - Bravo	O - Oscar	2 - Two
C - Charlie	P - Papa	3 - Three
D - Delta	Q - Quebec	4 - Four
E - Echo	R - Romeo	5 - Fife
F - Foxtrot	S - Sierra	6 - Six
G - Golf	T - Tango	7 - Seven
H - Hotel	U - Uniform	8 - Eight
I - India	V - Victor	9 - Niner
J - Juliet	W - Whiskey	0 - Zero
K - Kilo	X - X ray	. - Point
L - Lima	Y - Yankee	
M - Mike	Z - Zulu	

2.02.06 Explain the importance of knowing how to contact key personnel:

- The importance lies in getting knowledgeable people to the location where they are needed.
- The RCT must be aware of the location, situation, and personnel or equipment involved.
- The ability of the RCT to contact key personnel can reduce personnel injury, equipment damage, uncontrolled radiological release, unrestricted movement of controlled materials, and other important actions.

2.02.07 Identify requirement that apply to basic radio and telephone communications:

- Sender and receiver identification must be included in each radio or other message.
- The radio should be held about 3 to 4 in. in front of and slightly to the side of the mouth.
- Radios should be turned sideways so that you are not speaking directly into it.
- Speak slowly and in a normal tone of voice.
- Wait until the airway is clear before sending a message.
- Never interrupt another station unless you have an emergency.
- Always identify yourself by your name or position.
- When calling another unit and you receive no answer, call twice only, then clear the network.
- Keep conversations brief and confined to business.

2.03.01 Identify the effect of the following factors on analyzing radioactive samples with counting instruments:

1. *Self-absorption:* Absorption of the emitted radiation by the sample material.
   Part of the radiation is continuously self-absorbed by the sample media and does not reach the detector, so observed activity is lower than actual activity (e.g., liquid samples, oversaturated media).

2. *Backscatter:* Radiation traveling away from the detector is reflected, or scattered back, by material behind the sample (e.g., the planchet).

3. *Resolving time*: The time interval which must elapse after a detector pulse is counted before another pulse can be counted.

4. *Geometry:* Errors can occur if the geometry of the sample and detector is varied between calibration/samples.

5. *Random Decay*: The 5th source of general counting error is the random disintegration of radioactive atoms.

2.03.02 Identify two purposes for using counting statistics in sample analysis:

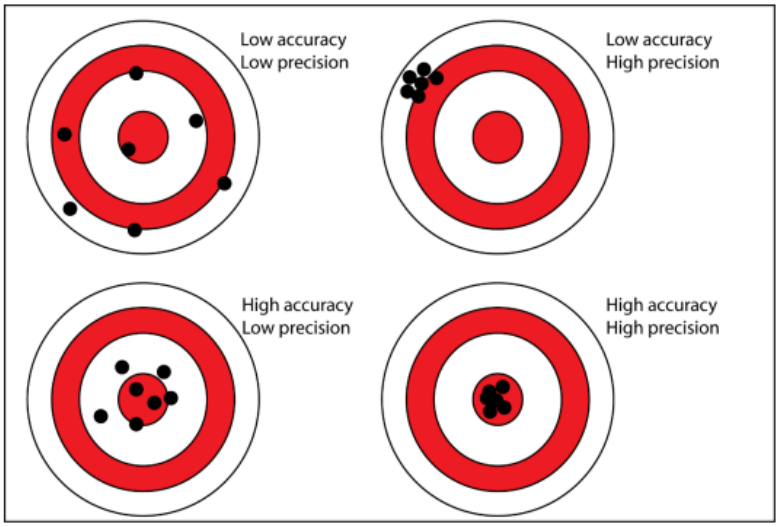
1. *Serves as a check on the normal function of nuclear counting equipment* – keeping a record of the statistics allow a set amount of fluctuation to be expected and excess fluctuation to be a sign of abnormality.

2. *To predict the inherent statistical uncertainty associated with a measurement* – this allows us to estimate the precision associated with that measurement.

2.03.03 Identify the definition of the following terms:

Accuracy: The quality of being exact or true.

Precision: The quality of being consistent.



Mean: The mathematical average value of all the values in the data set.

Median: The center value in a data set arranged in ascending order.

Mode: An individual data point that is repeated the most within a particular data set.

$$\bar{x} = \frac{\sum x_i}{n}$$

$\bar{x}$ = mean (pronounced *x bar*)

x_i = data point with index *i*

n = number of data points

$\sum$ = summation symbol $\Rightarrow \sum_{i=1}^{n} x_i = x_1 + x_2 + x_3 + \ldots x_n.$

Variance: The amount of scatter of data points around the mean (how much the data varies from the mean) – signified mathematically by sigma squared: σ^2

Standard deviation: The average difference between the data points and the mean value of the data points (the square root of the variance) – signified as sigma: σ

Confidence level: The percentage of the time you are confident that a given value will fall within a set range – Found by multiplying the standard deviation (σ) by the correct multiple (k).

Confidence interval: Interval or range of values created around the mean value by the desired confidence level.

Error	Confidence Level (%)	k
Probably	50	0.6745
Standard	68	1.0000
9/10	90	1.6449
95/100	95	1.9600
99/100	99	2.5750
99.7/100	99.7	3.0000

The above table represents the k values used to find the desired confidence level. (i.e., to find a confidence level of 95%, multiple the standard deviation (σ) by 1.9600.

Background: The radiation that enters the detector concurrently with the radiation emitted from the sample being analyzed.

Decision level: The value of a net observation at, or above, which a decision is made that a positive quantity of radioactive material is present.

Minimum Detectable Activity (MDA): The smallest activity in a sample that will yield a result equal to or above the DL. (95% of the time).

Minimum Detectable Concentration (MDC): Minimum detectable concentration (MDC) is the MDA divided by the sample volume and filter collection efficiency. Essentially, it is the MDA of an air sample (the lowest concentration of activity needed in a volumetric air sample that will yield results equal or above the DL 95% of the time).

2.03.04 Given counting data, determine the mode, median, or mean:

Mean: Average of all the data points in a data set.

Median: Middle data point or average of the two middle data points when in ascending order.

Mode: Data point repeated most often in a data set.

2.03.05 Given Counting data and formulas, calculate the Standard Deviation:

Must Perform- do the math.

$$\sigma = \sqrt{\frac{\sum_{i}^{n} (x_i - \bar{x})^2}{n-1}}$$

2.03.06 Given Counting data and formulas, Determine the confidence interval for the Stated confidence level:

Confidence Interval (CI) **= The** mean (average) counts (or count rate) +/- counting error.

Where “counting error” is the standard deviation (σ) multiplied to the desired confidence level (k).

Therefore,

CI = Mean counts (or count rate) +/- k multiplier (k) × Standard deviation (σ).

CI = $\bar{x}$ ± (k) (б)

So CI(95%) = $\bar{x}$ ± (1.96) (б)

Error	Confidence Level (%)	k
Probably	50	0.6745
Standard	68	1.0000
9/10	90	1.6449
95/100	95	1.9600
99/100	99	2.5750
99.7/100	99.7	3.0000

2.03.07 Objective REMOVED and number marked as "RESERVED" (for future use):

N/A

2.03.08 Objective REMOVED and number marked as "RESERVED" (for future use):

N/A

2.03.09 Given counting data, determine the background count rate and net sample/source count rate:

$$R_b = \frac{N_b}{T_b} \qquad R_n = R_g - R_b$$

2.03.10 State the purpose of determining efficiencies and correction factors for counting instruments:

- The fraction of articles emitted by the source that travels towards the window.
- The fraction emitted toward the detector window that reaches the window.
- The faction of the particles incident on the window that passes through the window and produces an ionization.
- The fraction scattered into the detector window.

2.03.11 Given counting data, calculate the efficiency and correction factor for a counting instrument:

Efficiency = $\boldsymbol{E = \frac{cpm}{dpm}}$ Correction factor = $\boldsymbol{CF = \frac{1}{E}}$

2.03.12 State the purpose of creating quality control chart for counting instruments:

A counting instrument's response changes over time. To ensure the counting instrument remains acceptable for use, a QC chart can be used as a daily log sheet. QC charts enable us to track the performance of the system while in use and help identify problems.

2.03.13 List the items that must be checked as part of receipt tests and daily source checks on counting instruments:

As part of the daily source check, the instrument and assembly are inspected for:

- Broken or loose knobs
- Defective cables or connectors
- Damaged or loose ac power cord
- Defective or damaged detector assembly
- Missing calibration sticker
- Expired calibration

2.03.14 State when to perform receipt tests and daily source checks on counting instruments:

The receipt test is typically performed for any of the following occasions:

- The instrument is being received from calibration.
- After maintenance or repair of the instrument.
- New sources are to be used for the tests.

Daily source checks: Every day when the instrument is in use.

2.03.15 Describe corrective actions to be taken if a deficiency is noted during a surveillance test or daily source check of a counting instrument.

If the count rate is not within established response limits, repeat the test with the same source / counting interval.

If the count rate from the second attempt is not within established limits, remove the instrument from service and notify supervision.

2.03.16 Given counting data and formulas, calculate the error present in a single count of a sample to the desired confidence level.

$$\sigma_s = \sqrt{\frac{R_g}{T_g}}$$

The above formula calculates the error present in a single sample count (single sample standard deviation).

If the background count/count time is also given, it will be included in the equation as follows:

$$\sigma_s = \sqrt{\frac{R_g}{T_g} + \frac{R_b}{T_b}}$$

T_g = sample count time

T_b = background count time

2.03.17 Describe methods to improve statistical validity of sample measurements performed with counting instruments:

1. Increasing the background count time reduces errors associated with background count rate.
2. The lower the activity, the longer the sample should be counted to make results more precise.
3. Sample count time cannot be longer than the background count time.

2.03.18 Given counting data and formula, calculate a DL, MDC, and MDA for a counting instruments:

$$DL\ (cpm) = 1.64 \times \sqrt{R_b\left(\frac{1}{T_b} + \frac{1}{T_g}\right)}$$

$$MDC\ \left(\frac{\mu Ci}{mL}\right) = \frac{2.71 + 3.29 \times \sqrt{2.71 \cdot 3.29 \times \left(1 + \frac{T_g}{T_b}\right)}}{(6.29e{+}10) E_c \cdot T_g \cdot T_s \cdot E_f \cdot F_c}$$

$$MDA\ (dpm) = \frac{2.71 + 3.29 \times \sqrt{R_b \cdot T_g\left(1 + \frac{T_g}{T_b}\right)}}{E_c \cdot T_g}$$

R_b = background count rate (cpm)
T_b = background counting time (min)
T_g = gross sample count time (min)

2.04.01 Define the following terms:

- *Absorbed dose (D):* The average energy imparted by ionizing radiation to matter in a volume element per unit mass of irradiated material; it is expressed in units of rad or gray (100 rad = 1 gray).
- *Radiation Weighting Factor (W_R)*: A modifying factor used to calculate the equivalent dose from the average tissue or organ absorbed dose. The absorbed dose is multiplied by the appropriate radiation weighting factor.
- *Whole Body (Q)*: For purposes of external exposure, the whole body comprises the head, trunk (including male gonads), arms above and including the elbow, and legs above and including the knee.
- *Extremity:* Includes the hands and arms below the elbows and the feet and legs below the knees.

- *Equivalent dose (H_T):* The product of the average absorbed dose ($D_{T,R}$) in rad or gray (Gy) in a tissue or organ (T) and a radiation (R) weighting factor (W_R).

$$H_T = (D_{T,R})(W_R)$$

- *Effective dose (E):* The summation of the products of the Equivalent dose (H_T) received by specified tissues or organs of the body and the appropriate tissue weighting factor (W_T). It includes the dose from radiation sources internal and/or external to the body.

$$E = \sum W_T \cdot H_T$$

- *Tissue Weighting Factor (W_T):* The fraction of the overall health risk, resulting from the uniform, whole body irradiation, to a specific tissue (T). The equivalent dose to tissue (H_T) is multiplied by the appropriate weighting factor (W_T) to obtain the effective dose (E) contribution from that tissue.

- *Committed Equivalent Dose ($H_{T,50}$):* The effective equivalent dose is the equivalent dose (H_T) calculated to be received internally by a tissue or organ over a 50-year period after the intake of a radionuclide into the body. It does not include contributions from radiation sources external to the body.

Weighting Factors For Various Tissues (10 CFR 835)

Organs or tissues, T	Weighting factor, W_T
Gonads	0.20
Red bone marrow	0.12
Colon	0.12
Lungs	0.12
Stomach	0.12
Bladder	0.05
Breast	0.05
Liver	0.05
Esophagus	0.05
Thyroid	0.05
Skin	0.01
Bone surfaces	0.01
Remainder[1]	0.05
Whole body[2]	1.00

- *Committed Effective Dose (E_{50})*: The sum of the committed equivalent doses ($H_{T,50}$) to various tissues or organs, each multiplied by the appropriate tissue weighting factor (W_T).

$$E_{50} = \sum W_T \cdot H_{T,50}$$

- *'Total Effective Dose (TED):* The sum of the Effective Dose (E) for external exposures and the Committed Effective Dose (E_{50}) for internal exposures.

- *Cumulative Total Effective Dose*: The sum of the total effective dose values recorded for an individual for each year of employment at a DOE site or facility effective January 1, 1989.

- *Annual Limit on Intake (ALI)*: The derived limit for radioactive material taken into the body of an adult worker by inhalation or ingestion in one year. ALI is the smaller value of intake of a given radionuclide in a year by the reference man that would result in a committed effective dose equivalent of 5 rems or a committed equivalent dose of 50 REM to any individual organ or tissue.

- *Derived Air Concentration (DAC):* For the radionuclides listed in appendix A 10 CFR 835, the DAC is the airborne concentration that equals the ALI divided by the volume of air breathed by an average worker for a working year of 2000 hours (assuming a breathing volume of 2400m^3).

- *Derived Air Concentration-hour (DAC-h):* The product of the concentration of radioactive material in air and the time of exposure to that radionuclide, in hours.

- *Bioassay:* The determination of kinds, quantities, or concentrations and locations of radioactive material in the human body.

- *In Vivo:* A direct bioassay measurement of radioactivity in living tissue (e.g., whole body count).

- *In Vitro:* Indirect bioassay measurement of radioactivity by means of representative sampling to estimate the radioactivity in tissue (e.g., urine samples).

- *Declared pregnant worker:* A woman who has voluntarily declared pregnancy to her employer in writing for the purpose of being subjected to the occupational dose limits to the embryo/fetus. She may revoke this declaration in writing at any time.

2.04.02 State the DOE limits for exposure to ionizing radiation for:

Type of Exposure		Annual Limit (rem)
General Employee	Whole body TED (internal + external)	5
	Lens of the eye	15
	Skin and extremities	50
	Any organ or tissue other than lens of the eye (internal + external)	50
Declared pregnant worker	Embryo/fetus (internal + external)	0.5 per gestation period
Minors occupationally exposed	Whole body TED (internal + external)	0.1
	Lens of the eye, skin, and extremities	10% of general worker limits
Members of the public	TED for members of the public exposed to radiation and/or radioactive material during access to a controlled area	0.1

- *Emergency exposures:* Operations managements will weigh actual and potential risks to rescue and recover individuals against the benefits to be gained. Performed by volunteers.

TED Limit (rem)	Activity Performed
5	All
10	Protecting major property
≥ 25	Lifesaving or protection of large populations

- *Planned special exposures:* Must have exceptional situation with no other alternative, management specifically requests the planned special exposure in writing, and joint written approval is received from DOE Headquarters.

2.04.03 State the administrative control levels (ALC's) for occupational exposure to ionizing radiation at the Hanford site.

A DOE ACL of 2,000mrem/y per person is established for all DOE facilities. Approval by the DOE Site Manager and/or appropriate secretarial officer or designee is required prior to allowing a person to exceed 2,000 mrem.

Table 5. H2C Administrative Control Levels

Annual Maximum Equivalent Dose (mrem)				Approval Required to Exceed This Level (approvals are sequential)
Whole Body	Skin and Extremity	Lens of the Eye	Any Organ or Tissue[d]	
500	15,000	4,500	15,000	Level 2 line manager & project RadCon manager (RCM)
1,000	22,500	6,750	22,500	Level 1 line manager & Tank Operations Contractor (TOC) RCM
1,500	30,000	9,000	30,000	Chief Operating Officer
2,000				DOE Site Manager
Age × 1,000 mrem = lifetime TED				Level 1 line manager & TOC RCM

2.04.04 State the conditions that require the monitoring of exposure to ionizing radiation with the following types of external dosimetry:

Primary dosimeters (Hanford Standard Dosimeter):

- 100 mrem annual effective whole-body dose
- 5 rem/y to the skin or extremities
- 1.5 rem/y to the lens of the eye
- Entry into HRA or VHRA
- Declared pregnant workers who are expected to receive from external sources an equivalent dose of 50 mrem or more to the embryo / fetus during the gestation period.
- Members of the public and minors who are likely to receive 50 mrem in one year.

Neutron dosimeters: Provided when a person is likely to exceed any of the thresholds above, and 10 percent or more of the dose is likely to be due to neutron radiation.

Multiple whole-body dosimeters: Used to assess non-uniform exposures when:

- Calculated effective dose (E) is expected to exceed the equivalent dose (H_T) by more than 30% and exceed 100mrem.
- Calculated effective dose (E) is expected to exceed the equivalent dose (H_T) measured by the reference dosimeter by more than 100mrem.

Extremity dosimeters: Used when:

- Equivalent dose to the extremities may result in 5 rem or more in a year.
- During specific jobs in non-uniform fields with large / unpredictable dose rates in which extremities may receive an equivalent dose more than 10 times greater than equivalent dose received by the chest and the extremities may receive an equivalent dose greater than 500mrem.

Supplemental dosimeters: Required when:

- Entering an HRA or VHRA
- A person could exceed 10 percent of an ACL from external radiation in one workday.
- Required by RWP or RCO
- A person's estimated exposure is ≥ 80 percent of the ACL.

Area monitoring dosimeters: Installed where people work at least 8hrs/month if immediately next to or within 15 meters of a posted area excluding RCA. Area dosimetry is not required for any posted radiological areas where personnel dosimetry is required.

2.04.05 Describe the theory of operation of a Thermoluminescent Dosimeter (TLD):

Thermoluminescence is the ability of some materials to convert energy from radiation to a radiation of a different wavelength, normally in the visible light range.

Excited electrons are trapped after being irradiated. When heated, the electrons release their energy as light. The light emitted is then measured and is relative to dose. There are two categories:

- Flourescence: Emits of light *immediately* when irradiated (not useful for TLD).
- Phosphorescence: Emits light *after* the irradiation period (delay time can be seconds to weeks or months). This is the principal of operation for TLDs.

2.04.06 Describe how a TLD reader measures the radiation dose from a TLD:

A typical TLD reader has a heater, a photomultiplier tube, and a meter/recorder.

Heating the thermoluminescence material causes the trapped electrons to return to the valence band. When this happens, energy is emitted in the form of visible light. The light output is detected and measured by a photomultiplier tube (PMT), and a dose is calculated.

2.04.07 List the advantages and disadvantages of a TLD compared to a film badge:

Advantages:

1. Able to measure a greater range of doses appropriate to the Hanford site
2. Doses may be easily obtained
3. They can be read onsite of being sent away for developing
4. Quicker Turnaround time for readout
5. Reusable
6. Longer shelf life (it can still be read after one year)

Disadvantages:

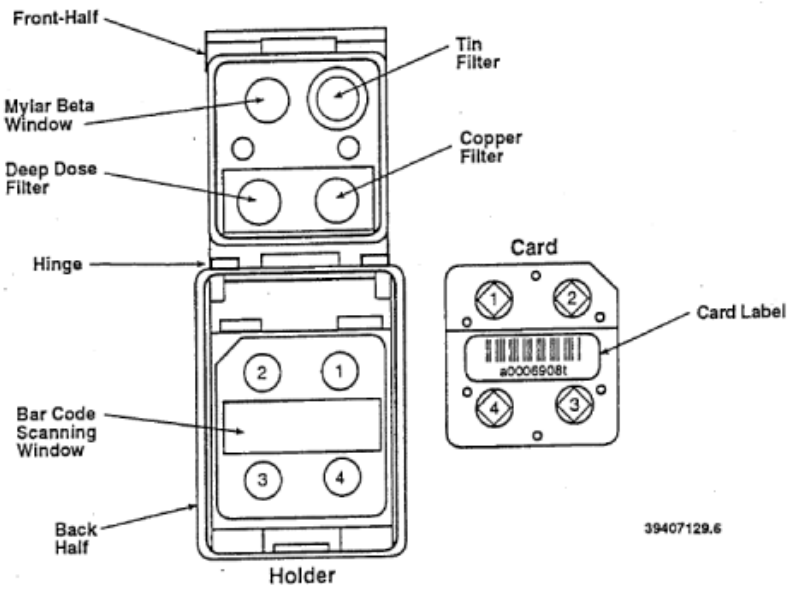
1. Each dose can only be read once, since the readout process zeroes the TLD
2. More expensive than film

2.04.08 Describe the elements and filters used in the following record dosimeters available on the Hanford site:

- *Hanford Standard Dosimeter (HSD):* Issued to all Hanford personnel who are assigned a dosimeter.

Table 10. Hanford Standard Dosimeter Elements

Position	Phosphor	TLD Thickness	Filters	Purpose
1	TLD-700	0.38 mm (100 mg/cm^2)	ABS[1] plastic & copper (333 mg/cm^2)	Eye dose determination (ß-γ)
2	TLD-700	0.38 mm (100 mg/cm^2)	ABS & PTFE[2] plastic (1,000 mg/cm^2)	Deep dose equivalent (ß-γ)
3	TLD-700	0.15 mm (40 mg/cm^2)	Mylar™ (7.5 mg/cm^2)	Shallow dose equivalent (ß-γ)
4	TLD-600	0.38 mm (100 mg/cm^2)	ABS plastic & tin (703 mg/cm^2)	Energy determination (ß-γ, + neutron detection)



- *Hanford Combination Neutron Dosimeter (HCND):* Consists of the following three components:
    1. Beta-photon TLD.
    2. An albedo neutron TLD.
    3. PNAD.

- *Area Dosimeter*: Issued to a location for monitoring ambient dose.

- *Extremity Dosimeter (finger ring):* Issued for specific tasks that possess the potential for significant extremity dose that may not be measured by the HSD or HCND. Designed to measure photon and beta exposure to extremities.

2.04.09 Describe the operation of the following supplemental dosimeters used at Hanford:

- *Electronic Dosimeters:*

    Thermo EPD: Semiconductor
    Personal radiation monitor that uses 3 silicon diode detectors that detect high energy gamma, low energy gamma, and beta radiations. It uses built in shielding and associated micro circuitry to detect, measure, and calculate the appropriate values.



- *Pencil Dosimeters:* A direct-reading ion chamber pocket dosimeter that uses the principle of discharging a pair of oppositely charged surfaces when air between them is subject to radiation. Now primarily only for emergency use and may be included in emergency kits at some facilities.

2.04.10 State the requirements & precautions for wearing and reading external dosimetry on the Hanford site:

- Dosimeters should be securely attached to prevent loss.
- The beta window must face outward from the body.
- Label on dosimeters that read “Front” is upright when read from the front.
- Wear on the trunk of body unless otherwise directed by RWP.
- Do not subject TLD to extreme heat or contaminate with grease or oils.
- Assigned TLDs can be taken home but daily/weekly issued dosimeters should not be taken home.

2.04.11 State the conditions that require the monitoring of occupational exposure to ionizing radiation with internal dosimetry:

1. Workers who are likely to receive 100mr committed effective dose (E_{50}) and/or 5Rem committed equivalent dose (H_{t50}) to any organ or tissue within one year from radionuclide intake.

2. Declared pregnant workers likely to receive an intake resulting in 50mrem or higher (>10% of the limit) to the fetus.

3. Minors and members of the public likely to receive 50 mrem/yr (>50% of the annual limit) committed effective dose (E_{50}).

Routine bioassays (i.e., whole body counts, chest counts, and/or urinalysis) should be required in RWPs where:

- Work involves respiratory protection and involves working directly with rad material.
- Work in HCA involving contact or disturbance of the contamination.
- Exposure to airborne activity such that the total exposure for a year would exceed 40 DAC-hr.
- As directed by the company based on procedures, technical basis, RadCon manual/tables.

2.04.12 Describe the following methods of internal dosimetry used on the Hanford site:

IN VIVO: Direct measurement of γ radiation emanating from the body (i.e., whole body count). No samples are required, and results are obtained quickly.

IN VITRO: Estimating the amount present in the body by analyzing excretions or secretions from the body (i.e., urinalysis, fecal analysis, nasal discharge, sputum (mucus from throat) samples, mouth smears). Requires sample submission and analysis, results can take up to one month depending on sample and testing.

2.05.01 State the difference between removable, transferable, and fixed surface contamination:

Contamination is defined as the presence of residual or unwanted radioactive material resulting from DOE activity in or on material or property.

Fixed: Surface contamination that is not easily transferred to personnel or equipment through normal contact.

- Fixed contamination does not include rad material present in a matrix, such as soil or cement.
- Fixed contamination is measured using direct survey techniques (commonly referred to as frisking or scanning). When non-removable (fixed) levels are to be recorded, the removable level must be subtracted from the total.

Removable: Surface contamination that can be removed from surfaces or transferred to personnel and equipment through non-destructive means such as normal contact, wiping, brushing, or washing.

Transferable: Refers to the total contamination levels, expressed in terms of dpm per a given area, on items such as shoes, shoe covers, vehicle tires, tools, or other equipment that has contacted contaminated soil.

2.05.02 Describe the of contamination surveys used to identify the presence of removable βγ or α contamination:

Contamination Surveys are performed to determine either:

- The absence of contamination
- The amount of contamination present

Technical smear: The required method used to **quantify** contamination values. At least one tech smear should be taken per square meter of surface area.

Large Area Wipe (LAW): used for area or items where contamination is known not to be present or has a low potential for contamination. It used to supplement tech smears and only used to qualify ("go/no-go") contamination presence. It is not approved for quantifying (determining and documenting) the amount of contamination.

2.05.03 State the purpose for performing a direct contamination survey:

- To assess Total and Fixed contamination values. (Fixed = Total – Removable)
- Clearing or releasing items from radiological controls.
- Assessing fixed contamination areas.
- Are needed in conjunction with smears to quantify the amount of radioactive material.
- Even though a tech swipe can indicate no removable contamination, it cannot very of the absence of rad material.

2.05.04 Describe how to perform a transferable contamination survey:

Performed by pressing a cloth or paper swatch slightly larger than 100 cm^2 into the soil of interest, using sufficient pressure to simulate normal walking conditions. Then directly measure the soil collected on the swatch with a field instrument capable of detecting Table 2-2 contamination values.

2.05.05 Convert direct reading measurements of cpm to dpm/100cm2 for documentation on RSR:

Contamination measurements of any type require conversion from cpm to dpm and should be reported in dpm.

dpm = (gross cpm – background cpm) × CF

CF = Correction Factor and is based on the instrument's efficiency. For a GM, the standard efficiency is 10 percent, so it has a correction factor is 10. (Efficiency of 10% means it only detects 10% of what is passing through the detector, multiplying by 10 accounts for this as 10% × 10 = 100%).

Alpha correction factors vary by company (H2C = 14% with CF = 7, HMIS and CPCCo = 10% with CF = 10).

A proper tech smear represents 100cm^2 of coverage so the result should be in dpm/100cm^2.

For direct surveys, the observed measurement will initially be in cpm/probe area. Converting this to dpm will give dpm/probe area; however, not all probes have 100cm^2 detector heads. For probes with 100cm^2 detector heads (i.e., 2360), the result will already be per 100cm^2. For all other probes, this is accomplished by adding the requisite number of adjacent probe areas to approximate 100cm^2. PAMs with a 54cm^2 probe area will require two adjacent probe areas. GMs with a 15.5cm^2 probe area will require six adjacent probe areas to approximate 100cm^2. The activity found on each of the required probe areas are then added together making a sum activity per 100cm^2 and can reported as dpm/100cm^2.

2.05.06 Describe the components of a radiological monitoring program for contamination control:

- Constant monitoring.
- Area and equipment surveys.
- External personnel surveys.
- Personnel internal monitoring and bioassay.

2.05.07 State the overall goal of a contamination control program and list actions that contribute to its success:

The goal underlying any effective contamination control program is to minimize CAs and to maintain contamination levels ALARA.

- Engineered controls (containment and confinement)
- Administrative controls (procedures, RWP, work instructions)
- PPE

2.05.08 State the basic principles of contamination control and list examples of implementation methods:

Good housekeeping is a prime factor in an effective contamination control program.

- Engineering controls. (ventilation, filtration, shielding, containment – glove bags, tents, sleeving)
- Access/administrative controls. (RWP, procedures, work instructions)
- Personnel protective measures. (PPE, glove changes, arm sleeves)
- Decontamination.
- Preventive methods. (fixative)
- Waste minimization.
- Minimization of mixed waste.

2.05.09 Describe the possible engineered control methods used for contamination control:

Engineered controls are a physical design feature in which components and systems such as containment, ventilation, filtration, or shielding are used to reduce airborne radioactivity, radiation levels, and the spread of contamination.

1. *Ventilation*. (Such the airflow is from non-CA to high CA, HEPA filters)
2. *Containment.* (Glove bags, tents, sleeving, windbreaks)
3. *Physical Design and control* (Facility design, selecting components that minimize radioactive buildup, etc.).
4. *Physical Access / Admin Controls.* (Two basic access control points)

2.05.10 Describe the use of protective clothing and contamination monitoring equipment:

The purpose of protective clothing is to prevent skin and personal clothing contamination.

- The type and extent of PPE required is determined by:
  - Type and form of contamination
  - Levels of contamination
  - Type of work being performed

- Whole Body Protection: Lab coats, Coveralls, Water resistant coveralls, Gore-tex suits, Welder coveralls, etc.

- Hand Protection: Surgical gloves, Rubber gloves, Neoprene gloves, Leather / Canvas / Cut Resistant gloves.

- Foot Protection: Shoe covers, Booties, Rubber knee boots.

- Respiratory Protection: Full-Face masks, Supplies air systems, SCBA.

2.05.11 Describe the general principles of decontamination, preventing and controlling contamination, waste minimization, and minimization of mixed waste:

- Line management is responsible for ensuring decon, where practicable, of facilities, tools, material, and equipment so that contamination can be minimized in the workplace.
    - TWDs should include provisions to control contamination at the source to minimize the amount of decontamination needed.
    - Work preplanning should include consideration of handling, temp storage, and decon of materials, tools, and equipment.
    - Decon activities should be evaluated and controlled to prevent the spread of contamination.
    - Water is the preferred decontamination agent.

- *Preventing / controlling contamination:*
    - Identify and repair leaks before they become a serious problem.
    - Establish adequate work controls before starting jobs.
    - While conducting pre-job briefings, discuss measures to help reduce or prevent contamination spread.
    - Change out gloves or protective gear as necessary to prevent cross-contaminating equipment.
    - Pre-stage areas to prevent spreading contamination from work activities.
    - Cover piping / equipment below work areas to prevent dripping contamination onto other areas.
    - Cover tools or equipment used during the job to minimize decon after the job.
    - Follow good work practices such as good housekeeping and cleaning up after jobs.
    - Place contaminated tools or equipment in plastic bags before moving outside of a CA.
    - When possible, wrap tools or equipment prior to entry to control contamination during use inside the CA.
    - Using tape to prevent contamination may leave sticky residue where contamination can later cling to.
    - Confine the spread of radioactive releases by a sound preventive maintenance program.
    - Minimize material taken into the CAs and control material taken out of CAs.

- *Rad Waste Minimization:* A radioactive waste minimization program should be in effect to reduce the generation of radioactive waste and spread of contamination from CAs, HCAs, and ARAs.
    - Restrict material entering RBAs to what is needed for performance of work.
    - Substitute recyclable or burnable items for disposable ones and reuse equipment when practicable.
    - Select consumable materials such as protective coverings and clothing that are compatible with waste processing systems, volume reduction, and waste form acceptance criteria.
    - Reserve an assortment of tools primarily for use in CAs, HCAs, and ARAs.
    - Maintain tools in a designated storage or distribution area or a contaminated tool crib.
    - Survey potentially contaminated material from CAs to separate uncontaminated from contaminated material.
    - Segregate known uncontaminated from potentially contaminated waste.
    - Segregate reusable items, such as protective clothing, respirators, and tools, at the SOP.
    - Minimize the number and size of RMAs.
    - Emphasize training in waste reduction philosophies, techniques, and improved methods.

- *Mixed Waste Minimization:* Restrict quantities of hazardous material, such as paints, solvents, chemicals, cleaners, and fuels from entering RBAs, and take measures to prevent inadvertent radioactive contamination of these materials.
    - Technical and administrative controls should be established to minimize the volume of mixed generated and the amount of radioactivity in such waste.
    - Materials suspected of being mixed waste should be identified and segregated as soon as practicable in the generating process to avoid combining mixed waste with other waste forms.

2.06.01 State the primary objectives of an air monitoring program:

- Measure the concentration of the radioactive contaminant(s) in the air by collection and analysis.
- Identify the type and physical characteristics of the radioactive contamination.
- Evaluate the potential hazard to the worker.
- Evaluate the performance of airborne radioactivity engineered controls such as containment and confinement.
- Demonstrate the airborne radioactivity released to the general environment is maintained ALARA and below the allowable limits established by the DOE.
- Determine whether the level of protection provided to the workers is sufficient to minimize the internal equivalent dose.
- Assess air concentration data to determine whether bioassay should be initiated to verify whether an exposure has occurred, and if so, to determine the magnitude of the exposure.

2.06.02 Describe the three physical states of airborne radioactive contaminants:

1. *Particulate*: Particulate contaminates are solid and liquid particles suspended in air. Solids are subdivided into fumes, dusts, and smokes. Liquids are subdivided into mists and fogs.

2. *Gases:* Gases are substances that exist in the gaseous phase when under normal conditions of temperature and pressure. Radioactive gases are usually treated as an external source of exposure.

3. *Vapors:* Vapors are the gaseous phase of a substance that would otherwise exist as a solid or liquid under normal conditions of temperature and pressure.

2.06.03 Describe the primary considerations to ensure a representative air sample is obtained:

- The airborne radioactivity concentration entering the sample line must be representative of the airborne radioactivity concentration in the air near the sampling device.

- The airborne radioactivity concentration entering the sample device must be representative of the airborne radioactivity concentration at the point of concern (the air workers are exposed to and/or breathe).

2.06.04 State the relationship between sample purpose and location for air sampling:

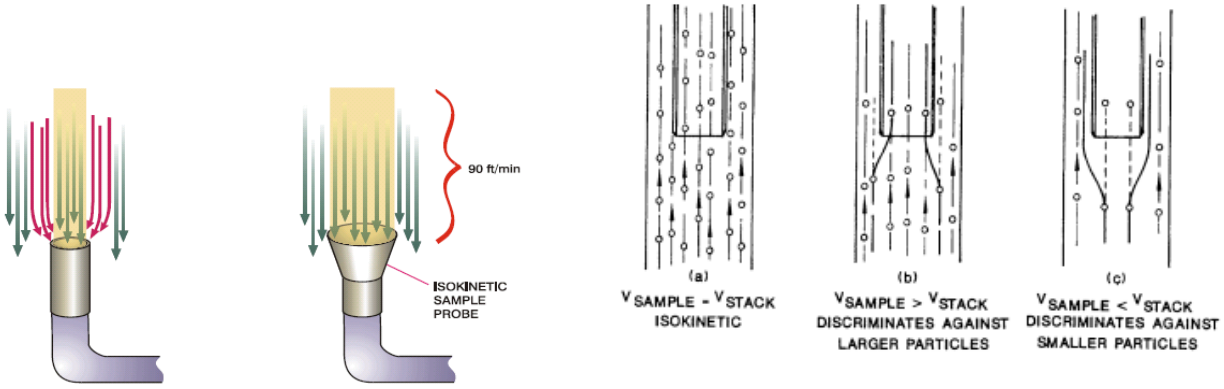
- Air samplers positioned to evaluate the effectiveness of a containment or confinement system should be placed near and downstream of the potential release point.

- When monitoring to determine the worker's potential intake, the sampler should be located near the worker's breathing zone (nose and mouth).

Purpose of Sampling/Monitoring	General Placement of Samplers/Monitors
Estimate worker's intake for calculating internal dose	Sampler located in worker's breathing zone, near nose and mouth (personal air sampling)
Identify area needing confinement control	Sampler in airflow pathway near actual or potential release point
Provide early warning of elevated airborne release	CAM placed between worker and release point
Test for leakage of radioactive materials from sealed confinement system and for posting	Sampler placed downstream of confinement/ control area
Determine total concentration from many potential release points	Downstream at exhaust point
Determine ARA status or respiratory protection adequacy	Sampler at worker's location/breathing zone
Special purposes, e.g., determining particle size	On a case basis, depending on airflow patterns

2.06.05 Define the term isokinetic sampling as associated with air sampling:

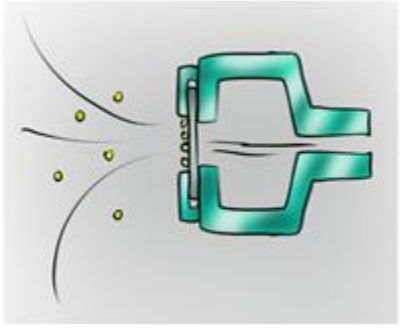
When the sample line velocity is equal to the system velocity at the sample Point, it is called Isokinetic sampling.



2.06.06 Describe basic sampling methods and the principle of operation for each general method of obtaining samples or measurements of airborne radioactivity concentrations:

- *Filtration:* Employs filtration of the air as the method of concentrating the airborne particulate contaminants (uses a filter).

Air Sampler head with filter installed collecting particulates:

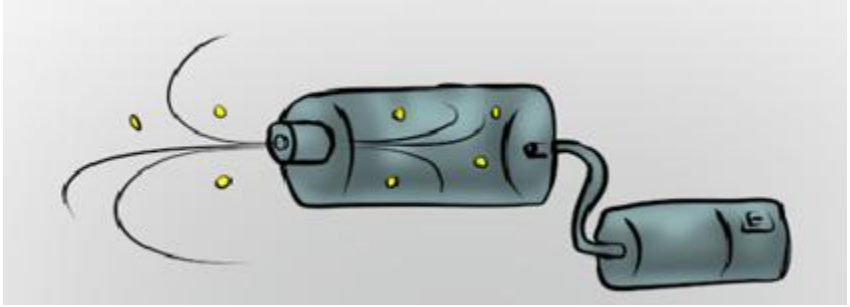


- *Volumetric:* Employs a sample container into which the sample is drawn and isolated for analysis (collects and stores a volume of the contaminated air in a container).

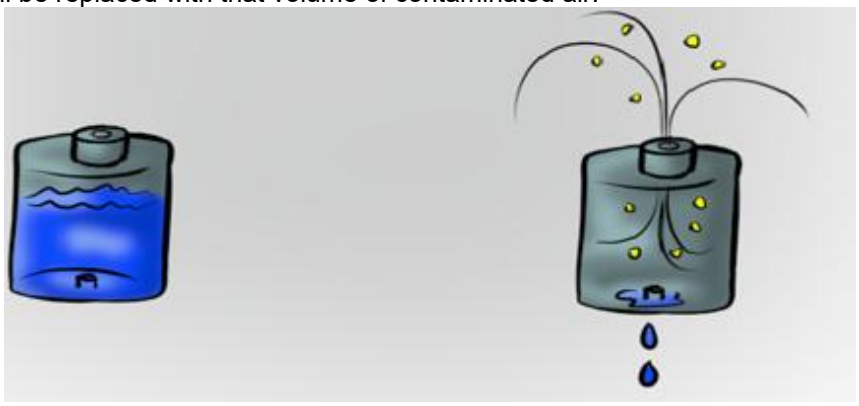
Vacuum all air out of a container then open the vacuumed container (which will suck in contaminated air):



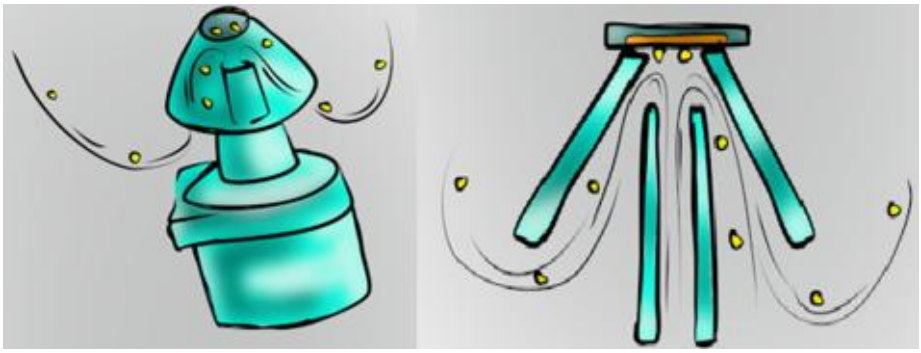
Traditional vacuum pump hooked up to a container (sucks air into the container):



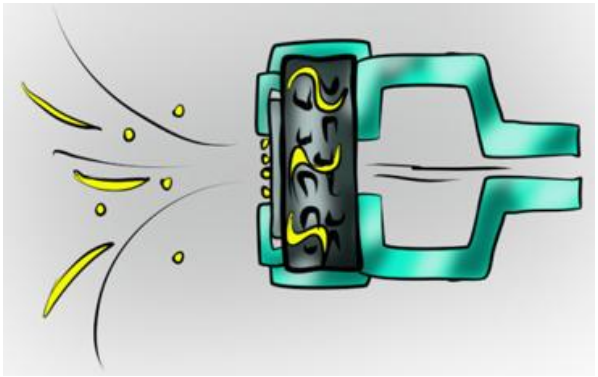
Fill a container with water. At the sample location, open the top and drain the water. The volume of the water will be replaced with that volume of contaminated air.



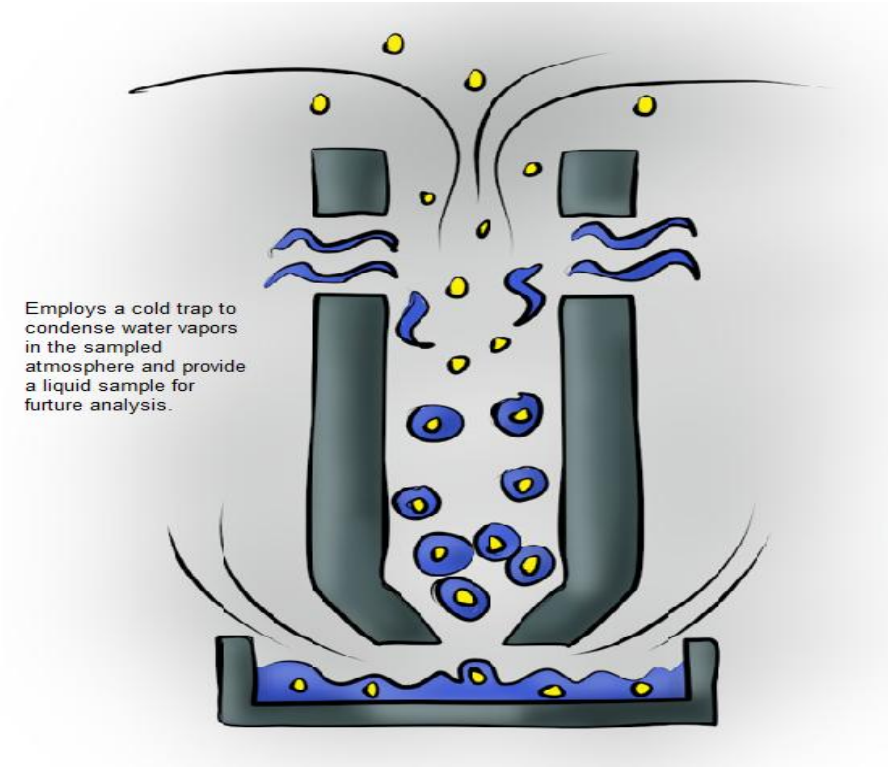
- *Impaction / Impingement:* Coat the collection side of the sample media with a thin layer of grease or adhesive. When the airflow is sharply deflected (forced to immediately change direction) the particles will be unable to follow the abrupt airflow change and end up trapped/collected on the sample media.



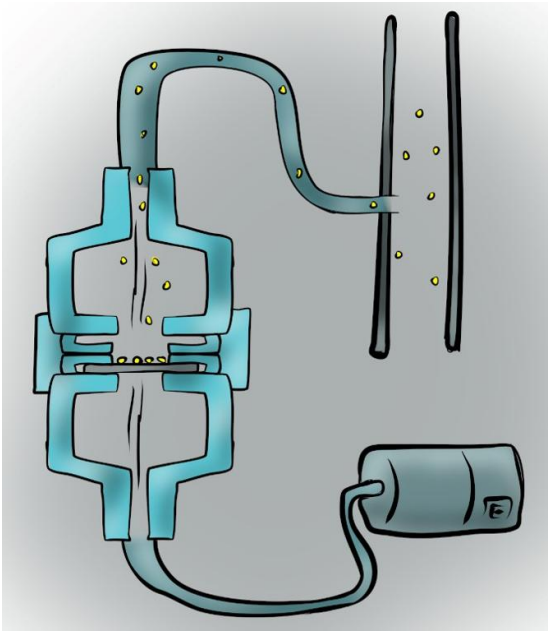
- *Adsorption:* Adsorption is the adhesion of a substance to the surface of another substance through a bonding process. This technique uses an air mover to draw air across adsorption media such as activated charcoal, silica gel, and silver-zeolite. This is used to collect organic vapors and non-reactive gases/vapors. Partiulates will need to be pre-filtered before the adsorption process.



- *<u>Condensation</u>:* A cold trap condenses water vapors and collects it as a liquid sample of the atmosphere.



- *<u>In line / Flow through</u>:* Employs an air mover or pathway to direct the sample air to also flow through or past a detection device.



2.06.07 Using the air sample equation, calculate an air sample concentration and determine the total derived air concentration:

Activity in uCi/ml = $\frac{R_n x\ K\ x\ C_f}{V\ x\ E\ x\ E_f}$

R_n = net count rate (cpm)
K = 1.6E -11
C_f = Filter geometry correction factor (1 unless otherwise specified)
V = Sample volume in ft^3 or (flow rate in cfm) x (time in min)
E = efficiency (cpm/dpm)
E_f = filter collection efficiency (Versapore, Fluoropore, and beta-gamma = 1; AKA alpha = 0.82)

$$\frac{Beta\ or\ alpha\ concentration\ in\ the\ breathing\ space\ \left(\frac{\mu Ci}{mL}\right)}{DAC\ value\ for\ the\ isotope\ \left(\frac{\mu Ci}{mL}\right)} = DAC\ fraction$$

TDAC = Alpha DAC fraction + Beta DAC fraction

Note: This is the expanded version of $\frac{N \times K}{V \times E}$ includes filter geometry C_f and collection efficiency E_f either of which may be given on the exam (when not given, they are assumed to equal 1).

2.06.08 List the factors that affect the accuracy of airborne radioactivity measurements and describe how these factors affect sample accuracy:

- Sample not representative of the atmosphere being sampled.
- Sample not representative of the air being breathed by worker.
- Incorrect or improperly installed sampling media.
- Malfunctioning, misoperated, or uncalibrated sampling device.
- Inaccurate or mis-operating of timing device, causing errors in the time value.
- Malfunctioning or uncalibrated flow meter.
- Mishandling of the sample media, causing cross-contamination or removal of sample material.
- Sample loading or humidity that causes changes in collection efficiency.
- Improper use or selection of analysis equipment.
- Inherent errors in the counting process due to sample geometry, self-absorption, resolving time, backscatter, and statistical variations.
- Mathematical errors during calculations due to mistakes and intermediate rounding.
- Incorrect making of the samples and inaccurate data recording.
    - Improper sample flow rate, which causes changes in collection efficiency.

2.06.09 Describe the five types of airborne radioactivity samplers / monitors:

1. *Personal air samplers:* Provide an estimate of the airborne radioactivity concentration in the air that the worker is breathing during the sampling period.

2. *High flow rate samplers:* Provide an estimate of the airborne radioactivity concentration at a particular location in a short time. Used to verify boundaries of ARA's and provide quick estimates for emergency response.

3. *Low flow rate samplers:* Provide an estimate of airborne radioactivity concentration averaged over a longer period at a particular location.

4. *Portable Continuous Air Monitors (CAM):* Provide an estimate of airborne radioactivity concentration averaged over time at a particular location and provide immediate readout and alarm capabilities for preset concentrations.

5. *Fixed continuous air monitors:* Provide an estimate of airborne radioactivity concentration averaged over time at a fixed location and provide immediate local and remote readout and alarm capabilities for preset concentrations.

2.06.10 Describe the general considerations for selection of an air sampling method:

The data obtained must be meaningful and accurate to assign radiological control measures.
General considerations include:

- Environmental conditions (humidity, temperature, flammability, dust, etc.)
- Physical characteristics (space requirements, power source, etc.)
- Energy and type of radiation.
- Expected concentration levels.
- Physical state of the contaminant.
- Type of survey required (breathing zone, general area, work area, etc.)
- Procedural requirements.

2.06.11 Define breathing zone air monitoring:

A form of air monitoring that is used to detect and quantify the radiological conditions of air from the general volume of air breathed by the individual (usually at a height of 1 to 2 meters).

2.06.12 Define "Personal Air Monitoring":

Sampling of air in the immediate vicinity (typically within 1 foot) of an individual's nose and mouth usually by a portable sample pump and collection tube (e.g., a lapel sampler) worn on the body.

2.06.13 Discuss the principles of air sampler placement:

- *Upwind:* The airflow pattern in the area is taking the airborne radioactivity from the source and moving it in a direction away from the air sampler causing the sampler to sample air that does not contain airborne radioactivity from the source.

- *Downwind:* The airflow pattern in the area is taking the airborne radioactivity from the source and moving it in a direction toward the air sampler causing the air sample to collect airborne radioactivity from the source.

- *Placement Factors:*
  - Purpose / type of sample
  - Location of workers in the field.
  - Direction of air exhaust
  - Airflow patterns
  - Presence of loose surface contamination
  - Identification of release points of source material.

- *Placement Considerations:*
  - Samplers should be placed for easy access.
  - Samplers should not interfere with the workers or the work.
  - Samplers should be positioned such that exhaust is pointed downwind.
  - Samplers should not be influenced by supply airflow; otherwise, the sampler would sample the supply air and not the workplace air.
  - Samplers on a horizontal surface should not have the discharge air directed at the surface where it may increase airborne radioactivity levels by resuspension.

2.06.14 Describe the RCT responsibilities if an initial air sample result show greater than the ARA posting criteria:

- Post the area accordingly and inform RadCon management.
- If work is in progress, evaluate whether work may continue; stop work if necessary.
- Note whether there are unprotected workers in the affected area.
- Evacuate workers who do not have appropriate respiratory protection.
- Determine whether the RWP is adequate for the new radiological conditions.
- If it is suspected that process-related radon daughters may be present in the workplace, transport the sample to a facility capable of determining the gross beta concentration within about 1 hour of sample collection.
- Record results on the RSR and logbook.
- Recount the sample after one decay period to determine the contributions from radon / thoron.

2.06.15 Describe the process for evaluating an air sample for radon/thoron.

- Identify beta to alpha ratios of sample(s) and evaluate to established ratios.
- Characteristic decay curves (compare the samples rate of decay).
- Gamma spectrometry.
- Pseudo Coincidence Counter Measurements (utilizing and cross-referencing multiple instrument types).

2.06.16 Describe the criteria requiring real time continuous air monitoring:

Real-time air monitoring equipment should be installed where increases in airborne radioactivity levels, should they occur, are likely to result in an exposure greater than 40 DAC-h in one week (>40DAC-h/week).

2.06.17 Calculate a CAM alarm setpoint (ASP) and state the maximum ASP value:

- The CAM ASP should be set at the lowest possible level without resulting in a significant number (one per month per CAM) of false alarms.

- The CAM ASP is determined by adding the calculated ASP to the background count rate.

$$ASP\ (cpm) = \frac{(DAC)(h)\left(60\frac{min}{h}\right)(E)(F)(PF)}{1.6E{-}11}$$

2.06.18 State the requirements for taking a cam out of service if it fails an operational or performance test:

When taking CAM out of service, RCT will:
1. Tag the CAM out of service with an appropriate service tag.
2. Notify facility management of the need for CAM repair, replacement, and/ or calibration.
3. Document CAM failure on the performance documentation
4. Provide specific detail in the “CAM Discrepancies” section or equivalent.
5. Prepare replacement CAM for service as needed.

2.06.19 Describe the application of ALARA principles in the decision to utilize or not utilize respiratory protection:

ALARA requires that the TED be minimized regardless of the source of exposure. Respiratory protection reduces internal dose but potentially increases stay time (which increases external dose).

2.07.01 Describe each of the following respirators and their associated limitations:

- *Air-purifying particulate-removing filter respirator:* **PF=50**
  - Removes particulates by filtration before they can be inhaled.
  - APRs do not supply oxygen, so they cannot be used in oxygen deficient atmospheres.
  - Powered APRs (PAPRs) are time limited by battery life.
  - No protection against gases or vapors.
  - Humidity may increase breathing resistance as filter becomes saturated.

- *Air-purifying, chemical cartridge and canister respirators for gases and vapors:* **(No Rad PF)**
  - Traps or reacts with specific vapors and gases to remove them from the air breathed.
  - Does not supply oxygen, so cannot be used in oxygen deficient atmospheres.
  - High humidity, temperature, and breathing rates associated with heavy physical exertion will shorten the life of the sorbent material.

- *Full face supplied air respirator:* **PF=1,000**
  - Supplied-air respirators (SAR) use a central source of breathing air that is delivered to the wearer through an air supply line or hose.
  - Since the airline respirator provides no protection if the air supply fails, it cannot be used in atmospheres that are IDLH or for emergency or rescues.
  - The trailing air supply hose limits mobility, so it may be unsuitable if frequent movement between separated workstations is needed.
  - The length of hose and pressure of the supply system can reduce the number of allowable users.
  - Control of the air quality is essential to avoid introduction of hazardous respiratory agents to the wearer's breathing zone.

- *Atmosphere-Supplying Respirator – Self-Contained Breathing Apparatus – SCBA:* **PF=10,000**
  - Allows the user to carry a respirable breathing supply that is not tethered to an external or stationary source.
  - SCBA has limited hair and is bulky/heavy (hampers movement).
  - Tank only lasts for 30, 45 or 60 minutes.

- *Combination Atmosphere-Supplying respirators:*
  - Combines the capabilities of SCBA and SAR by attaching the worker to an air-supply line but equipping them with a small air cylinder that enables the wearer to escape dangerous atmospheres in case primary air is interrupted. Note that this is not approved for IDLH atmospheres at Hanford.

2.07.02 Define the Following Terms:

- *Protection factor:* The ratio of the concentration of the contaminant in the atmosphere to the concentration of the contaminant inside the facepiece or hood.

$$PF = \frac{Concentration\ of\ contaminent\ in\ atmosphere}{Concentration\ of\ contaminent\ in\ respirator}$$

- *Assigned Protection factor;*

Type of Respirator	Quarter Mask	Half Mask	Full Facepiece	Helmet/Hood
Air-purifying (APR)	5	10	50	NA
Powered Air Purifying (PAPR)	NA	50	1,000	1,000
SCBA	NA	NA	10,000	10,000

- *Maximum use concentration:* Used to select air-purifying respirators (APR) with the proper protection level based on the APF (assigned protection factor) and exposure limit being used.

MUC = (APF) x (Maximum exposure limit)

2.07.03 State the general considerations and considerations for the nature of the hazard when selecting the proper respiratory protection equipment:

- Select and provide a suitable respirator based on the respiratory hazards to which the worker is exposed, considering workplace and user factors that affect respirator performance and reliability.
- Select a NIOSH-certified respirator, which must be used in compliance with the conditions of its certification.
- Identify and evaluate the respirator hazard in the workplace, including a reasonable estimate of worker exposures to respiratory hazards and an identification of the contaminant's chemical and physical form.
- Enough respirator models and sizes must be available so that respirator is acceptable for and correctly fits each user.

2.08.01 Define the following terms:

- *Sealed radioactive source:*
  A sealed radioactive source is a source manufactured, obtained, or retained for the purpose of using the emitted radiation. A known or estimated quantity of radioactive material is contained within a sealed capsule, sealed between layers of nonradioactive material or other means intended to prevent leakage or escape of the radioactive material.

- *Accountable Sealed Radioactive Source:*
  An accountable sealed radioactive source is a sealed radioactive source that has a half-life ≥30 days, and an isotopic activity equal to or greater than the corresponding value provided in the RadCon manual.

- *Exempt-quantity Sealed Radioactive Source:*
  An exempt-quantity sealed radioactive source is a sealed source with an isotopic activity less than the corresponding value for an accountable sealed radioactive source and is not considered a consumer product.

- ***Radiation generating device*:**
  A collective term for devices that produce ionizing radiation, including certain sealed sources that emit ionizing radiation, small particle accelerators used for single-purpose applications that produce ionizing radiation, and electron-generating devices that produce x rays incidentally. That is, it is a device that generates a measurable output of radiation.

  Sealed radioactive sources that can generate external radiation fields ≥ 100 mrem/hr at 30 cm from the accessible surface are also classified as RGD.

2.08.02 Describe the responsibilities of the RCT for radioactive sources:

- Follow the general source handling precautions sources as described in the company procedures.
- Perform receipt surveys.
- Using the radioactive source checkout log sheet to check sources in and out of their storage location.
- Performing the inventories and integrity testing following approved
- Procedures as directed by the source custodian.
- Taking appropriate actions if sources are leaking or missing.

- Document integrity testing results
- Transport sources in a labeled container to protect the source and avoid losing small sources.
- Avoid carrying sources in personal or protective clothing to preclude inadvertently taking a source offsite or disposing it as radiological waste.
- Handle only the edges and backs of sources to prevent damage to the active surface.
- Never leave a source unattended!

2.08.03 Describe the requirements for packaging, marking and labeling radioactive sources &/or their container:

Sealed sources with activities that exceed the values listed in the RadCon manual must be labeled with the radiation symbol and "Caution Radioactive Material" or "Danger Radioactive Material."

The label should provide sufficient information and include:

- Radionuclide
- Quantity of radioactive material
- Date of quantity estimate
- Serial number of the source or device

2.08.04 Describe the storage and posting requirements for radioactive sources:

Sources >5 mrem/ hr at 30 cm must be stored in a posted radiation area (RA) or an adequately shielded area within an RMA.

Storage rooms and/or source cabinets should be:

- Locked and posted
- Located to minimize damage from a fire
- Free of flammable substances
- Isolated from occupied areas or located in RMA or other appropriate radiological areas
- When stored in continuously occupied controlled areas, the radiation level at the closest approach is ALARA and is ≤ 0.5 mrem/ hr on average
- Gamma radioactive sources should be stored separately from locations where radiation detection / counting equipment is present

2.08.05 Describe the general precautions for radioactive sources:

Minimize exposure by using the following techniques:

- Inspect each source before use.
- Remove damaged sources from service. Do not directly handle any source if there is evidence of damage.
- Objects including fingers with or without gloves should never touch the active surface of unsealed sources.
- Protect the sources from becoming contaminated.
- Minimize handling time – use remote handling devices such as handles or tongs.
- Establish shielding.
- Use a smaller source when available.

2.08.06 Describe sealed radioactive source inventory requirements and the actions to be taken if a radioactive source is missing:

Source inventories should verify presence/condition and adequacy of all labels, locations, containers, and postings.

Accountable sources are inventoried at intervals not to exceed six months. Exempt-quantity sealed radioactive sources are inventoried annually.

If a source is missing, perform a search of the work area and promptly notify the source custodian, project / facility RCM, line management, and CTA/ POC.

2.08.07 State the requirements for integrity testing of radioactive sources:

Each accountable sealed radioactive source (except for gas only sources or tritium sources) is subject to a source leak test upon receipt, when damage is suspected, and at intervals not to exceed six months.

If source has been removed from service, then it is not subject to periodic testing.

2.08.08 Describe the actions required if a radioactive source is leaking:

An RCT is required to complete the following if they discover a leaking source:

1. Remove leaking source from service immediately and notify Source Custodian / RadCon management.
2. Wrap or contain the source, applying appropriate tags or labels.
3. Complete a RSR.
4. Initiate a corrective action report.

2.09.01 State the goals of an environmental monitoring program:

The following are the goals of an environmental monitoring program:

- Estimate human population doses
- Determine site impact
- Detect and quantify an unplanned release
- Meet legal or regulatory requirements
- Create and maintain a good public image
- Obtain pathway data
- Verify adequacy of RadCon measures
- Study air and water mixing patterns
- Non-industry applications

2.09.02 Define the term Critical Nuclide:

Among all the radionuclides that can be released from a site, a small group called *Critical Nuclides, are the specific radionuclides on a site that would, if released, cause the largest dose contribution to the population at risk near the site.*

2.09.03 Define the term Critical Pathway:

A *pathway* is any route that radioactive contamination can take from the source to a person where it becomes internally deposited or contributes to external dose.

A *Critical Pathway* is the route taken, from the point of release to body entry, of a *critical nuclide* that causes human exposure.

2.09.04 For each critical nuclide at Hanford, state the Critical pathway, Sample location, and Sample method:

Four Critical Nuclides at Hanford: **^{90}Sr, ^{239}Pu, ^{137}Cs, ^{241}Am**

Strontium-90 (^{90}Sr)	**Plutonium-239 (^{239}Pu)**
Critical Pathway: Soil/Foliar Deposition > Plants > Food Chain > *Bones (critical organ)* Sample Location: Ambient Air, Rainfall/Water, Soil, Plants Sample Method: Air, Water/Groundwater, Soil, Vegetation	Critical Pathway: Soil > Plants > Food Chain > *Lung (critical organ for insoluble Pu) or Bone (soluble Pu)* Sample Location: Ambient Air, Water, Soil, Plants Sample Method: Air, Water/Groundwater, Soil, Vegetation
Cesium-137 (^{137}Cs)	**Americium-241 (^{241}Am)**
Critical Pathway: Foliar Deposition > Plants > Food Chain > *Whole Body (critical organ)* Sample Location: Ambient Air or Bound to dust/powder, Rainfall/Water, Soil/Clay (earth) Sample Method: Air, Water/Groundwater, Soil	Critical Pathways: Pu facilities (PFP/WARP) trapped in plutonium matrix or from nuclear weapons testing fallout > *Bone (critical organ)* Sample Location: Ambient air, typically only found in labs or specific facilities, Nuclear Fallout (insignificant) Sample Method: Air sampling, Direct Dose/Contamination surveys, as appropriate if there is a release of material

2.09.05 Define the term Suspect Waste Site:

A *Suspect Waste Site* is any site that is thought for any reason to contain dangerous waste, hazardous waste, and/or radioactive waste. This term does NOT include sites already identified, only suspected sites.

2.10.01 Describe the types of activities that constitute radiological work:

Examples of radiological work, as defined in the RadCon manual, include:

- Work on contaminated or potentially contaminated facilities or systems
- Work in RBAs that are set up for contamination control
- Work that disturbs material fixing contamination in an FCA
- Work in RA, HRA, or VHRA
- Work that uses an RGD
- Work in a CA, HCA, or ARA
- Work in contaminated or potentially contaminated soil (including intrusive activities such as digging, excavating, or disturbing soil)
- Packaging or unpackaging radioactive material
- Work in any area where there is a high probability of radioactive material being present
- Work required RadCon hold points
- Work that would affect the radiological conditions of the facility, e.g., securing ventilation, moving, or changing shielding material, interrupting power to radiological monitoring systems or components.

2.10.02 State the requirements for, purpose of, and information found in an RWP:

RWPs are administrative controls that provide users with instructions for radiological safety.

RWPs are required for:

- Entering and performing work within radiological areas
- Handling materials with removeable contamination that exceeds the values of Table 2-2.
- Intrusive activity in a soil contamination area (SCA) or underground material area (URMA)
- Using vacuum cleaners or portable air handling equipment in areas set up to control removable surface contamination or airborne radioactivity

RWPs include the following information, as applicable:

- Description of work.
- Work area radiological conditions.
- Internal and external dosimetry requirements.
- Pre job briefing requirements.
- Radiation worker training requirements for entry.
- Protective clothing and respiratory requirements.
- Limiting radiological conditions that may void the RWP
- Personnel survey consideration
- Technical work document (TWD) number(s)
- Unique identifying number
- Date of issue and expiration.
- Authorizing signatures.

2.10.03 State responsibilities for initiating or using an RWP:

RWPs can be job specific or job general (routine). RWPs are based on actual or potential radiological conditions for the work to be performed; therefore, radiological surveys are routinely reviews to evaluate the adequacy of RWP requirements, and RWPs are updated if radiological conditions change to the extent that protective requirements need modification.

Signing an RWP (or RACing in under one) signifies the worker has read, understands, and agrees to comply with the RWP requirements.

2.10.04 Describe the purpose and use of void limits and action levels.

Void limits are rad conditions that, when verified to be present, cause an immediate cessation of work activities.

Void limits are reached due to:

- Personnel error.
- Radiological conditions were unknown and exceeded expectations.
- Changing radiological conditions from the work being performed exceeded expectations.

Action Levels define radiological conditions where the work is temporarily suspended until specific actions are taken, such as area decontamination to below specified levels, to ensure that radiological conditions, protective clothing, equipment, engineered controls, and monitoring are adequate to continue working, and to prevent reaching the limiting safe conditions.

2.10.05 Describe the process for revising RWP.

RWPs are valid for 12 month and require annual reviews. RWPs may periodically need revisions prior to the annual review. This can be due to unknown conditions discovered during work, changing conditions that require an increase/decrease in controls to continue work, or changes in work scope. These changes can range from an editorial change that does not affect controls, to a full re-write that changes the protection envelope.

- Make changes, use indelible blue or black ink if pen and ink changes allowed at facility.
- Change revision number and initial and date the change.
- Obtain all necessary approvals.
- Update RWP log, work packages, field logbook (if applicable), and any posted copies.
- Notify all affected personnel of the change and, if necessary, rebrief RWP and have everyone resign.

2.10.06 Given an RWP, state the following (from the RWP):

Given an RWP, be able to find and identify (on the RWP):

- Isotope(s) to be encountered
- Type(s) of radiation emitted
- Instrument range(s) needed for the job

When reviewing the RWP to select an instrument, the focus will be on the primary isotopes, the types of radiation to be expected, and the levels of radiation to be expected.

2.10.07 State the purposes, initiating conditions, & processes for conducting pre-job briefings & post-job ALARA reviews:

Pre-job *briefings* are conducted by the work supervisor with help from an RCT who reviews the RWP requirements. Pre-job briefings include all workers who will be involved in work activities that are not considered to be skill-of-the-trade. Procedures are verified, worker qualifications are verified, and emergency responses are discussed. RCT/worker needs and instructions are communicated and understood. At the end of the meeting, everyone should know what is expected, how to do it, and the conditions under which it is to be done.

Pre-job briefings are specifically held prior to work expected to exceed RadCon trigger levels such as:

- Estimated individual dose >250 mrem or collective dose for a task projected to be >1 person-rem
- Predicted airborne radioactivity concentrations exceeding 100 times the DAC or resulting in an integrated exposure in DAC-h specific to each facility
- Work area removable contamination >100 times the values in Table 2-2
- Entry into areas where whole body dose rates exceed 1 rem/h
- Entry into areas where extremity of skin dose rates are >10 rem/h

Post-job *meetings* (not required by RadCon manual) allow the opportunity to critique the work performance. Post-job meetings rely heavily on the input of each worker for information on how best to reduce exposure the next time the job is performed.

Post-Job *reviews* are performed as required by the applicable RadCon manual. Reasons include:

- An actual collective equivalent dose of 5 person-rem or greater
- Actual doses for a task outside the range of ± 25% of pre-job estimates of 1,000 person-mrem TED and/or 100 mrem TED per person
- Use of stop radiological work authority
- Identification of significant lessons learned
- A task results in a reportable radiological occurrence as specified in DOE O 232.2A, Group 6, "Contamination / Radiation Control," such as loss of sealed sources, spread of contamination above specified limits, or an occurrence requiring offsite medical aid for contaminated personnel

Post-Job *reviews* typically address the following (as applicable):

- Deviations from the original scope
- Any planned ALARA controls that could not be implemented as intended
- Time needed to perform the job
- Effectiveness of implemented exposure controls
- Any unplanned situations encountered during the job, especially those that caused increased personnel dose
- Lessons learned, both those to be implemented in future planning and those that presented an opportunity for improvement
- Actual vs. estimated collective dose
- Abnormal events / situations causing work to be stopped
- Actions taken to prevent recurrence of problems or situations.

2.10.08 State the purposes of radiological postings, signs, labels, & barricades and the RCT's responsibilities:

The purpose of radiological postings, signs, and labels is to identify items or areas that have (or have the potential to have) radiological hazards, identify the radiological hazard(s) present in an area, and help workers avoid inadvertently entering radiological area(s) and/or mishandling radioactive materials.

All access points into an area must be posted and applicable status boards should stay promptly updated.

Everyone is responsible for reading and following all the information specified on rad postings, signs & labels. Disregarding or unauthorized removal or relocation of radiological postings, signs, or labels, or barricades can lead to unnecessary or excessive radiation exposure and/or personnel contamination and may lead to disciplinary actions up to and including job termination.

If any material used to identify rad hazards is found outside an RCA, it should be reported to the RCO and an RCT will perform a survey of the sign, posting, or label and survey the area it was found.

2.10.09 Describe the following radiological postings criteria at Hanford and entry/exit requirements:

Criteria for Posting RCA, RBA, or RMA Areas:

AREA	POSTING CRITERIA	SIGN POSTING	ENTRY REQUIREMENTS	EXIT REQUIREMENTS
RCA	Any area which access is managed by or for DOE to protect individuals from exposure to radiation or radioactive materials	Each access point must be posted "Caution, RCA" whenever a radiological area or RMA exists inside the area	GERT required for unescorted access	None
RBA	At minimum, placed next to any entry or exit from a CA (or HCA/ARA not completely within a CA) and should surround RA, HRA, VHRAs so that RBA limits dose to <100mrem/year. Not required around inactive CAs where entry prohibited by posting or barricades.	"Caution, Radiological Buffer Area"	Escorted: GERT with qualified escort Unescorted: RadWorker I If RBA for radiation: Hanford Standard Dosimeter	If RBA for contamination: Hand and Foot Survey
RMA	Any area within an RCA, accessible to individuals, in which items or containers of radioactive material exist. Area may be excepted from posting "RMA" when the area is already posted as a higher radiological area.	"Caution, Radioactive Materials"	Escorted: GERT with qualified escort Unescorted: RadWorker I Personal dosimetry may be required as per RadCon operations	None, unless directed or specified by RadCon operations

Criteria for Posting Radiation Areas (No Exit Requirements for Radiation Areas):

AREA	DOSE RATE CRITERIA	SIGN POSTING	ENTRY REQUIREMENTS
Radiation Area (RA)	>5mrem/hr at 30cm but ≤100mrem/hr at 30cm	"CAUTION, RADIATION AREA" "Personnel Dosimeter Required" "RWP Required"	Escorted: GERT with qualified escort Unescorted: RadWorker I Personal Dosimeter Signature on RWP
High Radiation Area (HRA)	>100mrem/hr at 30cm but ≤500 rad/hr at 1m (100cm)	"DANGER, HIGH RADIATION AREA" "Personal Dosimeter, Supplemental Dosimeter, and RWP required"	RadWorker II -or- RadWorker I with H/VHRA access training Personal Dosimeter Signature on RWP A survey meter or dose rate device should be available
Very High Radiation Area (VHRA)	>500 rad/hr at 1m (100cm)	"GRAVE DANGER, VERY HIGH RADIATION AREA" "SPECIAL CONTROLS REQUIRED FOR ENTRY"	VHRA entry has the same requirements as HRA with additional measures implemented to ensure individuals are not able to gain unauthorized or inadvertent access.
Hot Spot	>5 times general area dose rate and >100mrem/hr on Contact	"CAUTION, HOT SPOT" Labeling of Hot Spots is not required in areas with general area dose rates >1rem/hr.	No entry requirements. Entry req's are established for the RA or HRA where the hot spot is located.

NOTE for HRA: One or more of the following controls must also be used for each entrance or access point to an HRA where radiation levels exist at or above 1rem/hr at 30cm:

- A control device that prevents entry to the area when high radiation levels exist or that, upon entry, causes the radiation level to be reduced below the level that defines HRA.
- A device that functions automatically to prevent use or operation of the radiation source while individuals are in the area.
- A control device with a conspicuous visible or audible alarm so that individuals entering the HRA and their supervisors are made aware of entry.
- ***Continuous direct or electronic surveillance that can prevent unauthorized access.***
- ***Entryways are locked and positive control over each entry is maintained.***

Criteria for Posting Contamination Areas:

AREA	CRITERIA	SIGN POSTING	ENTRY REQUIREMENTS	EXIT REQUIREMENTS
Contamination Area (CA)	Removeable Contamination levels (dpm/100cm^2) exceed the values in Table 2-2 but are ≤100 times those values	"CAUTION, CONTAMINATION AREA"	Escorted: GERT with qualified escort Unescorted: RadWorker II Signature on RWP Protective clothing Personal Dosimetry	Whole Body Survey
High Contamination Area (HCA)	Removeable Contamination levels (dpm/100cm^2) >100 times Table 2-2 values	"DANGER, HIGH CONTAMINATION AREA" "RWP Required"	RadWorker II Signature on RWP Protective clothing Personal Dosimetry	Contamination Survey
Airborne Radioactivity Area (ARA)	Airborne concentration exceeds or likely to exceed the DAC value in 10 CFR 835 (air is >1 DAC). or An individual in the area without respiratory could receive an intake >12 DAC-h in one week.	"CAUTION, AIRBORNE RADIOACTIVITY AREA" "RWP required"	RadWorker II Signature on RWP Protective clothing Respiratory as specified on the RWP Personal Dosimetry	Whole Body Survey
Fixed Contamination Area (FCA)	Areas outside a CA, HCA, or ARA where: Total Contamination exceeds values in Table 2-2 but Removable Contamination levels are less than values specified in Table 2-2 (Contamination is fixed onto a surface and cannot be removed by non-destructive means)	"CAUTION, FIXED CONTAMINATION"	No entry requirements	No exit requirements

Criteria for Posting Soil/Underground Contamination Areas:

Soil Contamination Area (SCA)	Radioactive material exists within top 15cm (6in) of soil such that: Direct contamination reading of soil exceeds Total Contamination values in Table 2-2 but Transferable Contamination levels do not exceed values in Table 2-2	"CAUTION, SOIL CONTAMINATION AREA"	Escorted: GERT with qualified escort Unescorted: RadWorker II Signature on RWP Personal Dosimeter Protective clothing as required by RWP	Exit Survey as required by RWP
Underground Radioactive Material Area (URMA)	Known or presumed radioactive material above naturally occurring background levels exist below the top 15cm (6in) of soil or below any impervious soil cover (e.g., asphalt, concrete)	"CAUTION, UNDERGROUND RADIATIOACTIVE MATERIAL" Should include any applicable warnings or special instructions.	GERT RadWorker II and RWP for intrusive activity Personal Dosimetry and/or Protective clothing as required by RWP	Exit Survey as required by RWP

2.10.10 Describe good practices and support equipment to use in setting up radiological areas:

Good practices to be considered when setting up radiological areas include:

- Establish walkways in low-dose areas.
- Avoid storing radioactive material near walkways or where personnel frequently work.
- Place boundaries close to the hazards without interfering with access to unaffected areas.
- Ensure the area is not so small that contamination is easily spread across the boundaries.
- Use drip trays or containment devices to prevent spreading contamination.
- Establish laydown areas for equipment to limit personnel safety hazards and / or radiation exposure.
- Set up SOP upwind of contamination hazards to support efficient removal of PPE. Post doffing instructions specific to the PPE being used.
- Post all accessible sides and entrance(s) to areas containing radiological hazards.
- Ensure boundaries are verified for contamination, radiation, and airborne radioactivity hazards.
- Ensure posting information and status boards are updated and correct.
- Ensure survey instruments are calibrated and operable and are close to the SOPs. The background radiation in the monitoring area needs to be low enough to allow for efficient detection of low-level contamination.
- Use PCM along with portable contamination survey instruments whenever possible. PCMs are more likely to detect contamination on individuals because personnel tend to survey too quickly and could pass over the contaminated area with the portable survey instrument without detecting contamination.
- Laundry and waste receptables should be placed for efficient use. Label receptacles to identify their contents.
- Avoid creating tripping hazards from wires, hoses, or cables, and anchor them to prevent movement across the CA boundary.
- Provide clothing (gloves and shoe covers) should be readily available in case of personnel contamination.
- Ensure a telephone or other communication device is available near the SOP or portable survey instruments.
- Provide material for changing radiological conditions, e.g., yellow and magenta rope, ribbon, or tape; signs and inserts; survey maps; additional protective clothing; spill kit.

2.10.11 State the requirements for removing or clearing material from RAD areas:

Each company has procedures for the radiological clearance of material and equipment. These are based on HNF-63871 (Statistical Basis for Radiological Contamination Surveys at the Hanford Site).

Material and equipment in CAs, HCAs, and ARAs cannot be **cleared** to RCA if:

- Removeable contamination levels on accessible surfaces exceed the values specified in Table 2-2, **or**
- Prior use suggests that the removable contamination levels on inaccessible surfaces are likely to exceed the values specified in Table 2-2

2.11.01 State the reasons to stop radiological work activities in accordance with Radiological control manual:

DOE-0343 gives all personnel the responsibility and authority to Stop Work.

Reasons that could result in Stop Work include:

- Any situation that could:
  - Place personnel or the public at risk or in danger
  - Adversely affect safe operation or cause damage to the facility
  - Result in release of rad or chemical affluents to the environment
- Rad specific:
  - Inadequate radiological controls.
  - Radiological controls not being implemented.
  - Radiological hold points not being satisfied.
  - Alarming dosimetry or unexpected dosimetry readings.
  - Dropped containers / spills.
  - Unexpected dose rates or surface contamination readings.

2.11.02 State the policy for procedure compliance and actions to be taken if a procedure or other technical work document cannot be performed as written:

Procedures form the basis for correct and safe operation of processes and facilities. Workers and their supervisors are held accountable for working in compliance with their procedures. If they cannot comply, Stop Work, notify supervision, and resolve the problem so work can proceed safely and in compliance with all requirements.

The word *shall* identifies elements that are required and must be fulfilled by the contractor.

The word *should* means the contractor has the responsibility to either directly follow the provision or demonstrate compliance through a technical equivalency.

No one can be directed to violate a procedural requirement outside of an emergency.

2.11.03 Describe the purpose, requirements, and general criteria of a hold point:

A RadCon hold point is a cautionary step in a work package that requires RadCon to perform some action or verification. The RadCon hold point requirements must be completed before work can continue.

Requirements:

- The RCT or applicable RadCon personnel performing the verification must sign and date the step in the work package before work peroceeds beyond that step.

General criteria for a hold point:

- Initial certification of containment tents or glove bags
- Verification of contamination levels prior to removing a certified glove bag from a component
- Actions taken to ensure airborne radioactivity levels do not exceed 100 times the applicable DAC values
- Action to asses changing radiological conditions and ensure controls are adequate for potential changes.

2.11.04 List four purposes of job coverage:

1. Keep radiation exposures ALARA and within limits and guidelines.
2. Reduce the creation and prevent the spread of contamination.
3. Reduce the creation of and prevent the spread of airborne radioactivity.
4. Minimize generation of radioactive waste.

2.11.05 State the difference between continuous and intermittent job coverage:

Continuous Coverage: Coverage sufficient to immediately influence or stop the work based on observed radiological conditions and/or work practices.

Intermittent Coverage: Coverage at frequent enough intervals to exclude any reasonable potential for unmonitored change.

2.11.06 Given example conditions, identify those that should require job coverage:

Some radiological conditions or types of jobs that could require RadCon job coverage are:

- Worker doses near administrative control levels (ACL).
- Radiation dose rates are high enough to potentially cause worker does to exceed ACLs in a short time.
- Radiation levels expected to increase significantly during the job.
- Entry to High Radiation Area (HRA).
- Potential for spreading contamination or causing airborne radioactivity.
- Potential for significant increase in contamination or airborne levels during job.
- First-time performance of a job.

2.11.07 Describe items that should be considered in planning job coverage:

- Determine the job details of what, when, and where; review the sequence of tasks to be performed and any special limitations / precautions.
- Obtain pre-job survey information; talk with RCTs who have previously covered the same or similar jobs to anticipate any problem areas.
- Review the area and system where the work will be performed or talk with workers to determine the potential radiological consequences of the tasks associated with the job.
- Ensure that instruments are operating properly and are protected from contamination.
- Collect supplies and equipment.
- Ensure workers have prescribed dosimetry, protective clothing, and respiratory protection equipment.
- If sufficient survey documentation is not available, surveys of areas may be necessary prior to planning work.
- Review all applicable work documents, RWP, ALARA documents, and procedures.
- Establish a method of communicating with workers during the job.
- Establish a method of communicating with and transferring samples to the RadCon count lab.
- Have the appropriate equipment available at the job site (air samplers, survey instruments, extra gloves, etc.).

2.11.08 Identify examples of information that should be discussed with workers during pre-job briefings:

- Scope of work to be performed
- Radiological conditions of the workplace
- Locations of low dose waiting areas and hot spots
- Procedural and RWP requirements
- Special RadCon requirements and hold points
- Radiological conditions such as contamination or radiation levels that may void the RWP
- Communications and coordination with other groups
- Provisions for housekeeping and final cleanup
- Emergency response provisions

2.11.09 Describe exposure control techniques that can be used to control worker and RCT radiation exposure:

- Wait in low dose areas while not performing the job.
- Periodically read or have workers read their dosimeters.
- Use workers allowable dose and area dose rate to calculate the time a worker can spend in the area.
- Use time tracking to ensure the worker leaves the area before their allowable time elapses.
- Use additional dosimetry as necessary when dose may differ for specific body parts (head, back, hands, etc.).
- Workers must leave temporary shielding in place unless they authorized by RadCon to remove it.
- When removing items from a cask, pit, riser, shield box, etc., survey the items while they are being raised rather than waiting until they are completely withdrawn to perform a survey.
- Keep workers from leaning across or over sources of radiation if work can be performed from another angle.
- Prevent workers from picking up sources with their hands (use tongs, tongs, buckets, bags, etc.).

2.11.10 Describe the in progress radiological surveys that should be performed under various rad conditions:

In-progress radiological surveys are performed as specified by the controlling technical work document/RWP or as deemed necessary by the lead RCT to ensure that conditions are satisfactory to protect the worker, public, and environment.

2.11.11 Describe site requirements for documenting in-progress radiological surveys:

All surveys are documented, whether results are positive or negative, *including surveys performed on protective clothing.* Failure to document a measurement is the same as saying you did not take the measurement. RadCon personnel maintain logs to document radiological occurrence, status of work activities, and information that should be communicated to all personnel. Make field notes of survey results for later documentation. All surveys should be documented with an appropriate map.

Specifically, attach a map (or photograph) to the RSR for all:

- Medium and high hazard radiological work activities.
- Low hazard radiological work activities where the survey resulted in radiological conditions that were not expected (exceeded RWP action or void level, contamination spread, unexpected dose or contamination encountered, etc.)
- Radiological work activities that required an air sample.

The map and photograph will contain the following information:

- Specific locations where dose rate measurements were made and the corrected dose rate(s), grab air sample location(s), and smear locations, as applicable.
- If the dose rate listed on the map applies to a line item, the map should list the dose rate according to the legend.

2.11.12 Explain actions that should be taken if surveys show that radiological conditions significantly different from that expected:

Should conditions develop that are a threat to the safety of the crew or to radiological control, the RCT should stop the progress of the job and have workers leave the area.

2.11.13 Describe contamination control techniques that can be used to limit or prevent personnel and area

contamination and/or reduce radioactive waste generation:

- Watch the workers. Point out and correct any work habits that could spread contamination.
- System components that are being repaired should be wiped down and drained before the system is opened.
- Ensure workers follow appropriate procedures for bagging, handling, and transferring material from a CA.
- Large items can be covered with plastic sheeting that is taped in place after all surfaces have been wiped.
- Do not allow bags of trash, tools, or used parts to accumulate in the work area or at the SOP or control point.
- Ensure that workers follow proper procedures to minimize contaminating tools and equipment.
- Watch for the movement of crane rigging, air or water hoses, electric leads, and extension cords into and out of contaminated area.
- Reduce the creation of radioactive waste (only take in and use what is necessary for the job).
- Control the removal of equipment from work areas.

2.11.14 Describe job coverage techniques that can be used to prevent or limit the spread of airborne radioactive material:

- Look for any actions that could create airborne radioactivity (opening systems, cutting, grinding, etc.).
- Take air samples during jobs in highly contaminated areas or with steps that could create airborne rad.
- Consider using a CAM during performance of a job that is likely to create airborne radioactivity.
- Real time air monitoring should be installed where unexpected increases in airborne could result in exposure >40 DAC-h in one week (example: failure of engineering controls).
- By ventilating enclosed areas with HEPA filters, airborne radioactivity generated in the area can be removed.
- In certain job situations, the use of respiratory protection can increase the total equivalent dose for the job. This is referred to as *dose expansion* and can occur when respirator usage causes increased time spent in rad areas.

2.11.15 Describe overall job control techniques in maintaining control of radiological work:

- Establish worker trust and confidence.
- During continuous job coverage, RCTs must keep workers within their line of sight either directly or via video.
- Keep in contact with workers. Remind workers RCTs are available to answer questions or make suggestions.
- Remind workers that casks, containers, or systems are not to be opened or work techniques changed without notifying RadCon.
- When an individual's work habit must be corrected, offer the correct method as advice, or help.
- Ask questions – different worker's experiences with the job may provide better approaches.
- Show a positive, helpful attitude towards workers.
- Do not overreact to situations when there is time for more moderate solutions.

2.11.16 State the responsibilities an RCT will and will not perform while supporting radiography operations:

The RCT will NOT:

- Use radiography devices.
- Use related handling tools.
- Act as a radiographer or radiographer's assistant.
- Conduct surveys of the source for the radiographer.
- Post areas for the radiographer.

The RCT will:

- Verify postings.
- Ensure safety of personnel not associated with the operation of the radiographic equipment.

2.12.01 Describe the general principles that apply to shipping radioactive material:

The bases behind the regulations governing packaging and shipping radioactive material are to keep radiation and radioactive material from affecting the environment during transportation and to keep the environment from affecting the integrity of the radioactive material.

2.12.02 List the primary agencies that govern the transport of rad material & where their regulatory requirements are applied:

1. *U.S. Department of Energy (DOE):* DOE O 460.2A, DOE/RL-2001-36, RadCon Manuals

2. *U.S. Department of Transportation (DOT):* 49 CFR (most rad requirements are in Part 173)

3. *U.S. Nuclear Regulatory Commission (NRC):* 49 CFR 172 Subpart 1

4. *U.S. Postal Service*

- Additional regulations come from U.S. Coast Guard, Federal Aviation Administration, State transportation agencies, and radiation health departments.

2.12.03 Define terms used in Department of Transportation (DOT) regulations:

- *A_1:*
  The A_1 value for a radionuclide is the quantity of that radionuclide that will result in a dose rate of 10 rem/h at 1m. It is the maximum activity of *special form* radioactive material permitted in a Type A package.

- *A_2:*
  The maximum activity of radioactive material, other than special form, low specific activity, or surface contaminated object, allowed in a Type A Package.

- *Normal Form:*
  Radioactive material that does not qualify as special form material.

- *Special Form:*
  Radioactive material that satisfies certain conditions such as being a solid, non-dispersible piece, or a sealed capsule containing radioactive material that can be opened only by destroying the capsule.

- *Type A Quantity:*
  A quantity of radioactive material, the aggregate radioactivity of which does not exceed A_1 for special form radioactive material, or A_2 for normal form radioactive material.

- *Type B Quantity:*
  A quantity of material greater than Type A quantity.

- *Exclusive use:*
  The sole use of a conveyance for which all initial, intermediate, and final loading and unloading is carried out according to the direction of the shipper or receiver.

- *Closed Transport Vehicle:*
  A vehicle equipped with a securely attached exterior enclosure that restricts access of unauthorized personnel to the radioactive material.

- *Highway Route Controlled Quantities.*
  A quantity within a single package that exceeds:
  - 3,000 times the A_1 value of the radionuclides for special form radioactive material, or
  - 3,000 times the A_2 value of the radionuclides for special form radioactive material, or
  - 27,000 Ci, whichever is least.

- *Limited Quantity:*

Quantity of class 7 (radioactive) material that does not exceed specified package limits.

- *Low Specific Activity (LSA):*
  Radioactive material with limited specific activity that is not fissile material or otherwise excepted.

- *Low-Toxicity Alpha Emitters:*
  Natural uranium, depleted uranium, natural thorium, ^{235}U or ^{238}U, ^{232}Th, ^{228Th}, and ^{230}Th when contained in ores or physical and chemical concentrations, and alpha emitters with half-lives of less than 10 days.

- *Mixed Lading (Non-Exclusive Use):*
  Shipments of radioactive material that are, can, or may be shipped on the same vehicle or temporarily stored at a terminal with other materials.

- *Offsite:*
  Any area outside the boundaries of a DOE site or any area within DOE site boundaries that the public has uncontrolled access to.

- *Onsite*:
  Any area within DOE site boundary where public access is controlled.

- *Radioactive material*:
  Any material containing radionuclides where both the activity concentration and the total activity in the consignment exceed specified values.

- *Radioactive material transportation*:
  The movement of RAD material by aircraft, rail, vessel, or highway vehicle.

- *Surface contaminated object*:
  A solid object that is not itself radioactive, but which has radioactive material distributed on any of its surfaces.

- *Transport Index (TI)*:
  Unit-less number placed on the label of a package to designate the degree of control exercised by the carrier. It is the maximum radiation level at 1 meter from the external surface of the Package. The TI value shall be rounded up to the next tenth.

2.12.04 Describe methods that may be used to determine the radionuclide contents of a package.

- *Facility characterization*:
  Primarily applicable to wastes and involves determining the quantity of radioactive material contained within a volume by comparing the amount of radioactive material entering and exiting a process. Example: A known volume of air passed through a HEPA filter, the difference in airborne concentration before and after the filter can be assumed to be retained within the filter.

- *Manufacturer's Information*:
  Determining the radionuclide content through the knowledge and control of the source of the material.

- *Measurement of gross radioactivity*:
  Based on dose rates from a container, correlated to the actual radionuclide in the material and actual radionuclide content and periodically verified.

- *Isotopic analysis*:
  Individual gamma-emitting radionuclides are directly measured using gamma spectroscopy.

2.12.05 Describe the required Radiation & Contamination surveys for packages & vehicles and state the applicable limits:

Each package must be designed and prepared so that under normal conditions the radiation level does not exceed 200 mrem/h on external surface of package and transport Index (TI) does not exceed 10 (10mr/h at 1m).

A package that exceeds 200mrem/h or a TI of 10 must be transported as an exclusive use shipment and the radiation levels may not equal or exceed the following during transportation:

- 200mrem/h on the external surface **of the package** unless the following conditions are met, in which case the limit is 1,000mrem/h:
  - The shipment is made in a closed transport vehicle.
  - The package is secured in vehicle so that its position stays fixed during transportation, and
  - There is no loading or unloading operations between beginning and end of transportation.

- 200 mrem/h at any point on the outer surface **of the vehicle**, including the top and underside of the vehicle.

- 10 mrem/h at any point 2 meters (6.6ft) from the outer lateral surfaces of the vehicle (excluding top or underside).

- 2 mrem/h in any normally occupied spot (e.g., the cab/driver's seat).

Package Survey Requirements:

The monitoring required by article 423 of the RadCon manual includes:

- Measurements of removable contamination levels
- Measurements of radiation levels

Package Contamination Surveys

- Table 2-2 removable contamination values are more limiting than 49 CFR requirements

Package Radiation Surveys

- Measure the contact radiation level at all points on the external surface of the package and record the highest level
- Measure the radiation levels at 1 meter from the surface of the package to establish the TI.

49 CFR 173 Contamination Control

- The level of removable contamination must be kept ALARA on the external surfaces of each package, conveyance, freight container, and overpack offered for transport, and the internal surfaces of each conveyance, freight container, and overpack in which inner packages or receptacles of radioactive materials are offered for transport.

Transport Vehicle Surveys

- Radiation and contamination surveys are performed when an exclusive use transport vehicle arrives at the site to ensure that the vehicle is not exceeding the DOT limits above.

Exclusive Use Closed Transport (49 CFR 173.441)

- ≤ 1,000 mrem/h on external surface of package
- ≤ 200 mrem/h on external surface of vehicle (including top and bottom)
- ≤ 10 mrem/h at 2 m from any points on the sides, front, and rear
- ≤2 mrem/h in normally occupied areas (diver and sleeper)

Exclusive Use Open Transport (49 CFR 173.441)

- ≤ 200 mrem/h contact with outer edges, upper surface of load, and bottom of vehicle
- ≤ 10 mrem/h at 2 m from the vertical plane projected from the outer edges
- ≤ 2 mrem/h in normally occupied areas (driver and sleeper)

Normal Exclusive Use Vehicle Being Returned to Service (49 CFR 173.443(c))

- ≤ 0.5 mrem/h on each accessible surface
- ≤ 240 dpm/cm^2 for beta-gamma emitters and low-toxicity alpha emitters; ≤ 24 dpm/cm^2 for all other alpha emitters

Empty "Radioactive Materials Use Only" Vehicles (49 CFR 173.443(d)(1))

- ≤ 10 mrem/h contact with interior surface
- ≤ 2 mrem/h at 1 m from interior surfaces

Non-Exclusive Use (49 CFR 177.842)

- ≤ 10 TI per package
- ≤ 50 sum of all package Tis
- Minimum distance required for TI = 50
  - 7 ft to occupied zones
  - 36 ft to undeveloped film (over 12 h in transit)

2.12.06 State the proper placement of placards on a transport vehicle:

- Do not over-label or placard a vehicle unnecessarily.
- Application of a placard when the hazard does not exist is a violation of regulations.
- Placards are required for any quantity of packages bearing Radioactive Yellow-III labels.
- Placards must be on all four sides of the vehicle.
- If a tractor is disconnected from the trailer, placards must be on all four sides of the trailer; otherwise, the front placard can be on the tractor.

2.12.07 State the inspection criteria that apply before a shipment and upon receipt:

Before shipment and upon receipt of radioactive materials, perform a visual inspection to identify discrepancies such as:

- Dents
- Package degradation
- Indication of leakage
- Flaking paint
- Crushed
- Wet or damaged
- Improper package closure
- Radiation levels exceeding limits on package, in vehicle cab, underneath vehicle, or other areas

Inspect documentation and labeling for:

- Improper packaging for the type or quantity of rad material
- Improper or missing markings, labels, or placards
- Incomplete or incorrect information on shipping papers
- Instructions were not provided to carrier for maintaining exclusive use of a vehicle

For all shipments other than by mail, shipping papers must adhere to the requirements of 49 CFR 172.200 through 172.204.

2.12.08 Describe the required procedures for shipment and receipt of radioactive material:

Performance and documentation of surveys follow normal RCT operating procedures (e.g., instrument operating procedures, radiological survey procedures, etc.).

Radiological material receipt surveys must be performed as soon as practical, but no more than 8 hours after the beginning of the working day following receipt of the package.

2.12.09 Explain how radioactive material categories are distinguished by their labels

Table 3. Category of Label to Be Applied to Radioactive Material Packages

Transport Index	Maximum Radiation Level at Any Point on the External Surface (mrem/h)	Label Category[2]
0[1]	≤ 0.5	White-I
> 0 but ≤ 1	> 0.5 but ≤ 50	Yellow-II
> 1 but ≤ 10	> 50 but ≤ 200	Yellow-III
> 10	> 200 but ≤ 1,000	Yellow-III[3]

[1] If the measured TI is ≤ 0.05, the value may be considered to be zero.

[2] Any package containing a highway route controlled quantity must be labeled Radioactive Yellow-III.

[3] Must be shipped under exclusive use provisions.

2.12.10 Describe the RCT's actions if a shipment is received that exceeds radiation or contamination limits:

If the delivering vehicle or package is above the applicable contamination limits, notify RadCon and line management. The delivering carrier, all intermediate carriers, and the shipper must be notified promptly so that potentially contaminated vehicles can be withdraw from service and surveyed. Loading docks and terminals through which the package passed must also be surveyed.

Note that Table 2-2 contamination limits are lower than DOT limits; if contamination levels are above Table 2-2 but below DOT limits, notification of the shipper and carrier is not required.

2.12.11 Describe the steps to be taken when opening a package containing radioactive material:

- The package should be moved to an appropriately posted radioactive material area or radiological area based on survey data obtained from the receipt survey.

- Responsibility for opening the package is with the line organization, but an RCT should be present to survey the contents and check for internal contamination as needed.

2.13.01 Describe the radiological control considerations for emergency response:

All normal procedures should be followed during an emergency. Activities such as instrument use, personnel surveys, air sampling, and contamination surveys will not change.

In all emergencies, there are actions for which all RCT's are responsible. These include:

- *Recognition:* Recognize that all emergencies have the same basic response philosophy.
- *Response:* Respond to all emergencies; Even in non-rad events, there is a need to verify assumptions.
- *Communication:* Ensure communications are clear.
- *Documentation*: Document time, actions taken, and decisions made.

2.13.02 List the initial and describe follow-up actions an RCT performs when responding to a CAM alarm:

Initial Response: (SWIMS)

- Stop operations that may be causing airborne radioactivity.
- Warn others to evacuate.
- Secure unfiltered ventilation.
- Contact RadCon and line management.

Supplementary Actions: (follow-up)

- Upon re-entry, don respiratory equipment and protective clothing based on conditions of the event.
- Evaluate the affected area by taking an air sample, measuring radiation levels, and checking for CAM malfunctions.
- Obtain additional air sample as necessary to determine boundaries and maintain access control.
- Identify isotopes to help identify the source and protective measures.
- Consider additional filtered ventilation to minimize personnel exposure and reduce the need for respiratory equipment.
- Measure and control surface contamination to minimize spread.
- Survey exhaust systems, ventilation filters, and ducts – have decontamination performed as necessary to minimize contamination spread.
- Evaluate the potential for internal exposure and contact the facility dosimetrist for proper internal dosimetry protocol.
- Interview personnel for information on any off-normal event that could have caused the alarm.
- Take air samples when operations resume to verify that the cause of the airborne activity has been corrected.

2.13.03 Describe the RCT response to a PCM alarm:

When a PCM alarms, workers should perform a second survey using the same PCM. If a second alarm occurs, the potentially contaminated individual should notify RadCon who should then:

- Obtain any necessary equipment (e.g., instruments, plastic bags, PPE, tags, etc.).
- Don PPE as needed before entering the PCM area.
- Set up boundaries to control area access.
- Question the individual to help determine the source and location of contamination.
- With portable instruments, perform a whole-body survey with emphasis on area that initiated the PCM alarm.
- If no contamination is detected, allow the individual to exit the area.
- If contamination is detected, follow the applicable decon procedure.
- Perform a radiological survey of the area around the PCM, the path traveled by individual, and the PCM, as needed.
- Initiate an investigation to determine the cause of contamination and any required follow-up with the employee.
- If no contamination is detected with portable instruments, perform a self-survey using a PCM and:
    - If the PCM alarms, contact your supervisor for guidance **–** the PCM may need to be taken out of service.
    - If the PCM does not alarm, the PCM is still operational.

2.13.04 Describe the RCT response to an abnormal dosimetry reading:

Personnel in a radiological area who discover that their dosimetry has an abnormal reading should:

- Inform co-workers in the area that their dosimetry has an abnormal reading.
- Place their work in a safe condition.
- Immediately leave the area, following all requirements for exit.
- Contact RadCon personnel.

RCT's will investigate and document the cause of the anomaly. The investigation and documentation should include:

- Source and type of radiation
- Other personnel dosimeter readings
- Dose rates and length of exposure
- Location relative to the source of exposure
- Shielding potentially involved

2.13.05 Describe the RCT response to ARM alarm:

When an ARM alarms:

- Workers should Stop Work activities, immediately exit the area, and notify RadCon personnel.
- RadCon personnel will evaluate the cause of the alarm and, for true alarms, estimate the dose.
    - This dose estimate, along with relevant workplace monitoring data, will be used to determine the need for additional actions.
- If personnel evacuated a contamination area, perform personnel contamination surveys, and obtain answers to the following question:
    - Did all personnel evacuate the alarm area?
    - What are the possible causes for the alarm?
    - What activities were taking place at the time of the event?

- Read supplemental dosimeter of evacuated personnel, if any, and note any high readings.
- A recovery plan may be needed that could require a new RWP. If dose rates exceed safe conditions or void limits, leave the area and notify management.
- If the ARM appears to be operating properly and there is no apparent cause for the alarm, source check the ARM.

2.13.06 Describe the RCT response to a dry or liquid radioactive material spill:

SWIMS:

- Stop the spill or stop working.
- Warn others.
- Isolate the area.
- Minimize exposure.
- Secure unfiltered ventilation.

As with all incidents, gather all available/relevant information and document accordingly. Be sure to include:

- Location, time, and date
- Sequence of events
- Actions taken
- Radiological conditions
- Status of the event

2.13.07 Describe the Incident Command System:

The *ICS (Incident Command System)* is a nationwide emergency management system with defined responsibilities and communication pathways that allow pre-designated and trained individuals to respond to events.

- The ICS consists of five components:
    1. Command
    2. Building Emergency Director (BED)
    3. Operations
    4. Logistics
    5. Planning

- The *Incident Commander (IC*) is responsible for the overall management strategy associated with the incident.

- The *BED (Building Emergency Director)* manages facility operations and personnel and ensures that emergency procedures are implemented.

- *RCT's* establish radiological conditions at the *ICP (Incident Command Post)*, including:
    - Contamination monitoring
    - Air monitoring
    - Dose rate monitoring
    - Helping with access control between the warm zone and cold zone

At the Incident Scene:

- The OSC coordinates all activities at the incident scene.
- The FOS reports to the OSC and is responsible for assisting in coordinating all facility operations directly applicable to mitigation of the incident.
- The Hazards Communicator is responsible for communicating data to the DOE/RL, EOC, and UDAC for further consequence assessment.
- The RHA coordinates and ensures accomplishment of RadCon functions throughout the incident scene.
- The Chemical Hazards Assessor is responsible for evaluating chemical hazards throughout the incident scene.
- The Staging Area Manager (SAM) manages the resource staging area.

2.13.08 Define the response levels associated with radiological emergencies.

Hazardous Material Operational Emergencies: The following events meet the emergency action level criteria and activate additional emergency plan implementing procedures:

- *Alert:* This emergency level is an event in progress or having occurred that involves an actual or potential substantial degradation of the level of safety at a facility. It does not extend beyond the facility boundaries.

- *Site Emergency:* An event that involves an actual or potential major failure of functions necessary for the protection of workers or the public. Any hazardous material is released beyond facility boundaries but not beyond Hanford Site boundaries. The TED at the facility boundary must be ≥1rem to declare a Site Emergency.

- *General Emergency:* An event that involves actual or imminent catastrophic reduction of the facility safety or security systems with potential for release of large quantities of hazardous materials to the environment (beyond Hanford Site boundaries). The TED at the Hanford Site boundary must be ≥1rem at the Hanford Site boundary to declare a General Emergency.

2.13.09 State the responsibilities of a facility RCT reporting to the staging area..

RCTs assigned to the Facility Emergency Response Organization (FERO) will respond to the designated area at their facility to wait for assignments.

Duties of the RCTs at the staging area include:

- If not already staged, bring emergency kits and instruments.
- Survey those who are in the segregation area or others who request a survey.
- Perform quick sort surveys.
- Perform habitability surveys including direct contamination, dose rates, and air samples, if applicable.
- Stay and assist at the staging area until their supervisor, lead, RHA, or SAM gives them direction.
- If you evacuate to a staging area and you are the affected facility, there should be no eating, drinking, or tobacco use (except for medically related needs).

2.13.10 State to whom at the Incident Command Post an RCT is to report.

RCTs at the ICP will report to a lead, supervisor, or RHA; when the RHA is not available, report to the OSC.

2.13.11 List the responsibilities of a facility RCT during a radiological event:

In all emergencies there are similar actions for which all RCTs are responsible. These include:

- Respond to all emergencies as assigned by your FERO.
- Ensure that communications are clear.
- Document actions taken and decisions made.

All emergencies have a basic response philosophy:

- Assess the hazards.
- Protect yourself and avoid becoming a complicating factor in the event.
- Look for ways to help the individuals involved.
- Look for ways to stop or mitigate the problem.
- Look for ways to determine the extent of the problem, setting up boundaries and control points as needed.
- Seek assistance as needed.
- Follow standard procedures.

2.14.01 Describe the preliminary actions and notifications required of the RCT for an individual suspected to be contaminated.

When an individual is suspected to be contaminated, RCT should:

- Obtain necessary instruments/PPE.
- Make a quick assessment of the condition of the individual and the possibility of spreading contamination.
- If injury is evident, immediately notify 911 or the shift office, depending on severity. Notify RadCon management.
- If there's no injury, perform a preliminary survey to assess the extent and location of any contamination.
- While conducting the survey, question the individual to understand the potential causes of contamination.
- If internal deposition is suspected or contamination is extensive, a second RCT may be required to record readings and collect smears (including nasal smears).
- Additional support may be necessary to control potentially compromised (contaminated) areas and/or assist with postings, traffic control, and further investigation.
- High levels of contamination on skin or clothes should be removed immediately to reduce dose.
- If tank waste is involved, immediate flushing is required due to chemical hazards.
- Retain removed contamination or contaminated articles for analysis by Dosimetry Operations.

2.14.02 Describe three factors that determine the actions taken in personnel decontamination.

When responding to a personnel contamination event, RCT's actions will be guided by three basic factors:

1. *Condition of the worker*
   - Give clear instructions to the worker ("do not touch yourself", etc.) and be mindful of the worker's concerns and psychological condition, which could deteriorate over time due to nervousness/anxiety.

2. *Location of the contamination*
   - Is contamination on the general skin surface?
   - Is the contamination located on or near a body orifice?
   - Is contamination located near a break in the skin?
   - Is there a skin condition present near the contamination?
   - Is contamination on the clothing?

3. *Activity of the nuclide(s) present*
   - Determination of the type of activity (alpha or beta-gamma) and save samples for laboratory analysis.

2.14.03 State good practices for storing samples waiting for analysis:

Good practices for storing samples for analysis:

- Retain until disposal is authorized by Dosimetry Operations.
- Package samples securely to prevent cross-contamination or loss of contamination.
- Label samples with identifying information: time, date, individual's name and Hanford identification number, survey number, and a brief event description.

2.14.04 Describe personnel survey, notifications, and contamination control practices:

Personnel survey: A detailed whole-body survey should be performed on all exposed surface areas of the individual including protective clothing if worn, personal/modesty clothing, and/or skin for both beta/gamma and alpha contamination. Starting at the head and proceeding to the feet, pay particular attention to the following areas:

- Nose and mouth (including nasal smears if necessary)
- Any known or potentially contaminated areas
- Skin folds
- Buttocks
- Hands
- Knees
- Feet

The whole-body survey should take several minutes.

The RCT should:

1. Verify that the instrument is properly in service (Cal date, battery, source check, etc.).
2. Adjust the audio as appropriate (consider turning down if the individual is in an emotional state)
3. Hold the probe about ¼ in from the surface being surveyed.
4. Move the probe slowly, about 2 in/s, over the surface.
5. If the count rate increases during the survey, pause over the area (based on instrument response time).
6. If the count rate increases to a valve greater than background, or alarms if using a PCM, contamination is confirmed.

Notifications: If contamination is found, notify RadCon supervision and line management.

Contamination Control: Contain the contamination as much as possible before allowing the person to move by:

- Removing and bagging shoes and/or covering feet with plastic shoe covers / booties.
- Covering their hands with gloves, preferably surgical gloves.
- Donning a clean set of PPE over contaminated clothes or wrapping the individual with another covering.

2.14.05 State factors to consider when deciding between decontamination at existing facilities vs. transport to more elaborate decontamination facilities or a hospital:

Consider the following before moving a contaminated individual from existing facilities to a more extensive decon facility:

- The ease with which the decontamination can be performed.
- The potential for spreading contamination or contaminating a wound.
- The additional dose received by the individual due to postponing the decon.

2.14.06 Describe the RCT's actions when clothing contamination is detected:

General RCT actions when clothing contamination is detected:

- Advise the worker to refrain from moving around or touching the contaminated area.
- Follow all procedures for decontamination.
- Contain or remove areas of gross contamination on clothing by tape pressing and bagging. Resurvey.
- Carefully remove and contain all contaminated clothing that is not decontaminated by normal decontamination procedures. Save contaminated clothing and all decontamination materials for assessment by Dosimetry.
- Perform a whole-body survey after removal of contaminated clothing to determine that there is no skin contamination.
- If contamination persists, consider moving to a decontamination facility.
- Assess the potential for internal deposition (airborne, puncture, etc.) by surveying the outside and inside of masks, surveying the facial area, and taking mouth or nasal smears.
- Decontamination of items such as jewelry, watches, or prosthetic devices should only be done with the owner's consent of each step.
- Be aware of liquids on permeable items that could cause contamination to soak deeper into the material or Dry Clean Only items that should not use soap and water.

2.14.07 State the precautions to observe when conducting decontamination:

Skin decontamination precautions:

- Prevent contaminated water from entering body openings (e.g., eyes, ears, nose, mouth, wounds).
- Use soap and tepid (lukewarm) water.
- Major medical care takes precedence over decontamination.
- Address modesty concerns ***before*** they become an issue. If there is a modesty issue, have skin decontamination performed by an RCT of the same gender.
- Use biological hazard bags and gloves when handling human biological samples.

2.14.08 Describe the RCT's actions when skin contamination is detected:

RCT actions when skin contamination is detected:

- Notify the RadCon supervisor, line manager, and Dosimetry Operations.
- Determine the condition of the skin (cuts, sores, abrasians, irritations, etc.).
- **Medical treatment of contaminated skin is reserved for medical personnel.**

The decontamination method for intact skin is determined by:

- Thickness of skin (palm vs. scalp)
- Location on the body (near eyes, etc.)
- Hair coverage (hair cutting/removal requires permission from the individual)
- Physical and chemical state of contaminants (liquid vs. solid, acid vs. base)
- If contamination is detected on or near a worker's face, obtain nasal smears to evaluate potential intake.

Stop decontamination efforts if the skin becomes irritated or the individual complains of discomfort.

2.14.09 Describe the general methods used for skin decontamination

- *Tape Press* (may be used but has potential to irritate skin and hair; if hair coverage is thick, use soap and water).
- *Tepid Water and Mild Soap* ("Lukewarm" water— cold water closes pores, but hot water can cause burns or discomfort. Stronger or more abrasive techniques/chemicals are only used by medical personnel).

Decon efforts may be continued if each application results in reduction in contamination levels and no skin irritation is seen. Discontinue decontamination any time the worker does not desire to continue.

2.14.10 List the information needed for dose estimates with skin contamination:

The RCT's primary responsibilities for dose assessments are to survey and notify.

The RCT's role includes measuring skin contamination and being aware of the following:

- Level of contamination
- Approximate size of the contaminated area
- Type of contamination involved
- Length of time the skin was contaminated

Any of the following conditions will require notification of Dosimetry Operations and potential bioassay follow-up of the individual:

- Facial, nasal, or other contamination is detected that shows potential for internal contamination.
- Airborne monitoring indicates the potential for intake >10 DAC-h (after respiratory protection factor is included).
- Upon direction of the radiological control organization when an intake is suspected.
- When a worker indicates a concern over the potential of an intake.

2.15.01 Describe the steps in the response process of an RCT to minor injuries occurring in radiological areas:

Minor injuries are injuries that do not threaten life, mobility, or long-term survival (shallow cuts, sprains, bruises, etc)

RCT response actions to any injury:

- Wear nonpermeable gloves (such as surgeon's gloves)
- Wear PPE to protect against biohazards and/or bloodborne pathogens
- Place all bio-hazard items including contaminated items in red "Bio-Hazard" marked bags.

RCT response process to minor injuries in a radiological area:

- Notify RadCon and line management, providing at minimum: worker's name/HID, condition, location.
- Request additional RCTs and/or medical assistance as necessary.
- Render first aid if needed as qualified.
- Exit any radiological areas using normal step-off procedures.
- Survey the wound as appropriate to determine if it is contaminated.
- Perform whole body survey.
- Escort to a decon station if contaminated or first aid station if not contaminated.
- Decontaminate if necessary/possible, but do not decontaminate the wound or broken skin. Decon should be limited to the areas around the wound and avoid direct contact with the injury.
- Seek medical assistance as needed. Notify medical and dosimetry as soon as possible.

2.15.02 List RCT responsibilities at the scene of a major injury in a radiological:

A major injury is any injury that could potentially lead to death, prolonged disability, or permanently diminished quality of life (fractured bones, severe burns, organ damage, chest pains, paralysis, unconsciousness, etc.)

RCT responsibilities at the scene of a major injury:

- **Lifesaving efforts take precedence over contamination control. Medical personnel will make the decision on what exist surveys will be required depending on the condition of the person.**
- Get help to the scene immediately (Call 911 and stay on the line until help arrives or told to hang up).
- Administer first aid as qualified.
- Survey the injured person.
- Interface with medical personnel.
  - Provide known information about the injury and any radiological conditions.
  - Understand that medical responders might not have adequate radiological training to recognize or understand radiological hazards.
  - Ensure awareness of radiological conditions including known contamination levels and/or dose rates within the area which could dictate how and when life saving actions occur (i.e., a dangerously high dose rate in the area may require the injured person to be moved before treatment or contamination levels may alter the method/type of medical treatment.

If an injured person is transported to a hospital via ambulance an RCT will:

- Accompany them in the ambulance if there is room and the paramedics do not object. Otherwise, an RCT will follow the ambulance in another vehicle or arrange to meet them at the hospital.
- Perform surveys of the ambulance and equipment in the ambulance
- Perform surveys of attending personnel, as well as self-survey
- Contain contaminated items by bagging or wrapping
- Wipe/remove loose contamination
- Control the spread of contamination by medical personnel (advising techniques but not interrupting)

2.15.03 Describe the required documentation after a major or minor injury in a rad area:

Documentation after injury in a radiological area:

- Complete a RSR to document your survey results.
- Complete a Personnel Contamination Data – Skin Form if they were contaminated.
- Complete a Personnel Contamination Data – Clothing Form if any personal effects were contaminated.
- Record event in the RadCon Logbook (if used) without putting any personally identifying information.

2.15.04 State the purpose of DTPA, the two prerequisites before DTPA can be administered, and how DTPA is eliminated from the body:

The purpose of DTPA is to remove alpha emitters from the body before they can deposit on bones or in the liver.

DTPA is a *chelating* agent. Chelation is a process in which a chemical (the DTPA) and metal ions (the radioactive ions in the body) chemically combine to form a soluble compound (together they become soluble – able to be dissolved, allowing removal by the urinary system). DTPA is eliminated from the body through the urine.

DTPA is classified as a drug and requires:

1. Doctors order.
2. Signed “informed consent form”.

2.16.01 Identify the following features and specifications for “Low Range Ion-Chamber” instruments:

EBERLINE RO-20

The Eberline RO-20 is an **air-vented ion chamber dose rate meter** used to measure beta and photon radiation.

Zero position works in conjunction with the Zero knob to electronically zero the meter.

The Batt 1 and 2 positions check the condition of the batteries.

Eberline RO-20:

- Detector type: Air-vented ion chamber
- Deep dose measurements may be obtained through the front, sides, and bottom of the instrument.
- Operator adjustable controls: Off, Batt 1, Batt 2, Zero, adjust range
- 5 ranges: 50R/h, 5R/h, 500mR/h, 50mR/h, 5mR/h
- Chamber walls and beta shield density-thickness: 400mg/cm^2
- Beta window density-thickness: 7mg/cm^2

RO-20 Specific Limitations / Characteristics:

Sensitive to changes in atmospheric pressure. The RO-20 is fully temperature compensated and not sensitive to changes in temperature within the specified temperature range (-22 to 122° F).

If the metal case touches grounded metal objects, reading may drop suddenly to zero (with recover after a delay).



	RO-20
Detector Type	Air-vented ion chamber
Radiation Detected/Measured	Beta, gamma, x ray
Ranges	0 - 5 mR/h 0 - 50 mR/h 0 - 500 mR/h 0 - 5 R/h 0 - 50 R/h
Energy Dependence	WC: Gamma >50 keV WO: Gamma >2 keV Beta >65 keV
Chamber Wall Density-Thickness	640 mg/cm^2 plastic inside 0.063 in. Al Total ≈ 1,000 mg/cm^2
Beta Shield Density-Thickness	1,000 mg/cm^2 (sliding window)
Alpha Window Density-Thickness	N/A
Beta Window (Thin End-Window) Density-Thickness	7 mg/cm^2 total (two layers of Mylar)
Chamber Sensitive Volume	220 cm^3
Temperature (±10 percent)	–22 - 122 °F
Response Time	mR/h ranges < 5 s R/h ranges < 2 s
Check Source	ICCS or LBS

2.16.02 Given a described source of radiation and a RO-20 correction factor chart,

calculate the equivalent dose rate:

Source size and distance relative to the detector can lead to only partial ionization of the fill gas, causing a reduced output signal. Geometry Correction Factors are used to mathematically determine the true dose rate.

An RO-20 can be used for dose rate measurements on contact. RO-20s (*and Ludlum 9-3's, which are not yet covered in this book),* are the only instruments with correction factors for on contact distance.

$$DDR = (WC \times CF\gamma) + \eta \qquad SDR = [(WO - WC) \times CF_{\beta}] + DDR$$

DDR = Deep dose rate,
SDR = Shallow dose rate,
WC = Window closed reading,
WO = Window open reading,
CFγ = Gamma (penetrating) correction factor,
CF_{β} = Beta (non-penetrating) correction factor,
η = Neutron meter reading.

RO-20 Correction Factors for Disk Sources:

Table 5. RO-20/RO-20AA. Contact and 1/8 in. Correction Factors for Disk Sources

Disk Diameter cm (in.)	Correction Factor for Contact Measurements[a]		
	For exposure rate on contact with source (report beta dose rate after correction in rem/h or mrem/h)		For exposure rate at 3 mm (1/8 in.) from source (report beta dose rate after correction in rem/h or mrem/h)
	Beta[a, b]	Gamma[a, c]	Beta[b]
1.27 (0.5)	250	54	180
2.54 (1.0)	46	26	55
5.08 (2.0)	15	8	13
7.62 (3.0)	6	5	6

Note:[d] When making measurements where the meter is >1 in. from the source and for beams ≥ 3 in., or for sources ≥ 18 in., the general correction factors below are used:

- 2 for WO beta readings
- 1 for WC gamma readings.

[a] Measured with source surface on contact with the beta shield guide rails

[b] Non-penetrating radiation correction factors based on ^{204}Tl ($beta_{max}$ = 0.763 MeV), except for the general beta-gamma correction factor of 1, which is based on $^{90}Sr/Y$

[c] Penetrating radiation correction factors based on ^{137}Cs with the source in contact with the beta shield

[d] Field geometry correction factors can be applied to objects with a smallest dimension ≥ 18 in., if dose rates are uniform across the surface.

Note: Disk correction factors may be used for measuring cylinders with the RO-20/RO-20AA.

1. Contact geometry corrections factors are applied to RO-20/RO-20AA measurements taken with the RO-20/ RO-20AA beta shield guide rails on contact with the source. These correction factors are listed above and on the instrument.
2. When determining correction factor values and the item of concern has two measurements, e.g., 5 in. and 2.5 in., choose the smaller measurement. When the diameter of the item is between two available values in the table, e.g., between 2 in. and 1 in., choose the smaller diameter.

RO-20 Correction Factors for Beams:

- Small-beam correction factors are used when the radiation exists as a beam that is too narrow to uniformly ionize the air in the chamber. This occurs when the beam diameter is <2-in coaxial to the chamber or <3-in perpendicular to the chamber. Beam correction factors are shown in Table 4:

Table 4. Beam Correction Factors for the RO-20/RO-20AA

Beam Diameter, cm (in.)	Correction Factor for Gamma Beams	
	Perpendicular to Chamber	Coaxial with Chamber
< 1.27 (0.5)	97	129
1.27 (0.5)	24	32
1.9 (0.75)	11	14
2.54 (1.0)	6	8
3.81 (1.5)	3	4
5.08 (2.0)	2	2
≥ 7.62 (3.0)	1	1

2.16.03 Identify the following features and specifications for "High Range" instruments:

EBERLINE RO-7

The RO-7 provides remote monitoring in high-range beta and gamma radiation fields. It consists of a digital readout, and three interchangeable detectors that may be connected by flexible cables of different lengths.

Eberline RO-7:

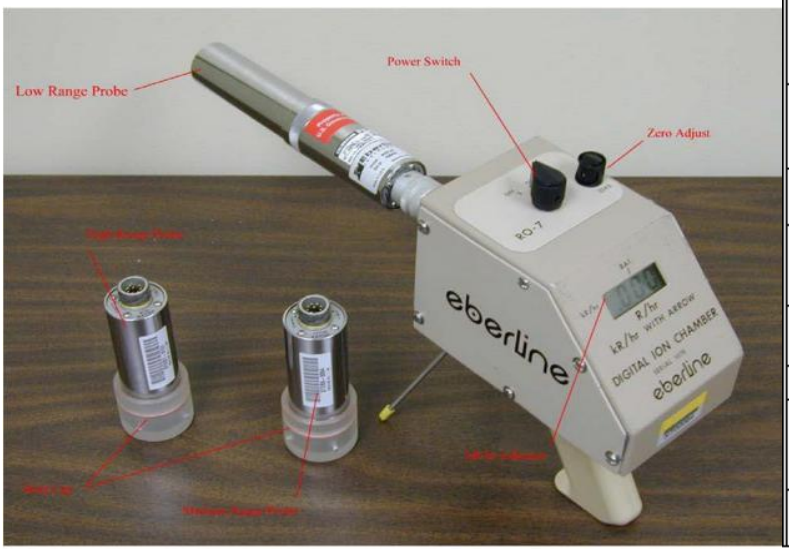
- Detector type: Air-filled ion chamber
- The RO-7 has a digital scaler and digital display capable of reading from 1mR/h to 19.9K R/h depending on the attached probe.
- ON/OFF are the only switch positions because the instrument identifies the detector and adjusts the readout accordingly. The zero knob will zero the LCD readout. A small switch operates the meter face light.
- Three interchangeable detectors:
    - **RO-7-LD**: 1 mR/h to 1.99 R/h
        - Low range, gamma only detector
        - Density-thickness of 800mg/cm^2
    - **RO-7-BM**: 100 mR/h to 199.9 R/h
        - Mid-range, beta-gamma detector
        - Probe chamber wall density-thickness: 800mg/cm^2
        - Mylar window on end of probe (cap removed) density-thickness: 7mg/cm^2
        - Lucite cap density-thickness: 1000mg/cm^2 on the end and 200mg/cm^2 on the sides
    - **RO-7-BH**: 10 R/h to 19.9k R/h
        - High-range, beta-gamma detector
        - Probe chamber wall density-thickness: 800mg/cm^2
        - Mylar window on end of probe (cap removed) density-thickness: 7mg/cm^2
        - Lucite cap density-thickness: 1000mg/cm^2 on the end and 200mg/cm^2 on the sides
- Interconnecting devices from detector to instrument:
    - 15-foot and 60-foot flexible cables
    - 2-foot and 5-foot rigid extensions
    - Stainless steel underwater hosing with 60 ft of cable

RO-7 Specific Limitations / Characteristics:

The response time of the basic instrument averages 2.5s to 90% of the final reading. Since the detector is air-vented, atmospheric temperature and pressure changes affect the instrument reading. The instrument response will remain within ±10% within the specified temperature range (-20 to 160° F).

Keeping the mid-range or modified mid-range detectors in radiation fields >200 R/h or low-range in fields >2 R/h for extended periods can permanently damage the detector regardless of instrument being on.

The detector should be kept dry and out of high humidity environments to prevent leakage currents.



	RO-7
Detector Type	Air-vented ion chamber (three)
Radiation Detected/Measured	Beta, photon
Ranges	Three detectors: LD: 1 mR/h - 1.999 R/h BM: 100 mR/h - 199.9 R/h BH: 10 R/h - 19.99 kR/h
Energy Dependence	Low: Photon >50 keV Mid/High: Photon >25 keV Beta >200 keV
Chamber Wall Density-Thickness	400 mg/cm^2 phenolic housed in 1.5 mm Al (800 mg/cm^2 total)
Beta Shield Density-Thickness	LD: None BM/BH: 1,000 (end) 200 (side) mg/cm^2
Alpha Window Density-Thickness	N/A
Beta Window (Thin End-Window) Density-Thickness	7 mg/cm^2 total
Chamber Sensitive Volume	LD: 50 cm^3 BM/BH: 7 cm^3
Temperature (±10 percent)	–20 - 160 °F
Response Time	2.5 s - 90 percent of final reading on all ranges
Check Source	ICCS or LBS

WILLIAM B. JOHNSON EXTENDER

The Johnson extender detector consists of two GM tubes at the end of a telescoping probe that extends to a length of 12 ft. The centerlines of the detector are marked on the probe with colored lines. The forward tube has a white band that matches the three applicable meter face scales. The rear tube has a yellow band on the housing that matches the four ap-applicable meter face scales.

William B. Johnson Extender:

- Detector type: GM (consists of two GM tubes)
- Instrument has a switch on front of body that controls light and sound ON/OFF.
- Instrument has a second switch on side of body with 10 positions: Off, Battery Check, Battery Check, and 7 ranges (4 yellow, 3 white).
- 7 instrument ranges:
    - White (low-range): 0-1,000 uR/h, 0-10 mR/h, 0-100 mR/h
    - Yellow (high-range): 0-1000 mR/h, 0-10 R/h, 0-100 R/h, 0-1000 R/h

Johnson Extender Specific Limitations / Characteristics:

Can be used to measure dose rates in gamma radiation fields only. Instrument normally responds to gamma but will respond to beta and low energy gamma when low-range cap is removed. This instrument is NOT to be used for setting personnel dose rates of <5 R/h unless RadCon supervision approves it.

Do not extend the detectors rapidly. Not allowing pressure to equalize slowly may damage the detector. Do not twist while collapsing or extending.





	Extender 2000W
Detector Type	GM (two)
Radiation Detected/**Measured**	Beta, **gamma, x ray**
Ranges	Yellow scale: 0 - 1,000, 0 - 100, 0 - 10 R/h, 0 - 1,000 mR/h White scale: 0 – 100, 0 - 10 mR/h, 0 - 1,000 µR/h
Energy Dependence	Low Range: Gamma >65 keV Beta >150 keV High Range: Gamma >83 keV
Chamber Wall Density-Thickness	N/A
Beta Shield Density-Thickness	N/A
Alpha Window Density-Thickness	N/A
Beta Window (Thin End-Window) Density-Thickness	Low: 2 - 3 mg/cm^2 High: N/A
Chamber Sensitive Volume	Low: 2.5 cm^3 High: 0.12 cm^3
Temperature (±10 percent)	–4 - 140 °F
Response Time	Ranges 1, 2, 5: 1.5 s Ranges 3, 4, 6, 7: 4 s
Check Source	ICCS or LBS

2.16.04 Identify the following features and specifications for "Neutron Detection" instruments

THERMO-FISHER SCIENTIFIC RadEye N/NL

The RadEye is a portable survey instrument designed for measuring equivalent dose rate from neutrons. The detector consists of a ^{3}He proportional counter contained in an electronic unit which is encased in a polyethylene moderator with a boron-loaded cover plate to reduce the effects of scatter. The electronic unit can be separated from the polyethylene moderator for maintenance.

Thermo Fisher Scientific RadEye N/NL:

- Detector type: ^{3}He proportional
- Maximum range of ≈ 60,000 mrem/h
- Responds to a neutron energy range of thermal 0.025eV – 10MeV.
- Responds negligibly to gamma fields ≥1 R/h and is insensitive to alpha and beta radiations.
- Operator-adjustable controls: Knob to loosen/open cover plate, Up/Down arrows, ON button, Speaker button.

RadEye Specific Limitations / Characteristics:

- Not affected by temperature shock
- Operable over the humidity range of 10-95% relative humidity & protected against water jets.
- Immune to EMI found in industrial environments and is not known to be susceptible to RF interference.
- Battery life is ≈ 500 h under normal use conditions.
- Responds negligibly to gamma radiation in fields of 1 R/h

	RADEYE N/NL
Detector Type	^{3}He
Radiation Detected/**Measured**	Neutron
Ranges	0.04 mrem/h - 60 rem/h
Energy Dependence	0.025 eV - 10 MeV
Chamber Wall Density-Thickness	N/A
Beta Shield Density-Thickness	N/A
Alpha Window Density-Thickness	N/A
Beta Window (Thin End-Window) Density-Thickness	N/A
Chamber Sensitive Volume	Unknown
Temperature (±10 percent)	–4 - 122 °F
Response Time	Allow a minimum of the time from the table below or 60 s for a standard < 0.5 mrem/h minimum detectable dose rate. In no case should the count time be < 5 s.
Check Source	Neutron check source or check point



2.16.05 Identify the following features and specifications for "Scintillation Detection" Instruments:

BICRON MICRO REM METER

The Bicron Micro-REM meter uses an energy-compensated, tissue-equivalent NaI(Tl) (thallium-activitated sodium iodide) organic scintillator to provide a flat energy response calibrated in rem.

Bicron Micro-REM meter:

- Detector type: NaI(Ti) Energy compensated, tissue equivalent, organic scintillator.
- Responds to Gamma energies 40 keV – 1.2 MeV.
- Insensitive to α, β, and neutron radiation.
- Operator adjustable controls: Off, Bat, HV, adjust range
- 5 ranges: ×1000, ×100, ×10, ×1, ×0.1
- Instrument operating range: 0 to 200,000 μRem/hr.

Bicron Micro-REM Specific Limitations / Characteristics:

The high voltage supply (HV) must be checked prior to each use and periodically during extended use.

Typically, only the ×1, ×10, ×100 scaled are used due to the inability to source check the ×0.1 and ×1000 ranges.

Useable at temperatures from -4 to 122 °F and useable for all ambient humidity and pressure conditions at Hanford.



	Micro Rem Meter
Detector Type	Sodium iodide scintillation NaI(Tl) 1 × 1 in.
Radiation Detected/**Measured**	Low-level photon energy-dependent
Ranges	0 - 20, 0 - 200, 0 - 2,000, 0 - 20,000, 0 - 200,000 μR/h
Energy Dependence	Variable – calibrated for energy to be measured
Chamber Wall Density-Thickness	Tissue equivalent
Beta Shield Density-Thickness	N/A
Alpha Window Density-Thickness	N/A
Beta Window (Thin End-Window) Density-Thickness	N/A
Chamber Sensitive Volume	N/A
Temperature (±10 percent)	–4 - 122 °F
Response Time	Fast: 3 s for 90 percent full meter deflection Slow: 11 s for 90 percent full meter deflection
Check Source	μrem check source

NOTE: The Ludlum 194 Microrem meter uses a plastic scintillator detector, only has 3 switch positions (OFF, LOW range, and HI range), has an audible alarm, and has a reset button to quickly reset zero. These facts are stated as "newer instrument" comparisons to the Bicron Microrem but the 194 is otherwise not explained or addressed in detail.

2.16.06 Discuss the factors that affect selection of a portable radiation survey instrument and identify appropriate instruments for external radiation surveys:

Selecting the proper instrument is based on instrument specifications and measurements needed. Factors include:

- *Proper instrumentation for the job:*
  - Determine whether external radiation surveys and/or contamination monitoring is needed.

- *Type of data required*:
  - Instruments that read R/h, mR/h, rem/h, mrem/h for radiation surveys
  - Instruments that read out counts per minute for contamination.

- *Types of Radiation Detected/Measured:*
  - Ion chambers have a flat response for games and are closest to being tissue equivalent. Also useful for measuring beta although a correction factor may be needed since they are calibrated with gamma sources (e.g., RO-20).
  - Gas proportional detectors distinguish between alphas and betas and often discriminate against (reject) gammas and betas unless the beta window is not too thick (e.g., RadEye).
  - ZnS scintillation detectors only detect alphas (e.g., PAM, 43-93/2360).
  - NaI scintillation detectors are used to detect gammas (e.g., Bicron Micro-REM).

- *Detector Window/Shielding/Density-thickness:*
  - Many external radiation survey instruments incorporate a moveable beta shield to discriminate between beta and gamma radiations. The beta shield also attenuates low-energy gammas.
  - Instruments are designed to measure the dose received by the extremities, skin, and the whole body (using correction factors as appropriate).
    - The RO-20 has a detector wall with density-thickness of 1,000 mg/cm^2 which corresponds to whole-body dose and a beta window with density-thickness of 7mg/cm^2 which corresponds to skin (and extremities when assessed at a depth of 0.007cm).

- *Intensity of the radiation to be measured:*
  - Alpha and low-energy beta will not penetrate the window of most external radiation instruments.
  - GM detectors over-respond to low-energy gammas.
  - Most instruments under-respond to high-energy neutrons.

- *Environmental factors:*
  - Extreme temperatures, High humidity, Magnetic fields, Atmospheric pressure, Radioactive gas

Density-thickness is an important component of many instruments. If the isotope(s) of concern are known, the *Chart of the Nuclides* and the *Beta Range-Energy Curve* graph (below) can be used to determine if an instrument has an appropriate density-thickness.

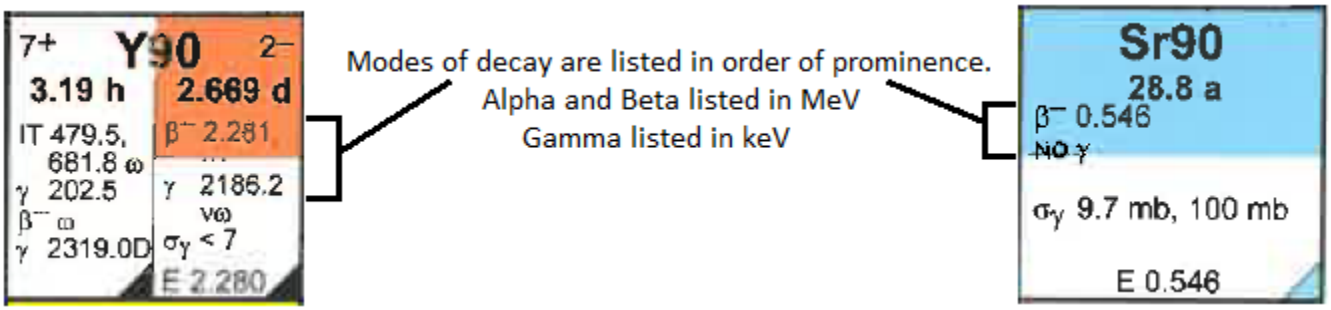
In other words, it should be determined whether the radiation emitted from the isotope(s) can penetrate the instrument's detector chamber wall, beta window, beta shield, or supplemental dosimeter. If it cannot attenuate through the instrument's shielding, it will not reach the detector and cannot be read by the instrument.

To use the *Beta Range-Energy Curve:*

- Find the isotope(s) of concern in the *Chart of the Nuclides.*
- Locate the beta (β^-) energy value within the isotope's info square.
- Find the beta energy value on the Y-axis on the *Beta Range-Energy Curve*
- Follow it across and see where it intersects with the energy-range curve (curved line)
- Follow straight down from the intersection point to find the value for the density-thickness (in mg/cm^2).

Example used in the book (^{90}Y and ^{90}Sr):

- Found on the nuclide info squares, the beta energy values for ^{90}Y and ^{90}Sr are 2.281 and 0.546, respectively.
- Finding where those values intersect with the energy-range curve gives density-thickness values of 1,100 mg/cm^2 and 170 mg/cm^2, respectively.
- Based on this, it can be determined that if an instrument with a chamber wall density-thickness of 800 mg/cm^2 is used (RO-7), ^{90}Y beta emissions will penetrate the chamber wall while ^{90}Sr beta emissions will not.



Note: How to read the information of each nuclide/isotope square can be found on *Chart of the Nuclides* book, page 44.

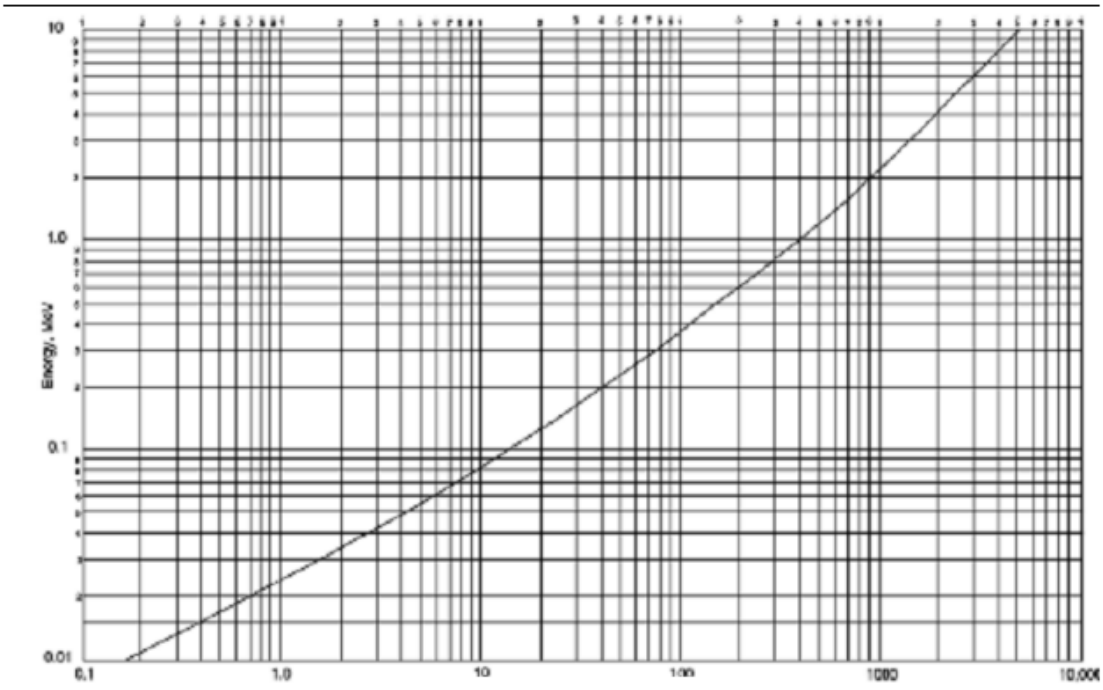


Figure 17. Beta Range-Energy Curve

Note: Y-axis represents beta energy in MeV; X-axis represents density-thickness of instrument/shielding

2.16.07 Describe the preoperational checks that are performed prior to using a survey instrument.

Preoperational checks performed prior to using a survey instrument:

- Calibration Sticker
- Physical Damage
- Battery check
- Zero adjustment
- Source check

2.17.01 List the factors that affect selection of a portable contamination monitoring instrument:

Factors to consider when selecting contamination instruments:

- Type of radiation to be measured.
- The energy of the radiation to be measured.
- The intensity of the radiation (activity levels)
- Interference from mixed radiation fields.
- Background radiation levels.
- Environmental factors (radioactive gases, temperatures, etc.).
- Procedural requirements.

2.17.02 Describe the following features & specifications for commonly used count rate meter probes used for βγ & or α surveys:

GM "PANCAKE" PROBES (EBERLINE HP-210 AND HP-260)

The Eberline Model HP-210/-260 hand probes are sensitive beta detectors that use a thin-window "*pancake"* GM detectors. The difference between the HP-210 and HP-260 is the inclusion of shielding on the HP-210.

Eberline Model HP-210/-260:

- Detector type: Sealed GM pancake detector
- Mica window density-thickness: 1.4 to 2.0 mg/cm^2
- Responds to alpha >3 MeV, beta >100 keV, and gamma >6keV. Only measures beta.

Eberline Model HP-210/-260 Specific Limitations / Characteristics:

Humidity and temperature do not affect the detector because it is sealed at a pressure lower than atmosphere.

The mica window is fragile, and it cannot function with a broken mica window.

Note: The "HP-210" shielded probe is available with different high-density housings:

- HP-210AL (Aluminum)
- HP-210L (Lead)
- HP-210T (Tungsten, the mostly commonly used version at Hanford)

	HP-210/260
Detector Type	GM pancake
Radiation Detected/ **Measured**	Alpha/**beta**/gamma
Ranges (cpm)	N/A
Energy Dependence	Alpha >3 MeV Beta >100 keV Gamma >6 keV
Efficiency (%)	Beta (std) Gamma < 1 Alpha 0 - 20
Window Density-Thickness (mg/cm^2)	1.4 - 2.0
Fill Gas	Halogen-quenched Ar
Interfering Radiation Sensitivity	≈ 3,600 cpm/mR/h (^{137}Cs)
Shielding	Tungsten (HP-210)
Shielding Reduction	4:1
Window Material	Mica
Detector Active Area (cm^2)	15.5
Temperature (°F)	N/A
Response Time (s)	N/A
Operating Voltage	≈ 900 V dc
Dead Time (µs)	50
Sources	N/A



Figure 1. Eberline Model HP-260



Figure 2. Eberline HP-210 Probe

LUDLUM 43-93 PROBE (100cm^2 PROBE FOR 2360)

The Ludlum 43-93 is a dual alpha and beta scintillator probe.

Ludlum Model 43-93 Dual Alpha-Beta Probe:

- Detector type: A thin silver-activated zinc sulfide (ZnS(Ag)) scintillator for alpha and a thin plastic scintillator for beta.
- Mylar density thickness: 1.2 mg/cm^2
- Responds to alpha, beta, gamma, and neutron. Only measures alpha and beta.

Ludlum Model 43-93 Specific Limitations / Characteristics:

The 43-93 is calibrated (married) with its associated count rate meter. Exchanging probes invalidates calibration.

The thin mylar window is easily damaged and it cannot function properly with a broken mylar.

Varying levels of cross-talk will occur between the alpha and beta channels.



	43-93
Detector Type	ZnS(Ag) – alpha Plastic scint. – beta
Radiation Detected/ **Measured**	**Alpha/beta**/gamma/ neutron
Ranges (cpm)	N/A
Energy Dependence	Alpha >3 MeV Beta >100 keV
Efficiency (%)	^{99}Tc = 15, ^{90}Sr/Y = 20 ^{239}Pu = 20 10 (std) CF = 10
Window Density-Thickness (mg/cm^2)	1.2
Fill Gas	N/A
Interfering Radiation Sensitivity	28,000 cpm/mR/h γ 350 cpm/mrem/h η
Shielding	N/A
Shielding Reduction	N/A
Window Material	Mylar
Detector Active Area (cm^2)	100
Temperature (°F)	–7 - 122
Response Time (s)	N/A
Operating Voltage	200 - 2000 V dc
Dead Time (µs)	≤ 5
Sources	N/A

2.17.03 Describe the following features and specifications for commonly used count rate instruments:

PAM (PORTABLE APLHA METER) body and detector

The PAM is a portable alpha count rate meter that consists of a count rate meter body (called the Bicron Surveyer X) and an alpha scintillation detector.

PAM (Portable-Alpha-Meter):

- Detector type: ZnS(Ag) scintillation
- Commonly used with a 54cm^2 probe head. A 100cm^2 probe head is available.
- 2 layers of mylar with a density-thickness of 0.58mg/cm^2 (54cm^2 probe) or 0.82mg/cm^2 (100cm^2 probe)
- Detects and measures only alpha >4 MeV
- Operator adjustable controls: Off, Batt, HV and a separate switch for speaker On/Off
- 3 ranges: ×1, ×10, ×100
- Max range: 100,000 cpm

PAM Specific Limitations / Characteristics:

Has a radioactive source attached to the side (typically 0.85 nCi of ^{230}Th).

Not affected by temperature shocks or ambient humidity but may be sensitive to electromagnetic fields.

The probe is calibrated (married) to its associated count rate meter.



	PAM
Detector Type	ZnS(Ag) scintillation
Radiation Detected/ Measured	**Alpha**
Ranges (cpm)	0 - 1,000 0 - 10,000 0 - 100,000
Energy Dependence	Alpha >4 MeV
Efficiency (%)	16 (std) CF = 6
Window Density-Thickness (mg/cm^2)	0.82 (100 cm^2) 0.58 (54 cm^2)
Fill Gas	N/A
Interfering Radiation Sensitivity	>1 R/h γ >500 mrad/h β
Shielding	N/A
Shielding Reduction	N/A
Window Material	Mylar (2 layers)
Detector Active Area (cm^2)	54 or 100
Temperature (°F)	–22 - 122
Response Time (s)	0 - 1,000: 12 0 - 10,000: 5 0 - 100,000: 2
Operating Voltage	≈ 1200 V dc
Dead Time (µs)	N/A
Sources	Response: 0.85 nCi ^{230}Th Source: 0.15 µCi ^{230}Th

EBERLINE E-140 GM (Body)

The Eberline E-140 Count Rate Meter is a beta contamination instrument.

Eberline E-140 Count Rate Meter:

- Detector type: The normal detector is a pancake GM probe (HP-210 or HP-260).
- Operator adjustable controls: Off, Batt, adjust range
- 3 ranges: ×1, ×10, ×100
- Max range: 100,000 cpm
- Responds to alpha, beta, gamma. Only measures beta.

Eberline E-140 Special Limitations / Characteristics:

Even in mixed fields, the response is mainly beta due to limited range of alpha and poor interaction of photons.
Battery life is ~400 h.
Not affected by temperature shocks but good practice is to allow it to acclimate to ambient temperatures.

Because of long resolving times, high count rates cause a decrease in the counting efficiency and measured counts may be as much as 40% below actual count rate. Another resolving time issue is that if radiation events occur at too high a frequency, a string of pulses may prevent the detector from resolving and producing a measurable pulse. If a measurable pulse does not occur, the electronics will show it as no radiation detected.



	Eberline E-140N
Detector Type	HP-210/-260 GM
Radiation Detected/ **Measured**	Alpha/**beta**/ gamma
Ranges (cpm)	0 - 1,000 0 - 10,000 0 - 100,000
Energy Dependence	Alpha >3 MeV Beta >100 keV Gamma >6 keV
Efficiency (%)	Beta (std) Gamma < 1 Alpha 0 - 20
Window Density-Thickness (mg/cm^2)	1.4 - 2.0
Fill Gas	Halogen-quenched
Interfering Radiation Sensitivity	≈ 3,600 cpm/mR/h (^{137}Cs)
Shielding	Tungsten (HP-210)
Shielding Reduction	4:1
Window Material	Mica
Detector Active Area (cm^2)	15.5
Temperature (°F)	–40 - 140
Response Time (s)	Variable: 2 - 10
Operating Voltage	≈ 900 V dc
Dead Time (µs)	50
Sources	Response: 0.3 µCi U_{nat} Source: 1 µCi $^{90}Sr/Y$

LUDLUM MODEL 3 GM (Body)

The Ludlum Model 3 is similar to the Eberline E-140 but has several notable differences:

- Smaller in size and weight
- 500,000 cpm max range instead of 100,000 cpm max range
- Has automatic dead time compensation and anti-saturation circuitry
- Four ranges instead of three
- Has knob for selecting a scaler count time of 0.1, 0.5, 1, 2, or 10 min

Ludlum Model 3 Count Rate Meter:

- Detector type: The normal detector is a pancake GM probe (HP-210 or HP-260).
- Responds to alpha, beta, gamma. Only measures beta.
- Operator adjustable controls: Off, Batt, adjust range
- 4 ranges: ×0.1, ×1, ×10, ×100
- Max range: 500,000 cpm

Ludlum Model 3 Specific Limitations / Characteristics:

Temperature range of -4 to 122 °F and battery life of ~2000 h.

In the 500,000 cpm range, dead time losses can be as much as 95 percent.

Like the E-140, if incident radiation occurs at a high rate, a string of pulses will occur, preventing the GM detector from completely recovering. The Model 3 is electronically compensated to reduce that effect.



	Ludlum Model 3
Detector Type	HP-210/-260 GM
Radiation Detected/ Measured	Alpha/**beta**/ gamma
Ranges (cpm)	0 - 500 0 - 5,000 0 - 50,000 0 - 500,000
Energy Dependence	Alpha >3 MeV Beta >100 keV Gamma >6 keV
Efficiency (%)	Beta (std) Gamma < 1 Alpha 0 - 20
Window Density-Thickness (mg/cm²)	1.4 - 2.0
Fill Gas	Halogen-quenched
Interfering Radiation Sensitivity	≈ 3,600 cpm/mR/h (^{137}Cs)
Shielding	Tungsten (HP-210)
Shielding Reduction	4:1
Window Material	Mica
Detector Active Area (cm²)	15.5
Temperature (°F)	–4 - 122
Response Time (s)	Variable: 4 - 22
Operating Voltage	≈ 900 V dc
Dead Time (µs)	50
Sources	Response: 0.3 µCi U_{nat} Source: 1 µCi ^{90}Sr/Y

LUDLUM MODEL 26/26-1 GM

The Model 26 and 26-1 are GMs with unapproved dose rate capabilities. The detector and readout electronics are combined into a single lightweight instrument that do not require a separate body and probe.

The 26 and 26-1 differences from E-140 and Model 3:

- Uses the same style GM pancake probe for detection
- Smaller in size and weight, combined unit does not require separate body and probe
- Has a digital readout with automatic electronic scaler, no needle face or manual range control

Model 26/26-1 GM:

- Detector type: It uses a standard pancake GM probe with a stainless steel screen.
- Responds to alpha, beta, gamma. Measures beta (dose rate is disabled, see Limitations below).
- Operator adjustable controls: “OK” (Turns unit On/Off and toggles audio while unit On) and “M” (Toggles Mode between Rate, Max, and Count). The center button “U” (Units) is disabled (see Limitations below).
- Range cpm: 0.00cpm up to 99.9 Kcpm
- Has built in diagnostic tests that will alert users of instrument failures and overloads.

GM Model 26/26-1 Specific Limitations / Characteristics:

The Model 26-1 is capable of measuring dose rates via electronic discrimination and unit conversion. However, only ion-chambers are approved for true dose rates at Hanford. Therefore, the “U” (Units) button to switch between displaying cpm and mR/h has been Disabled and is not functional at the Hanford site.

Temperature range of -40 to 158 °F and battery life of ~1,000 hours.





2360 (Body)

The Ludlum 2360 Scaler/Ratemeter Data Logger incorporates both analogue and digital functions. It can display alpha and/or beta radiation date in either a scaler mode (digital) or a ratemeter mode (analog).

Ludlum 2360:

- Detector type: ZnS(Ag) scintillator for alpha and plastic scintillator for beta (43-93 probe).
- Responds to alpha, beta, gamma, and neutron. Only measures alpha and beta.
- The 2360 has five external user-operated switches:
  - Eight position count time (MIN) switch: 0.1, 0.5, 1, 2, 5, 10, 60, PC
  - Six position rotary switch: Off, Batt, ×1000, ×100, ×10, ×1
  - Three position radiation type toggle switch: alpha only, alpha + beta, beta only
  - Three position toggle switch: Reset – [null] – Read HV
  - A pushbutton switch on the handle starts a scaler count for the count time selected.

2360 Special Limitations / Characteristics:

The 2360 has an internal digital scaler. By source checking any scale, the detector and all electrical circuits of the count rate meter are verified; it is unnecessary to perform a source check on each of the ranges.

The response time for the 2360 is 7 seconds on ×1 and ×10 scales and 2s on the ×100 and ×1000 scales.

An overload alarm (with red light illumination) occurs when the analog linear meter reaches a predetermined setpoint or a light leak is detected.

The 2360 speaker emits separate tones for alpha and beta radiation.



Ludlum 2360
(see 43-93)
Alpha/beta/gamma/ neutron
×1 = 0 - 500 ×10 = 0 - 5,000 ×100 = 0 - 50,000 ×1,000 = 0 - 500,000
Alpha >3 MeV Beta >100 keV
^{99}Tc = 15, ^{90}Sr/Y = 20 ^{239}Pu239 = 20 10 (std) CF = 10
1.2
N/A
28,000 cpm/mR/h γ 350 cpm/mrem/h η
N/A
N/A
Mylar
100
–7 - 122
×1, ×10 = 7 ×100, ×1000 = 2
500 - 1500 V dc
≤ 5
Exterior-mounted Pu/Cs response source; 100 cm^2 check source

2.17.04 Describe the following features and specifications for PCM's:



	PCM-1B
Detector Type	Gas flow proportional
Radiation Detected/ **Measured**	Alpha/**Beta**/Gamma
Energy Dependence	Alpha >3 MeV Beta >200 keV
Efficiency (%)	@ 3 in. = 11 - 15 Foot/hand ≈ 20
Window Density-Thickness (mg/cm^2)	0.29
Fill Gas	P-10 (Ar/CH_4)
Number of Detectors	15
Gamma Sensitivity (cpm/mR/h)	105,000 (^{137}Cs)
Window Material	Mylar (1 layer)
Detector Active Area (cm^2)	503 each
Temperature (°F)	32 - 122
Operating Voltage (V dc)	≈ 1,400
Sources (dpm)	5,000 ^{137}Cs 500 ^{239}Pu (if equipped)



	HFM-7/7A	HFM-6/6A
Detector Type	Gas flow proportional	Gas flow proportional
Radiation Detected/ **Measured**	**Alpha/Beta**/Gamma	Alpha/**Beta**/Gamma
Energy Dependence	Alpha >3 MeV Beta >200 keV	Alpha >3 MeV Beta >200 keV
Efficiency (%)	≈ 20	≈ 20
Window Density-Thickness (mg/cm^2)	0.29	0.29
Fill Gas	P-10 (Ar/CH_4)	P-10 (Ar/CH_4)
Number of Detectors	6	6
Gamma Sensitivity (cpm/mR/h)	105,000 (^{137}Cs)	105,000 (^{137}Cs)
Window Material	Mylar (1 layer)	Mylar (1 layer)
Detector Active Area (cm^2)	Hand: 323 each Foot: 503 each	Hand: 323 each Foot: 503 each
Temperature (°F)	32 - 122	32 - 122
Operating Voltage (V dc)	≈ 1,400	≈ 1,400
Sources (dpm)	5,000 ^{137}Cs 500 ^{239}Pu	5,000 ^{137}Cs 500 ^{239}Pu (if equipped)



	PCM-2
Detector Type	Gas flow proportional
Radiation Detected/ **Measured**	**Alpha/Beta**/Gamma
Energy Dependence	Alpha >3 MeV Beta >200 keV
Efficiency (%)	@ 3 in. = 11 - 15 Foot/hand ≈ 20
Window Density-Thickness (mg/cm^2)	0.29
Fill Gas	P-10 (Ar/CH_4)
Number of Detectors	34
Gamma Sensitivity (cpm/mR/h)	105,000 (^{137}Cs)
Window Material	Mylar (1 layer)
Detector Active Area (cm^2)	503 each
Temperature (°F)	–33 - 113
Operating Voltage (V dc)	≈ 1,400
Sources (dpm)	5,000 ^{137}Cs 500 ^{239}Pu (if equipped)



	Argos 5
Detector Type	Gas flow proportional
Radiation Detected/ **Measured**	**Alpha/Beta**/Gamma
Energy Dependence	N/A
Efficiency (%)	Alpha 10 Beta 20
Window Density-Thickness (mg/cm^2)	0.8
Fill Gas	P-10 (Ar/CH_4)
Number of Detectors	25
Gamma Sensitivity (cpm/mR/h)	N/A
Window Material	Aluminized Mylar
Detector Active Area (cm^2)	Total area = 14,475
Temperature (°F)	32 - 104
Operating Voltage (V dc)	≈ 1,600
Sources (dpm)	5,000 ^{137}Cs 500 ^{239}Pu (if equipped)

2.18.01 List the factors that affect the operator's selection of a portable air sampler/filter:

Monitoring of airborne radioactivity is performed:

- Where an individual is likely to receive an exposure of ≥40 DAC-h in a year, or
- As necessary to characterize the airborne radioactivity hazard where respiratory protective devices for protection against airborne radionuclides have been prescribed.

Criteria for selecting a portable air sampler/filter include:

- Purpose of the air sample
- Type of radiation emitted by the airborne contaminant
- Physical state of the airborne contaminant (including dust loading potential)
- Type and duration of job being performed
- Collection time and counting equipment available
- Availability of electrical power
- Physical space available for the sampler
- Environmental conditions
- Required analysis

2.18.02 Describe the specifications, application, and orientation of air filters used on the Hanford site:

Site facilities perform air monitoring for both alpha- and beta-emitting isotopes. Not all filters are equal for analysis of different types of radiation. Two primary filters are used (both have a collection efficiency >95 percent for 3 µm particles):

- *Versapore-3000*:
  - Chemically reactive and can therefore be used for destructive analysis
  - Useful for analysis of alpha and beta emissions
  - Sample collection occurs on smooth side of filter
  - Typically used for: All environmental samples, CAMs, routine fixed head monitoring

- *Fluoropore FS (FSLW 047 000)*:
  - Not appropriate for destructive analsys
  - Collection efficiency is 99.9 percent
  - Recommended for alpha spectrometry
  - Sample collection occurs on smooth side of filter
  - Typically used for: Grab samples, CAMs

For both filters, it is important that the smooth side is oriented as the collection side. Determining the smooth side can be accomplished using one or more of the following methods:

- Flashlight/Light Method
- Feel Method
- Divider Interaction

2.18.03 Identify the physical and operating characteristics, features and the limitation(s) of high volume portable air samplers:



Figure 1. Staplex High Volume Air Sampler

Staplex High Volume Air Sampler:

Operational Characteristics:

- Self-cooling
- Calibrated from 0 – 70 cfm. Flow rate sticker on side is specific for the collection method.
- Methods of sampling:
  - Filtration
  - Adsorption (charcoal)
  - Impaction (AKI head).

Physical Characteristics:

- 110V ac fan motor with On/Off switch
- Can attach intake screen with 4-inch filter holder or AKI head
- Tripods available or may be hung for optimum positioning
- Portable (weighs 10 lbs)

Limitations:

- Maximum 20 min run time per AKI sample (inappropriate for long term continuous sampling)
- May create airborne area due to exhaust
- Potential “crawler” while in operation due to high torque generated by the fan
- Cannot be used in explosive atmospheres

When used with an AKI head, particles enter from the rear of the sample head and must make a 180° turn at the collection point to exit/flow back out (escape) the sampler head without being collected. Heavier particles cannot make this turn while lighter particles, such as radon daughters, can. For this reason, the Staplex high vol sampler with AKI head reduces up to 95 percent of radon and thoron particles and is the recommended setup for ARA down-post air samples.



Figure 5. RADēCO H809

RADeCO H809-V1:

Operational Characteristics:

- Adjustable flow rate 1 – 8 cfm
- 110 – 225V ac
- Can use with hose extension for remote sampling
- Equipped with a rotameter airflow indicator
- Methods of sampling:
    - Filtration
    - Adsorption (charcoal or silver zeolite)

Physical Characteristics:

- Uses 47mm particulate and/or silver zeolite (for iodine) filters
- Three position switch (High, Off, Variable) and flow adjustment knob
- Portable (weighs 10 lbs)

Limitations:

- Cannot be used in explosive atmospheres
- Inappropriate for long term sampling
- Can create airborne radioactivity if exhaust too close to contaminated surfaces



Figure 6. RADēCO H810

RADeCO H810:

Operational Characteristics:

- Microprocessor-controlled airflow totalizer can be programmed to sample a preselected volume of air or a preselected time and then shuts off.
- Flow rate set at calibration but ranges from 8 – 12 cfm. Flow rate displayed on screen during operation.
- 120V ac
- Will sample up to 99,999 ft^3 or 99,999 L
- Display indicates cumulative volume, flow rate, and elapsed time during operation
- Data memory can store operational parameters up to 10 years
- Can be used with a hose extension for remote monitoring
- Methods of sampling:
    - Filtration
    - Adsorption (charcoal or silver zeolite)

Physical Characteristics:

- Sample head uses 47mm particulate and/or silver zeolite filters
- Portable (weighs 8.5 lbs)
- Instrument panel has Start, Stop, Units, Set, Enter, Clear, and Numeric keys for user operation

Limitations:

- Cannot be used in explosive atmospheres



Figure 7. HI-Q CF-900

HI-Q CF-900:

Operational Characteristics:

- Equipped with a rotameter airflow indicator
- Adjustable flow rate 1 – 5 cfm
- 110 – 125V ac
- Can be used with a hose extension for remote sampling
- Methods of sampling:
    - Filtration
    - Adsorption (charcoal or silver zeolite)

Physical Characteristics:

- Sample head uses 47mm particulate and/or silver zeolite filters
- Portable (weighs 9.5 lbs)
- Instrument panel has On/Off switch, flow adjustment knob, fuse holder, rotameter and timer (some models)

Limitations:

- Filtration
- Adsorption (charcoal or silver zeolite)

2.18.04 Identify physical & operating characteristics & limitation(s) of low volume air pumps/samplers.

Low volume air sampling systems are used for long-duration area air monitoring. Because of the low flow rates, some considerations should be given to short duration jobs (<30 minutes) where the small volume of air requires much longer count times to achieve the necessary minimum detectable concentration (MDC).

The following pertain to low volume air samplers:

Operational Characteristics of Typical Low Volume Air Pumps:

- Flow rate maintained constant by regulator
- Requires 110V ac power supply

Physical Characteristics of Typical Low Volume Air Pumps:

- Sample heads designed to accept 47mm media for particulates, iodine, or tritium, depending on head.
- Common components are carbon vane vacuum pump, constant flow air regulator, and a flow meter.
- Grounded three-conductor power cord is supplied

Limitations of Typical Low Volume Air Samplers/Pumps:

- Pumps may seize due to normal wear on carbon vanes and heating during use; may require several attempts to restart
- Flow meter should be as close as possible to the sampling head to minimize errors in measurements

---



Figure 8. Eberline RAS-1

Eberline RAS-1:

Typical Features:

- Rotameter flow meter with a flow rate range of 0.5 – 3.5 cfm
- Requires a 5 A power supply for the On/Off switch
- Screw-in type particulate filter holder, a “clamshell” type iodine filter holder, power switch
- Weight: 35 lb

Sampling considerations:

- Filter paper must cover the intake screen
- Charcoal/silver zeolite cartridge holder must have a good seal
- Check the flow rate after turning on and before turning off

---



Figure 9. Gooseneck

HI-Q Environmental Low Volume Sampling Systems ("Gooseneck"):

Typical Features:

- Rotameter with a flow rate range of 1 – 5 cfm
- Quick-disconnect particulate filter holder power switch, timer, variable height adjustment, contained on a cart for easy movement
- Weight: 45 lb

Sampling Considerations:

- Filter paper must cover intake screen
- Check flow rate after turning on and before turning off the sampler pump

2.18.05 Identify the steps for a preoperational checkout of a portable air sampler:

Preoperational checks for portable air samplers:

1. Verify that the air sampler has a current calibration sticker.
2. Ensure the correct filter is used for the application.
3. Physical damage
   - Power cord in good condition
   - All O-rings in place and not cracked
   - No missing or broken controls or cracked flowmeter
4. Working condition
   - Verify proper operation by checking sound, sight, smell, and feel
   - Ensure sampler is producing needed airflow
   - Verify controls are operable
   - Selected filters and/or cartridges are correct based on type(s) of contamination to be sampled/analyzed
   - Load filters and/or cartridges in proper orientation prior to sampling

2.18.06 Describe the criteria for and the physical and operating characteristics and the limitations of βγ CAMS:

Real-time air monitoring is used in areas where unexpected increases in airborne radioactivity levels could result in an exposure to an individual exceeding 40 DAC-h in one week. This exposure is representative of a Committed Effective Dose (CED) of 100 mrem.

Filters in CAMs are normally changed weekly or more frequently depending on dust loading and facility needs. CAMs can be permanently located or deployed for a specific task.

General Limitations for beta-gamma CAMs:

- Low airflow – must be placed near or downwind of suspected source
- Poor response to low-energy beta
- Lead shield is ineffective for high-energy gamma; place CAM in low background area
- Responds to radon, thoron, and daughters that can produce fluctuating background
- Not portable – can weigh up to 500 lbs.



Figure 10. Eberline AMS-3 Series

The Eberline AMS-3 uses both physical shielding and electronic subtraction to detect and measure airborne concentrations of beta-emitting radionuclides even in the presence of ambient gamma fields.

Eberline AMS-3:

Physical Characteristics:

- Uses two GM thin-window pancake tubes each with a surface area of 15.5cm^2
  - One GM monitors the activity on the 47mm collection filter and the other GM monitors gamma background inside the chamber which is electronically subtracted.
- Responds to alpha >3 MeV, beta >40 keV, and gamma >6 keV. Measures beta and accounts for gamma.
- Has 3 alarm indicators: Rotating red beacon, Red alarm light, and Bell (audio).

Operating Characteristics:

- Requires 120V ac
- Requires receipt test after receiving from calibration to verify operation
- Has three Alarm Set Point (ASP) related controls:
  - Push to Set (Display alarm set point)
  - Set Control (Adjust alarm set point)
  - Acknowledge switch (Acknowledges active alarm; Resets audible alarm but visual alarm will remain active until count rate drops below ASP).

AMS-3 Limitations:

- Temperature range of 20 – 104 ^{o}F, but an optional heater can lower the range to 0 ^{o}F
- Nearby mobile communication transceivers (e.g., radios, cell phones) may create instrument response
- Nearby electric spark sources may cause response interference

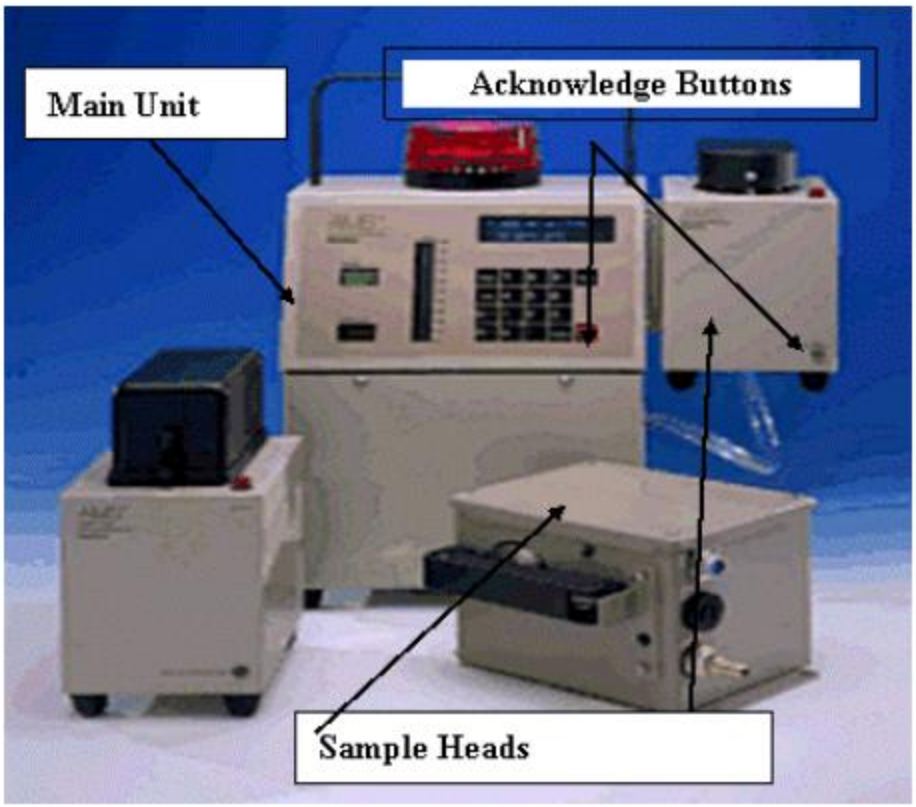


Figure 11. AMS-4 Beta CAM

The Eberline AMS-4 is microprocessor controlled and is the replacement for the AMS-3 family of CAMs as they need replacement.

Eberline AMS-4:

Physical Characteristics:

- Equipped with a radial inlet sampling head, or an inline head for stack monitoring, within which are two beta detectors. The sampling head has a swing out door that holds the filter paper.
- Responds to alpha >3 MeV, beta >100 keV, and gamma. Measures beta and accounts for gamma. Does not respond to neutron radiation.
- Has two identical sealed gaas porpoertional detectors.
    - The radial entry head detector is filled with argon-carbon dioxide (Ar-CO_2)
    - The inline head detector is filled with neon-carbon dioxide (Ne)-CO_2
    - The detector in the door counts the beta emissions and the second detector counts background gamma radiation for subtraction.
- Can be used with a detector-to-processor cable length up to 1000 ft.

Operating Characteristics:

- Configured with five user-modifiable activity Alarm Set Points (ASP)
    - Slow Alarm (checked every minute)
    - Fast Alarm (checked every 5 seconds)
    - Net Alarm (alarms if current beta net count rate exceeds the net ASP; checked every 5 seconds)
    - Stack Alarm (alarms if current release rate is equal or greater than ASP; checked every minute)
    - DAC-h Alarm (alarms if current total DAC-h reading is equal or greater than ASP; checked every minute)
    - Any of the alarms can be disabled by setting that alarm's ASP to 0

- The AMS-4 performs internal checks for system failures and will illuminate a malfunction light if it detects an issue with any of the following:
    - Min Flow Rate, Max Flow Rate, Min Beta Count Rate, or Max Beta Count Rate

Specifications and Limitations:

- Designed for indoor use and temperature range of -4 – 104 ^{o}F
- Designed for use at atmospheric pressure and should not be used within an operating vacuum or pressure vessel
- Nearby mobile communication transceivers (e.g., radios, cell phones) may create instrument response
- Nearby electric spark sources may cause response interference

2.18.07 Identify the physical and operating characteristics and the limitations of α CAMS:

Alpha CAMs are not normally used to determine low DAC concentrations due to inherent limitations when trying to detect alpha-emitting radionuclides; however, the collection media can be separately analyzed using low level alpha counting equipment to achieve the necessary MDC.

General Limitations for Alpha CAMs:

- Does not give an accurate quantitative alpha measurement, but warns of increased activity
- Dust buildup and radon-thoron activity affect efficiency



Figure 12. Eberline Alpha Air Monitor Model 5

Eberline Alpha-4/5/5A:

Physical Characteristics:

- Alpha CAMs use a 25 mm-diameter solid state diffused junction (semiconductor) detector to monitor activity collected on a 47 mm-diameter filter paper.
- Maximum sensitivity under laboratory conditions is ~4 DAC-h for ^{239}Pu. Typical operating alarm levels range from 8 – 40 DAC-h under field conditions.
- Workplace CAMs have a visual beacon and an audible alarm bell. Radiation alarms are typically non-latching (the alarm will self-clear when the count rate drops below the ASP).
- Semi-portable

Operating Characteristics:

- Requires 120V ac
- Has no internal radioactive sources
- Responds to alpha radiation exclusively due to detector design; however, also responds to naturally occurring alpha-emitting airborne particulate such as radon and thoron progeny.
    - Uses pulse height discrimination to differentiate between naturally occurring activity and nuclides of interest.
    - Beta and gamma are rejected by the thin active area of the detector.

Specifications and Limitations:

- Intended for indoor use with a temperature range of 19 – 129 ^{o}F
- Nearby mobile communication transceivers (e.g., radios, cell phones) may create instrument response and cause spurious alarms
- Particularly susceptible to strobe lights or light sources near the air inlet that can transmit (reflect) light into the detector
- CAMs located downstream of HEPA filters may not collect enough naturally occurring radioactivity on the filter to continuously provide a detector signal which may initiate failure alarms more often than acceptable.



Cranberra Alpha Sentry (and Alpha Sentry Sampling Head):

Physical Characteristics:

- Alpha Sentry CAM consists of a sampling head and controller. Up to eight sampling heads can be controlled from a single controller unit and can be configured with a wireless network.
- The air inlet is covered with a diffusion screen that removes more than 95 percent of unattached radon and radon progeny. This screen is most effective in treated air environments (e.g., laboratories, downstream of a HEPA filter).
- Semi-portable

Operating Characteristics:

- Provides alarm signals to indicate elevated levels of alpha emitting airborne radioactivity. The alarm signals include audible and visual alarms and actuating remote alarm relays.
- Provides a series of fault alarms that include:
    - High background, Low and high airflow, Detector bias power supply failure, Door open, No data acquisition, Excessive energy calibration shift, and Communication failure between control unit and sampling head
- Alarms may be latching or non-latching, depending on facility configuration

- By default, pressing Stop Alarm will silence the audible alarm but leave the visual alarm active (can be disabled through CAM software).

Specifications and Limitations:

- Flouropore membrane filter is recommended; Glass fiber filters discouraged due to self-absorption.
- Sensitive to visible light; Light leaks will generate spurious counts across the entire spectrum. Particularly susceptible to strobe lights.
- Operating temperature range 32 – 131 ^{o}F; however, the LCD screen may not function above 95 ^{o}F
- Influenced by electromagnetic and microwave fields.
- Influenced by radiofrequency fields and is sensitive to electrostatic shock.
- Appropriate for ambient humidity (0 – 95 percent) and pressure seen on the Hanford site. Should not be used in condensing environments.

2.18.08 Describe the routine CAM operational checks:

Response limits are set for each CAM when it is received from calibration.

Daily CAM checks include the following verifications:

- Positive airflow within tolerance
- CAM power on
- Non-zero response to background
- Chart recorder/meter operational and adequate paper supply
- Vacuum lines secure
- No debris or obstructions blocking airflow
- Counting light on
- Failure light off
- Sample plunger fully seated
- CAM calibration current
- CAM ASP is set & the Background Subtraction switch is on

Weekly source checks:

- Perform a source Check of the CAM each week to verify that the measured response is within the limits set for the source upon receipt from calibration.

Monthly alarm function test:

- Test each CAM in service for alarm operation. CAM alarms are initiated by placing a radioactive source into the filter holder.

2.19.01 Describe the features and limitations for commonly used laboratory counters or scalers:

Stand-alone counters or scalers measure gross activity, while spectroscopy systems perform spectral analysis to identify and quantify activity of specific nuclides.

National Institute of Standards and Technology (NIST)-traceable sources are used to check the systems. Common sources are ^{239}Pu for alpha counters, ^{90}Sr for beta counters, and mixed nuclide sources for gamma spectroscopy systems.

Procedures generally include:

1. Perform a background count.
2. Perform a source check.
3. Count a sample.
4. Perform background subtraction.
5. Divide by time to determine cpm.
6. Correct to 4π efficiency to determine dpm.
7. Record the data.
8. Maintain and update quality control (QC) charts.

Sample counting times will vary to obtain desired MDA or MDC values. The longer a sample is counted, the lower the MDA/MDC values. Background count times must be equal to or greater than sample count times.



Eberline SAC-4		Laboratory counter
	Detector type	ZnS(Ag) scintillator
	Detector shield	None
	Detector window	None
	Radiation measured	Alpha
	Controls	Timer: 0.1 - 50 min
	Source check	NIST-traceable ^{239}Pu

Eberline Scintillation Alpha Counter SAC-4:

Characteristics:

- The Eberline SAC-4 is a ZnS(Ag) scintillation alpha counter.
- Responds to and measures alpha >3 MeV.
  - A discriminator is set during calibration to ignore pulses from beta radiation.
- The detector and associated electronics are housed in a single unshielded box (no shielding).
- Uses an NIST-traceable ^{239}Pu check source.

Precautions/Limitations:

- Sample height is set during calibration and the efficiency values on the calibration sticker are valid for that position only. That sample height must be used for accurate analysis results.
- Accepts samples up to 2in diameter × 3/8 in thick; thicker samples result in too much self-absorption.



Ludlum Model 2929		Laboratory counter
	Detector type	ZnS(Ag) scintillator
	Detector shield	43-10-1: None; 43-78-5: Lead
	Detector window	None
	Radiation measured	Alpha, beta
	Controls	Timer: 0.1 - 990 min
	Source check	NIST-traceable ^{239}Pu and ^{90}Sr/Y

LUDLUM 2929 Alpha-Beta Dual Scaler:

Characteristics:

- The Ludlum 2929 is an ac-powered, two channel scaler that can be equipped to detect and measure alpha and beta radiation.
    - When equipped with a dual alpha-beta detector, the 2929 measures a sample for both gross alpha and gross beta with a single sample count.
- All 2929 detectors are ZnS(Ag) scintillators
- Responds to alpha >3 MeV and beta >100 keV. Beta response is energy-independent.
- Ludlum 2929 does not have specific check source requirement. Source checks typically use electroplated ^{239}Pu and ^{90}Sr disk sources.

Precautions/Limitations:

- Sample holder must be pushed in completely to trip a microswitch, enabling the detector voltage.
- Do not place samples with high activities (e.g., >25,000 dpm beta-gamma or >1,000 dpm alpha) in sample holder. Samples should be surveyed with portable instruments prior to placing in 2929.
- Must use the efficiency values used during calibration.
- Must use the correct sample holder or jig that it was calibrated with or geometry will be incorrect, resulting inaccurate analysis results.
- Intended for indoor use.
- When moving to another location, allow 2929 to reach equilibrium with ambient temperature (~1 hour).

2.19.02 Describe the specifications for low background automatic counting systems:



Tennelec		Low-background automatic
	Detector type	Gas flow proportional (P-10 gas) Larger guard detector for background
	Detector shield	Lead
	Detector window	80 µg/cm^2 Mylar
	Radiation measured	Alpha, beta
	Source check	^{239}Pu and ^{90}Sr

Tennelec:

Characteristics:

- The Tennelec is a low background automatic counting system used primarily to count smears or air filters.
- Has two detectors:
    - The primary detector is a gas flow proportional counter with 80 µg/cm^2 Mylar window used to count both alpha and beta. It is surrounded by lead shielding to reduce background.
    - A second larger gas proportional counter, the guard detector, is used to count gamma background. Together with the lead shielding, rejects up to 99 percent of background gamma.
    - Both detectors use P-10 gas (90 percent argon and 10 percent methane).
- Controlled by a desktop computer and associated software.
- Has an automatic sample changer with a dual stack that can handle up to 50 samples. One stack holds samples to be counted and the other holds samples that have been counted.



Liquid Scintillation Counter		
	Detector type	Liquid scintillation
	Detector shield	None
	Detector window	None (light-tight housing)
	Radiation measured	Low energy from ^{3}H or ^{14}C
	Source check	Set of 10 ^{3}H calibration sources

Liquid Scintillation Counters:

Certain nuclides (Tritium and ^{14}C) emit such low-energy betas that even a thin layer of air will stop them from reaching and being read by most detectors. Detecting this radiation is achieved with a liquid scintillation (LSC) system.

An LSC uses a *cocktail* (a sample immersed in a liquid scintillating medium) to maximize detection efficiency for low-energy betas. As the immersed sample emits low-energy betas into the liquid scintillating medium, the emitted betas are converted into light photons. The instrument sample chamber, which contains a photo-multiplier tube (PMT), is light tight allowing the instrument to count the light photons emitted from the cocktail.

LSC systems are typically used to count low-energy beta emitting samples from swipes and water samples.

2.19.03 Describe the features and limitations for commonly used γ spectroscopy systems:



Gamma Spectroscopy System		
	Detector type	Germanium
	Detector shield	Several inches of steel or an equivalent amount of lead
	Detector window	Usually none; 0.5 mm-thick Be or C fiber window for low-energy analysis
	Radiation measured	Gamma, x from 5 keV - 2 MeV
	Source check	NIST-traceable

Gamma Spectroscopy System:

Gamma spectroscopy mostly uses germanium detectors because the resolution obtained in these detectors allows different gammas with similar energies to be distinguished (resolved).

Characteristics:

- Gamma spectroscopy systems use an n-type HPGe (High-Purity Germanium) coaxial photon detector
- Detects gamma and x rays in the energy range of 5 keV – 2 MeV.
  - The peaks in the spectrum are close together, so high resolution is required to distinguish the peaks.
- Typically shielded by several inches of steel or an equivalent amount of lead.
  - The best shielding is steel or lead manufactured prior to World War II.
- Most HPGe detectors do not have a detector window.
  - Some low-energy HPGe detectors have a thin beryllium or carbon fiber window to allow gamma energy identification to ~5 keV.
- Energy and efficiency calibrations are obtained using NIST-traceable sources.
  - These are mixed sources that contain several gamma-emitting nuclides.